

Advanced Process Manager Configuration Forms

**AP88-600
09/08**

Implementation
Advanced Process Manager - 2

Advanced Process Manager Configuration Forms

AP88-600

Release 681

09/08

NOTE

This document was last updated for R681.
For information about those parameters introduced post R681,
refer to the PM Family Parameter Reference Dictionary.

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About This Publication

This publication supports **TotalPlant** Solution (TPS) System network Release 681. TPS is the evolution of TDC 3000. This manual consists of configuration forms. These forms are designed to facilitate the collection and organization of data for subsequent entry into the TPS system through the Universal Station.

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United States and Canada	Honeywell Solution Support Center	Honeywell TAC, MS P13, 2500 West Union Hills Drive, Phoenix, AZ, 85027	Phone: 1-800 822-7673 602- 313-5558 (In Arizona) Facsimile: (602) 313-3293
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Elsewhere	Call your nearest Honeywell office.		
World Wide Web	Honeywell Solution Support Online: http://www.honeywell.com/ps		
Training Classes	Honeywell Automation College: http://www.automationcollege.com		

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ADVANCED PROCESS MANAGER CONFIGURATION FORMS

APM CONFIGURATION FORM INSTRUCTIONS

FILLING OUT PAPER FORMS
ENTERING DATA FROM THE FORMS INTO THE SYSTEM

Form Name	Form ID
UCN Node Configuration	AP88-601A
APM Node Specific Configuration	AP88-601B

This publication also contains the following point forms in numerical sequence. They are listed here in alphabetical order for convenience.

Analog Input Data Point	AP88-602
Analog Output Data Point	AP88-603
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High Low Average Regulatory PV Algorithm	AP88-612
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PID Position Proportional Reg. Control Algorithm	AP88-633
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Totalizer Regulatory PV Algorithm	AP88-625
Variable Deadtime With Leadlag Reg. PV Algorithm	AP88-626

APM Configuration Form Instructions

FILLING OUT PAPER FORMS

Where to Find Advanced Process Manager Parameter and Data Point Help Information

Refer to the *APM Parameter Reference Dictionary* and to the *APM Control Functions and Algorithms* publications in the *Implementation/Advanced Process Manager – 2* binder.

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Universal Station's configuration display where values that have been written on the form and parameter options that have been circled on the form will be keyed in and selected, respectively.

APM Analog Input Point

Unit:	Tagname:	Engineer:	Date:
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Point Assignment Display

NAME Tag Name	_____	range = Not applicable default = Blank
PNTFORM Point Form	FuLL Component	Point is fully displayed and alarmed Pt. Is partially displayed but not alarmed

If the point is of the form **Component**, parameters within boxes that have **thick, bold borders** do not apply.

Lines indicate **what parameter/s or Parameter-option/s apply** when an option is selected for a previous parameter.

N/A if PNTFORM = Component.

PVSRCOPT PV Source Option	Only Auto	PV Source selection is not available and Field wiring or memory fetch supplies PV. Selection from PVSOURCE is available. And Operator/CL-program may supply PV.
PVSOURCE PV Source	Auto Man Sub	Field wiring or memory fetch supplies PV PV is supplied by Operator or program A value is substituted by a CL program

N/A if NMODE = None

NMODATTR Normal Mode Attribute	None Operator Program	Parameter NMODE sets "normal" mode Operator sets "normal" operating mode Program sets "normal" operating mode
-----------------------------------	------------------------------------	---

If parameters within a box do not apply when some other parameter option is selected, that box is indicated by

Lower-level alarm trip points must have a value specified before higher-level alarm trips points apply. The lines show the scheme.

Alarming Display N/A if PNTFORM = Component

PVHITP PV High Alarm Trip Point	_____	range = PVLOTP - PVHHTP, & NaN default = NaN
PVHIPP PV High Alarm Priority	NoAction Journal Low High Emergency Printer JnlPrint	
PVLOTP PV Low Alarm Trip Point	_____	range = PVLLTP - PVHITP, & NaN default = NaN
PVLOPR PV Low Alarm Priority	NoAction Journal Low High Emergency Printer JnlPrint	
PVHHTP PV Hi Hi Alarm Trip Point	_____	range = PVHITP to PVEUHI, & NaN default = NaN
PVHHPR PV High High Alarm Priority	NoAction Journal Low High Emergency Printer JnlPrint	
PVLLTP PV Low Low Alarm Trip Point	_____	range = PVEULO - PVLOTP, & NaN default = NaN

ENTERING DATA FROM THE FORMS INTO THE SYSTEM

Select **NETWORK INTERFACE MODULE** from Engr's Main Menu to get NIM BUILD TYPE SELECT MENU

UCN NODE CONFIGURATION Form AP88-601A	APM NODE SPECIFIC CONFIGURATION Form AP88-601B	PROCESS POINT BUILDING Forms AP88-602-632	NIM LIBRARY CONFIGURATION Form AP88-640
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Select a point type from this POINT BUILD SELECT MENU

Reg Ctl	AI/AO	DI/DO	Array	DevCtl	Pulse	Logic	Numeric	Proc Mod	Flag	Timer	Reg PV
---------	-------	-------	-------	--------	-------	-------	---------	----------	------	-------	--------

Select an algorithm type depending upon what is circled on the form for parameter CTLALGID

Pid	PidFf	PidErfb	RatioCtl	IncrSum	Switch	PosProp	AutoMan	RampSoak	ORSel	Pid PosPr
-----	-------	---------	----------	---------	--------	---------	---------	----------	-------	-----------

Select an algorithm type depending upon what is circled on the form for parameter PVALGID

DataAcq	FlowComp	MidOf3	HiLoAvg	Summer	VdtLdLag	TotaLizr	GenLin	CalcuLtr
---------	----------	--------	---------	--------	----------	----------	--------	----------

APM UCN NODE Configuration

AP88-601A

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	Engineer:	Date:
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UCN Node Configuration Display

NOTE

This form is intended primarily for use in configuring each PM on your system by selecting PM for NODETYP below. This same form can be used to configure each NIM on your system by selecting NIM for NODETYP (some different targets will appear as shown). If your system has other UCN devices (PMS, APMs, HPMS, LMs, SMs), the NIMs can also be configured using the forms contained in the documentation for those devices.

NTWKNUM Network Number Of The NIM Being Configured		<i>range = 1 to 20</i> <i>default = Not applicable</i>														
NODENUM UCN Address Of The Node Being Configured		<i>range = 1 to 64</i> <i>default = Not applicable</i>														
NODETYP Type Of Node On The UCN	<table style="width: 100%;"> <tr> <td style="width: 50%;">NotConfig</td> <td style="width: 50%;"></td> </tr> <tr> <td>Nim</td> <td> </td> </tr> <tr> <td>PM</td> <td> </td> </tr> <tr> <td>APM</td> <td> </td> </tr> <tr> <td>HPM</td> <td> </td> </tr> <tr> <td>LM</td> <td> </td> </tr> <tr> <td>SM</td> <td> </td> </tr> </table>	NotConfig		Nim		PM		APM		HPM		LM		SM		No UCN item configured at this address Network Interface Module Process Manager Network Advanced Process Manager Network High Performance Process Manager Network Logic Manager Network Safety Manager Network
NotConfig																
Nim																
PM																
APM																
HPM																
LM																
SM																
LOADSCOP Load Scope	NimOnly NimAndPm	Configured data goes to the NIM only Configured data goes to NIM, HPM, PM, APM, LM, or SM														
NMSGTXT Number of Message Text Items (see next page for MSGTXT ports)		<i>range = 0 to 15</i> <i>default = 0</i>														
TIMESYNC SOE Time Synchronization	Disable Enable	<i>range = Not applicable</i> <i>default = Disable</i>														
NODEASSN Node Assignment	ThisNim RemotNim	NIM assigned to report alarms and perform checkpointing for network specified in NTWKNUM Additional NIM on a network that already has "ThisNIM" defined														
PV State Text for all DigComp type points																
MOVPTXT PV-Moving Text For All DigComp Points		<i>range = Not applicable</i> <i>default = "MOVING"</i>														
BADPTXT Bad-PV Text For All DigComp Points		<i>range = Not applicable</i> <i>default = "BAD"</i>														

APM UCN Node Configuration

Status Message Text Configuration Display

N/A if NMSGTXT =0

MSGTXT(1) Status Message Text #1		<i>range = Eight Character String default = Blank</i>
MSGTXT(2) Status Message Text #2		<i>range = Eight Character String default = Blank</i>
MSGTXT(3) Status Message Text #3		<i>range = Eight Character String default = Blank</i>
MSGTXT(4) Status Message Text #4		<i>range = Eight Character String default = Blank</i>
MSGTXT(5) Status Message Text #5		<i>range = Eight Character String default = Blank</i>
MSGTXT(6) Status Message Text #6		<i>range = Eight Character String default = Blank</i>
MSGTXT(7) Status Message Text #7		<i>range = Eight Character String default = Blank</i>
MSGTXT(8) Status Message Text #8		<i>range = Eight Character String default = Blank</i>
MSGTXT(9) Status Message Text #9		<i>range = Eight Character String default = Blank</i>
MSGTXT(10) Status Message Text #10		<i>range = Eight Character String default = Blank</i>
MSGTXT(11) Status Message Text #11		<i>range = Eight Character String default = Blank</i>
MSGTXT(12) Status Message Text #12		<i>range = Eight Character String default = Blank</i>
MSGTXT(13) Status Message Text #13		<i>range = Eight Character String default = Blank</i>
MSGTXT(14) Status Message Text #14		<i>range = Eight Character String default = Blank</i>
MSGTXT(15) Status Message Text #15		<i>range = Eight Character String default = Blank</i>

APM Configuration

IO Module Configuration Table

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Engineer:	Date:
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IO Module Configuration Display – N/A if NTWKNUM = 0

For each IO Module, select the IO Module type (IOMTYPE) that applies. If there is a redundant IOM, check the Redun blank. If the type selected is AO or DO, select a failure option (FAILOPT). If the type is STIM OR LLMUX, select an ac Line Frequency (FREQ6050).

****CAUTION – Use this part of the form only if PKGOPT on page 2 = Redun_IO****

MOD NUM	IOM-A FILE	IOM-A CARD	Redun √	IOM-B FILE	IOM-B CARD																		
						IOMTYPE										FAILOPT		IOMTYPE		FREQ6050			

APM Configuration

IO Module Configuration Table

	Engineer:	Date:
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IO Module Configuration Display – N/A if NTWKNUM = 0

For each IO Module, select the IO Module type (IOMTYPE) that applies. If there is a redundant IOM, check the Redun blank. If the type selected is AO or DO, select a failure option (FAILOPT). If the type is STIM OR LLMUX, select an ac Line Frequency (FREQ6050).

****CAUTION – Use this part of the form only if PKGOPT on page 2 = Redun_IO (Continued)****

MOD NUM	IOM-A FILE	IOM-A CARD	Redun √	IOM-B FILE	IOM-B CARD	IOMTYPE										FAILOPT				IOMTYPE		FREQ6050	
						REDUN																	
						↓																	
21.	3	11	–	4	11	None	HLAI	LLAI	DI	PI	SI	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ	
22.	3	12	--	4	12	None	HLAI	LLAI	DI	PI	SI	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ	
23.	3	13	--	4	13	None	HLAI	LLAI	DI	PI	SI	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ	
24.	3	14	--	4	14	None	HLAI	LLAI	DI	PI	SI	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ	
25.	3	15	--	4	15	None	HLAI	LLAI	DI	PI	SI	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ	
26.	5	01	--	6	01	None	HLAI	LLAI	DI	PI	SI	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ	
27.	5	02	--	6	02	None	HLAI	LLAI	DI	PI	SI	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ	
28.	5	03	--	6	03	None	HLAI	LLAI	DI	PI	SI	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ	
29.	5	04	--	6	04	None	HLAI	LLAI	DI	PI	SI	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ	
30.	5	05	--	6	05	None	HLAI	LLAI	DI	PI	SI	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ	
31.	5	06	--	6	06	None	HLAI	LLAI	DI	PI	SI	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ	
32.	5	07	--	6	07	None	HLAI	LLAI	DI	PI	SI	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ	
33.	5	08	--	6	08	None	HLAI	LLAI	DI	PI	SI	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ	
34.	5	09	--	6	09	None	HLAI	LLAI	DI	PI	SI	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ	
35.	5	10	--	6	10	None	HLAI	LLAI	DI	PI	SI	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ	
36.	5	11	--	6	11	None	HLAI	LLAI	DI	PI	SI	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ	
37.	5	12	--	6	12	None	HLAI	LLAI	DI	PI	SI	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ	
38.	5	13	--	6	13	None	HLAI	LLAI	DI	PI	SI	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ	
39.	5	14	--	6	14	None	HLAI	LLAI	DI	PI	SI	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ	
40.	5	15	--	6	15	None	HLAI	LLAI	DI	PI	SI	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ	

APM Configuration

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IO Module Configuration Table

	Engineer:	Date:
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IO Module Configuration Display – N/A if NTWKNUM = 0

For each IO Module, select the IO Module type (IOMTYPE) that applies. If there is a redundant IOM, check the Redun blank. Fill in the file and card location(s). If the type selected is AO or DO, select a failure option (FAILOPT). If the type is STIM or LLMUX, select an ac Line Frequency (FREQ6050).

****CAUTION – Use this part of the form only if PKGOPT on page 2 = Nodfault****

MOD NUM	IOM-A FILE	IOM-A CARD	Redun ↓	IOM-B FILE	IOM-B CARD	IOMTYPE										FAILOPT				IOMTYPE		FREQ6050	
						REDUN	HLAI	LLAI	DI	PI	SI	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ	
1.			—	2	06	None	HLAI	LLAI	DI	PI	SI	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ	
2.				2	07	None	HLAI	LLAI	DI	PI	SI	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ	
3.				2	08	None	HLAI	LLAI	DI	PI	SI	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ	
4.				2	09	None	HLAI	LLAI	DI	PI	SI	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ	
5.				2	10	None	HLAI	LLAI	DI	PI	SI	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ	
6.				2	11	None	HLAI	LLAI	DI	PI	SI	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ	
7.				2	12	None	HLAI	LLAI	DI	PI	SI	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ	
8.				2	13	None	HLAI	LLAI	DI	PI	SI	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ	
9.				2	14	None	HLAI	LLAI	DI	PI	SI	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ	
10.				2	15	None	HLAI	LLAI	DI	PI	SI	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ	
11.				4	01	None	HLAI	LLAI	DI	PI	SI	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ	
12.				4	02	None	HLAI	LLAI	DI	PI	SI	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ	
13.				4	03	None	HLAI	LLAI	DI	PI	SI	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ	
14.				4	04	None	HLAI	LLAI	DI	PI	SI	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ	
15.				4	05	None	HLAI	LLAI	DI	PI	SI	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ	
16.				4	06	None	HLAI	LLAI	DI	PI	SI	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ	
17.				4	07	None	HLAI	LLAI	DI	PI	SI	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ	
18.				4	08	None	HLAI	LLAI	DI	PI	SI	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ	
19.				4	09	None	HLAI	LLAI	DI	PI	SI	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ	
20.			—	4	10	None	HLAI	LLAI	DI	PI	SI	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ	

Continued on next sheet

APM Configuration

IO Module Configuration Table

	Engineer:	Date:
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IO Module Configuration Display – N/A if NTWKNUM = 0

For each IO Module, select the IO Module type (IOMTYPE) that applies. If there is a redundant IOM, check the Redun blank. Fill in the file and card location(s).
If the type selected is AO or DO, select a failure option (FAILOPT). If the type is STIM OR LLMUX, select an ac Line Frequency (FREQ6050).

****CAUTION – Use this part of the form only if PKGOPT on page 2 = Nodfault (Continued)****

MOD NUM	IOM-A FILE	IOM-A CARD	Redun ✓	IOM-B FILE	IOM-B CARD	IOMTYPE										FAILOPT				IOMTYPE		FREQ6050					
						REDUN																					
21.						None	HLAI	LLAI	DI	PI	SI	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ					
22.						None	HLAI	LLAI	DI	PI	SI	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ					
23.						None	HLAI	LLAI	DI	PI	SI	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ					
24.						None	HLAI	LLAI	DI	PI	SI	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ					
25.						None	HLAI	LLAI	DI	PI	SI	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ					
26.						None	HLAI	LLAI	DI	PI	SI	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ					
27.						None	HLAI	LLAI	DI	PI	SI	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ					
28.						None	HLAI	LLAI	DI	PI	SI	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ					
29.						None	HLAI	LLAI	DI	PI	SI	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ					
30.						None	HLAI	LLAI	DI	PI	SI	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ					
31.						None	HLAI	LLAI	DI	PI	SI	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ					
32.						None	HLAI	LLAI	DI	PI	SI	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ					
33.						None	HLAI	LLAI	DI	PI	SI	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ					
34.						None	HLAI	LLAI	DI	PI	SI	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ					
35.						None	HLAI	LLAI	DI	PI	SI	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ					
36.						None	HLAI	LLAI	DI	PI	SI	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ					
37.						None	HLAI	LLAI	DI	PI	SI	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ					
38.						None	HLAI	LLAI	DI	PI	SI	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ					
39.						None	HLAI	LLAI	DI	PI	SI	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ					
40.						None	HLAI	LLAI	DI	PI	SI	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ					

APM Analog Input Point

Unit:	Tagname:	Engineer:	Date:
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Point Assignment Display

NAME Tag Name	_____	<i>range</i> = Not applicable <i>default</i> = Blank
NODETYPE Node Type	APM PM, HPM, LM or SM	An APM Point Is Being Configured If a PM, HPM, LM, or SM is being configured, go to the appropriate forms for those devices.
PNTFORM Point Form	FuLL Componnt	Point is fully displayed and alarmed Pt. is partially displayed but not alarmed

PTDESC Point Descriptor	_____	<i>range</i> = Not applicable <i>default</i> = Blank
EUDESC Engineering Units Descriptor	_____	<i>range</i> = Not applicable <i>default</i> = Blank
KEYWORD Keyword Descriptor	_____	<i>range</i> = Not applicable <i>default</i> = Blank
ASSOCDSP Associated Display	_____	<i>range</i> = Not applicable <i>default</i> = Blank
\$CDETAIL Point Custom Display	_____	<i>range</i> = Not applicable <i>default</i> = Blank
UNIT Process Unit Point Is Assigned To	____	<i>range</i> = Two Alphanumeric Characters <i>default</i> = Not applicable
NTWKNUM NIM's UCN That This Point Is In	____	<i>range</i> = 1 to 20 <i>default</i> = Not applicable
NODENUM APMM's Address On The UCN	____	<i>range</i> = 1 to 64 <i>default</i> = Not applicable
MODNUM I/O Module Number	____	<i>range</i> = 1 to 40 <i>default</i> = Not applicable
SLOTNUM Slot Number	____	1-16 (For HLAI or STIM module) 1-8 (For LLAI or PI module) 1-32 (For LLMUX module)

N/A if PNTFORM = Componnt.

PRIMMOD Primary Module ID	_____	<i>range</i> = Not Applicable <i>default</i> = Null
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PNTMODTY Point's IO Module Type	HLAI -> pg 2-A STIM -> pg 3-B PI -> pg 5-C LLMUX -> pg 2-D LLAI -> pg 4-E	High Level Analog Input Smart Transmitter Intf. Module Pulse Input Module Low Level Multiplexer Low Level Analog Input
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APM Analog Input Point

Unit:	Tagname:	Engineer:	Date:
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A

HLAI Only

PV Configuration Display

SENSRTYP Sensor Type	1 5_V P4_2_V 0_5_V Slidwire	+1 to +5 volts .4 to 2 volts 0 to 5 volts Slidewire Resistance Device
PVCHAR PV Characterization	Linear SqrRoot	Linear Squareroot
Btherm Rtherm DinRtd Etherm RPtherm JisRtd Jtherm Stherm NicklRtd Ktherm Ttherm CopprRtd <td> Thermocouple Type: B, E, J, K, R, RP, S, or T Resistance Temperature Device Type: Din, Jis, Nickel, or Copper </td>	Thermocouple Type: B, E, J, K, R, RP, S, or T Resistance Temperature Device Type: Din, Jis, Nickel, or Copper	
PVTEMP Thermal EUS For PV	DegreesC DegreesR DegreesF DegreesK	C = Celcius; F = Farenheit R = Rankin; K = Kelvin
SLWSRCID Slidwire Voltage Slot ID (number)	1 - 16 default = 1	

D

LLMUX Only

PV Configuration Display (continued)

NOTE: PV configuration display of LLMUX also applies to RHMUX IOP, except that RTD is not supported by RHMUX IOP

SENSRTYP Sensor Type	Thermcpl RTD 0_100_MV	Thermocouple Resistance Temp. Device 0 to 100 millivolts
PVCHAR PV Characterization	Linear	Linear
Btherm Rtherm DinRtd Etherm RPtherm JisRtd Jtherm Stherm NicklRtd Ktherm Ttherm CopprRtd <td> Thermocouple Type: B, E, J, K, R, RP, S, or T Resistance Temperature Device Type: Din, Jis, Nickel, or Copper </td>	Thermocouple Type: B, E, J, K, R, RP, S, or T Resistance Temperature Device Type: Din, Jis, Nickel, or Copper	
PIUOTDCF Open T/C Detect	Off On	Open Thermocouple Detection (LLMUX-Thermcpl only)
PVTEMP Thermal EUS For PV	DegreesC DegreesR DegreesF DegreesK	C = Celcius; F = Farenheit R = Rankin; K = Kelvin
TCRNGOPT Thermocouple Range Option	Normal Extended	Use PVEXEULO's normal range table Use PVEXEULO's extended range table

APM Analog Input Point

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Unit:	Tagname:	Engineer:	Date:
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B

This page - STIM Only

PV Configuration Display

STITAG Transmitter Tag ID		Device name stored in the Smart Xmtr. <i>default</i> = Blank
SENSRTYP Smart Transmitter Sensor Type		Differential Pressure Xmtr. Gauge Pressure Xmtr. Absolute Pressure Xmtr. Magnetic Flow Xmtr. Temperature Xmtr. Linear Square root
PVCHAR PV Characterization Option		

Btherm	B type thermocouple	NickIRtd	Nickel type Res. Temp. Device
Etherm	E type thermocouple	PT200Rtd	Point 200 Resistance Temp. Dev.
Jtherm	J type thermocouple	PT500Rtd	Point 500 Resistance Temp. Dev.
Ktherm	D type thermocouple	CU10Rtd	Copper 10 Resistance Temp. Dev.
Rtherm	R type thermocouple	CU25Rtd	Copper 25 Resistance Temp. Dev.
Stherm	S type thermocouple	RTD_OHMS	Ohms Resistance Temp. Dev.
Ttherm	T type thermocouple	RHRAD	RH Radiomatic
Ntherm	N type thermocouple	W5W26TC	C type thermocouple
DinRtd	Din type Res. Temp. Device	W3W25TC	D type thermocouple
JisRtd	Jis type Res. Temp. Device	NINIMOTC	Nickle/nickle/Mol. thermocouple

STI_EU STI Engineering Units		Applies if – <u>SENSRTYP = SFM</u>		<u>SENSRTYP = STT AND</u> <u>PV CHAR =</u>	
<u>SENSRTYP = SPT_DP/GP/AP</u>		CM_HR	BRL_DAY	CUB_M	• Linear — { MV V
INH20	MBAR BAR	GAL_HR	CM_SEC	LITRES	
MMHG	G_SQCM	LIT_HR	PRCNT	MLITRES	• RTD_OHMS - OHMS
PSI	KG_SQCM	CC_HR	BLANK	FL_OZ	
KPA	MMH20	CM_MIN	LBS	FEET	• All Other — { DEG_C DEG_F DEG_K DEG_R
MPA	INHG	GAL_MIN	KG	METERS	
		LIT_MIN	TONS	MM	
		CC_MIN	GRAMS	INCHES	
		CM_DAY	OZ	KG_CUM	
		GAL_DAY	GAL	G_CUM	
		KGAL_DAY	BRL	LBS_CUFT	
				LBS_CUIN	

DECONF DE Configuration Mode	PV PV_SV PV_DB PV_SV_DB	Broadcast PV 3.6 time/sec. Broadcast PV 3.2 times/sec., SV .46 X/sec. Broadcast PV 2.7 times/sec. plus data base B/cast PV 2.4 X/Sec., SV .34 X/Sec+ d/base
URL Upper Range Limit		<i>range</i> = See Parameter Ref.Guide <i>default</i> = NaN
URV Upper Range Value		<i>range</i> = See Parameter Ref.Guide <i>default</i> = NaN
LRV Lower Range Value		<i>range</i> = See Parameter Ref.Guide <i>default</i> = NaN
DAMPING Xmtr Filter Lag Time		SPT=0-32, STT=0-102.3, SFM=0-100 (discrete increments); <i>default</i> = 0.0

APM Analog Input Point

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Applies if SENSRTYP=STT

PIUOTDCF Open TC Detection Enable Option	Off On	Thermocouple fault detection Range=Off or On.
CJTACTION Int. Cold Jct. Compensation	On Off	On=Internal cold Jct. compensation active Off=External cold Jct. compensation active

PV Configuration Display (continued) – Applies To All Point Types Except As Noted

E

LLAI Only

PV Configuration Display

SENSRTYP Sensor Type	1_5_V 0_5_V RTD Thermcpl 0_100_MV	+1 to +5 volts 0 to 5 volts Resistance Temp. Device Thermocouple 0 to 100 millivolts
PVCHAR PV Characterization	Linear SqrRoot	Linear Square root
	Btherm Etherm Jtherm Ktherm Rtherm RPtherm Stherm Ttherm	Thermocouple Type: B, E, J, K, R, RP, S, or T
	{ DinRtd JisRtd NicklRtd CopprRtd	Resistance Temp. Device Type: Din, Jis, Nickel, or Copper
PVTEMP Thermal EUS For PV	{ DegreesC DegreesR	C = Celcius; F = Farenheit R = Rankin; K = Kelvin
TCRNGOPT Thermocouple Range Option	Normal Extended	Use PVEXEULO's normal range table Use PVEXEULO's extended range table

Applies if SENSRTYP = Slidewire or 0-100 mV

PVRAWHI PV Raw High Limit	_ _ _ _	range = default = units =	0 to 100 mv 0 - 100 millivolts	Slidewire 0.0 - 1 Ratio
PVRAWLO PV Raw Low Limit	_ _ _ _	range = default = units =	0 to 100 mv 0 - 100 millivolts	Slidewire 0.0 - 1 Ratio

Applies if PVCHAR = LINEAR or SQRROOT

INPTDIR Analog Input Direction	Direct Reverse	PVCALC increases as PVRAW increases PVCALC decreases as PVRAW increases
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PVEUHI PV High Range In EUs	_ _ _ _	range = PVEULO to PVEXEUI default = 100.0
PVEULO PV Low Range In EUs	_ _ _ _	range = PVEXEULO to PVEUHI default = 0.0
PVFORMAT PV Decimal Point Format	DO D1	D0 = 9999. D1 = 999.9
	D2 D3	D2 = 99.99 D3 = 9.999

PVTV	_ . _ . _ . _ .	N/A if PNTFORM = Compnnt. <i>range</i> = PVEXEULO-PVEXEUHI, & NaN <i>default</i> = NaN
PV Target Value in Eus		
OVERVAL	_ _ .	<i>range</i> = 0 to 100 <i>default</i> = 25
Overview Display Value		

APM Analog Input Point

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Alarming Display – N/A if PNTFORM = Componnt

As you enter alarm trip points and alarm priorities in the next section please refer to the following alarm priority definitions and the note.

NoAction	Alarm is not reported to the system and not annunciated (R670 & later)
Journal	Alarm is historized, not reported to US, not annunciated
Low High	Alarm is historized, reported to US, and annunciated
Emergency	Alarm is historized, reported to US, and annunciated
Printer	Alarm is historized, reported to US, and annunciated Not
JnlPrint	historized, reported to the printer, not annunciated Alarm
	historized, reported to the printer, not annunciated

Note: If you do not enter a trip point value for a given pair of trip-points and alarm-priorities, do not circle an alarm priority option for that pair (i.e., no trip point, no alarm priority).

Alarming Display – N/A if PNTFORM = Componnt.

\$AUXUNIT	_ _	<i>range</i> = Any configured LCN Unit ID <i>default</i> = Null (- -)
Auxiliary Unit		
\$ALDLOPT (R620 and later)	None	
Alarm Delay Option	OffDlyLo OffDlyHi OffDlyEm OnDlyLo OnDlyHi OnDlyEm	Off Delay Lo Off Delay Hi Off Delay Emergency On Delay Lo On Delay Hi On Delay Emergency
\$DLBADAL (R681 and later)	None	<i>range</i> = Offdelay, None <i>default</i> = None
Delay for BADPV	Offdelay	
\$DLTIME (R620 and later)	_ _ _	<i>range</i> = 0 to 9999 seconds <i>default</i> = 0 seconds
Delay Time		
\$REALARM (R660 and later)	_ _ _	<i>range</i> = 0 to 9999 minutes <i>default</i> = 0 minutes
Re-alarming Interval in minutes		
PVALDB	Half One Two Three	Four Five EU
PV Alarm Deadband In %		
PVALDBEU	_ _ _ _	<i>range</i> >= 0.0 <i>default</i> = 1.0
PV Alarm Deadband in EU		
PVHITP	_ _ _ _	<i>range</i> = PVLOTP - PVHHTP, & NaN <i>default</i> = NaN
PV High Alarm Trip Point		
PVHIPR	NoAction Journal Low	High Emergency Printer JnlPrint
PV High Alarm Priority	(R670 and later)	
\$HISELCT (R681 and later)	Cutout	<i>range</i> = Cutout, Allow <i>default</i> = Cutout
PV High Selective Cutout	Allow	
PVLOTP	_ _ _ _	<i>range</i> = PVLLTP - PVHITP, & NaN <i>default</i> = NaN
PV Low Alarm Trip Point		

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Unit:	Tagname:	Engineer:	Date:
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Alarming Display - N/A if PNTFORM = Compnnt

PVLOPR PV Low Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint
\$LOSELCT (R681 and later) PV Low Selective Cutout	Cutout Allow			<i>range</i> = Cutout, Allow <i>default</i> = Cutout			
PVHHTP PV Hi Hi Alarm Trip Point	_ _						

APM Analog Input Point

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Alarming Display - N/A if PNTFORM = Compnnt

ALENBST Point Alarm Enable Status	<i>Enable</i>	<u>Displayed</u>	<u>Logged</u>	<u>Reported to EIP</u>
	Disable	Yes	Yes	Yes
	Inhibit	No	Yes	Yes
		No	No	No

Applies if PNTMODTY = LLAI or LLMUX and if the SENSRTYP = THERMCPL or RTD

\$OLDEVDS Open-Line Detect Event Disable	<i>Enable</i>	Allows BADPV alarms when open line detected.
	Disable	Disables BADPV alarms when open line detected.

Applies if PNTMODTY = HLAI, STIM, PI, or if SENSRTYP not THERMCPL or if SENSRTYP not RTD

\$OLDEVDS Open-Line Detect Event Disable	<i>NOOTDETC</i>	Value must be selected. Point operation is not affected by this function.
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N/A if PNTFORM = Component.		
PRIMMOD		<i>range</i> = Not Applicable
Primary Module ID	_.	<i>default</i> = Null
PNTMODTY	AO	Standard AO IOP
Point Module Type	AO_16	High Density AO IOP

APM Analog Output Point

Unit:	Tagname:	Engineer:	Date:
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Operating Configuration Display

OPDIR(AnalOut) AO Direct/Reverse Action	Direct Reverse	For final OP: 0%=>4ma; 100% =>20ma For final OP: 0% =>20ma; 100% =>4ma
OPCHAR Output Characterization Option	On ☐ Off	I/O coordinates define OP char. curve I/O coordinates do not apply
\$OLDEVDS Open-Line Detect Event Disable	Enable Disable	Allows OUTPUTFL soft failure event when open line detected in output. Disables OUTPUTFL soft failure event when open line detected in output.
\$OPTOL Output Tolerance in %		range = 0.0 to 106.9 default = 0.0
OUTIND (R620 and later) Reverse Output Indicator	Direct Reverse	4 - 20mA = 0 – 100% 20 – 4mA = 0 – 100%

OPIN0 Input Coordinate # 0	-6.9	range = Fixed at -6.9 percent default = Not applicable
OPIN1 Input Coordinate # 1		range = >prev. coord, < next coord default = Not applicable
OPIN2 Input Coordinate # 2		range = >prev. coord, < next coord default = Not applicable
OPIN3 Input Coordinate # 3		range = >prev. coord, < next coord default = Not applicable
OPIN4 Input Coordinate # 4		range = >prev. coord, < next coord default = Not applicable
OPIN5 Input Coordinate # 5	106.9	range = Fixed at 106.9 percent default = Not applicable
OPOUT0 Output Coordinate # 0	-6.9	range = Fixed at -6.9 percent default = Not applicable
OPOUT1 Output Coordinate # 1		range = >prev. coord, < next coord default = Not applicable
OPOUT2 Output Coordinate # 2		range = >prev.coord, < next coord default = Not applicable
OPOUT3 Output Coordinate # 3		range = >prev.coord, < next coord default = Not applicable
OPOUT4 Output Coordinate # 4		range = >prev.coord, < next coord default = Not applicable
OPOUT5 Output Coordinate # 5	106.9	range = Fixed at 106.9 percent default = Not applicable

APM Analog Output Point

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Unit:	Tagname:	Engineer:	Date:
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Mode Configuration Display – N/A if PNTFORM = Componnt.

RCASOPT Remote Cascade Option	None Ddc	No cascade mode <u>of any type</u> is allowed In Cas, AM point controls this point's OP
NMODE AnalOut Point's Normal Mode	None Man Cas	No configured "normal" operating mode Manual is the configured "normal" mode If RCASOPT equals Ddc, Cascade may be configured as the "normal" mode.
N/A if NMODE = None		
NMODATTR Normal Mode Attribute	None Operator Program	Parameter NMODE sets "normal" mode Operator sets "normal" operating mode Program sets "normal" operating mode
MODEPERM Mode Permissive	Permit NotPerm	Operator can change this point's mode Optr. cannot change this point's mode
\$MODESEL Mode Selection List	Man Cas Prog	Man and Cas mode, and Program mode attribute are allowed for this data point.

CTLALGID Control Algorithm Identifier	<i>Null</i> AutoMan	No affect Auto Manual
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APM Auto Manual CTL Algorithm

Unit:	Tagname:	Engineer:	Date:
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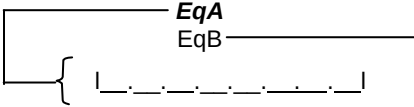
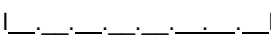
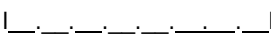
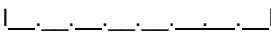
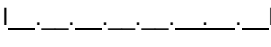
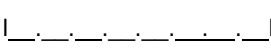
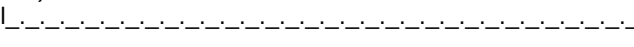
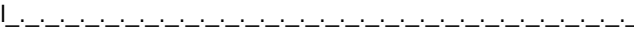
PV Configuration Display – N/A if PNTFORM = Component

OVERVAL		<i>range = 0 to 100</i>
Overview Display Value		<i>default = 25</i>

Mode Configuration Display

	No_Shed	Shed Hold	ShedLow	Shed High	Shed Safe
BADCTLOP Bad PV Mode Shed Option					
RCASOPT (Automan) Remote Cascade Option					
SHEDTIME Remote Cascade Shed Time					
SHEDMODE Shed Mode					
NMODE (RegCtl) RegCtl Point's Normal Mode					
NMODATTR (N/A if NMODE = None) Normal Mode Attribute					
MODEPERM Mode Permissive					
\$MODESEL (R630 and later) Mode Selection List					
EXTSWOPT External Mode Switching					

Reg Control Auto/Man Algo Display

CTLEQN Control Equation Type		$CV = X1 + B + BI$ $CV = X1 + (K * X2) + BI$ <i>range</i> = Any Real number <i>default</i> = 0.0
B Output Bias		<i>range</i> = Not applicable <i>default</i> = 1.0
K Gain Constant For X2 Input		<i>range</i> = ≥ 0.0 , NaN <i>default</i> = 0.0
RATE1 Rate At Which Bias Ramps Down		<i>range</i> = $\geq XEULO$ <i>default</i> = 100.0
XEUHI X Input High Range in EUs		<i>range</i> = $\leq XEUI$ <i>default</i> = 0.0
XEULO X Input Low Range In EUs		<i>range</i> = 0 – 1 for EqA; 1-2 for EqB <i>default</i> = 0 for EqA; 1 for EqB
NOCINPTS #Of Control Input Connections		
CISRC (Control-Input-Connection Source)	CISRC (point.parameter)	<u>Destination Parameter (CIDSTN)</u>
<u>Input#</u> (if NOCINPTS>0)		
1		X1 (EqA); X2 (EqB)
2		X1 (EqB)

APM Auto Manual CTL Algorithm

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Output Configuration Display

NOCOPTS # Of Control Output Connections		range = 0 to 4 default = 1
CODSTN (Control-Output-Connection Destination)		
<u>Output #</u>	<u>CODSTN (point.parameter)</u>	
1		
2		
3		
4		
OPHILM Output High Limit In Percent		range = OPLOLM - 106.9 & NaN default = 105.0
OPLOLM Output Low Limit In Percent		range = -6.9 - OPHILM, & NaN default = -5.0
\$OPTOL Output Tolerance in %		range = 0.0 to 106.9 default = 0.0
OUTIND (R620 and later) Reverse Output Indicator	Direct Reverse	4 - 20mA = 0 – 100% 20 - 4mA = 0 – 100% Note: With R670 and later, when this parameter is set to Reverse, the values of the displayed output and the related output parameters are changed. Refer to the definition of the OUTIND parameter in the PM Family Parameter Dictionary, XPM09.
SAFEOP Safe Output Limit In Percent		range = -6.9 to to 106.9 & NaN default = NaN
OPMCHLM Output Minimum Change In %		range = >=0.0, and NaN default = 0.0
OPROCLM OP Rate Of Change Limit In %		range = >=0.0, and NaN default = NaN

APM Auto Manual CTL Algorithm

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Alarming Display - N/A if PNTFORM = Componnt

As you enter alarm priorities in the next section please refer to the following

NoAction	Alarm is not reported to the system and not annunciated (R670 & later)
Journal	Alarm is historized, not reported to US, not annunciated
Low	Alarm is historized, reported to US, and annunciated
High	Alarm is historized, reported to US, and annunciated
Emergency	Alarm is historized, reported to US, and annunciated Not
Printer	historized, reported to the printer, not annunciated Alarm
JnlPrint	historized, reported to the printer, not annunciated

\$AUXUNIT Auxiliary Unit	_ _ _	<i>range = Any configured LCN Unit ID default = NaN</i>					
\$DLBADAL (R681 and later) Delay for BADCTL	None Offdelay	<i>range = Offdelay, None default = None</i>					
DLTIME (R620 and later) Delay Time	_ _ _ _	<i>range = 0 to 9999 seconds default = 0 seconds</i>					
\$REALARM (R660 and later) Re-alarming Interval	_ _ _	<i>range = 0 to 9999 minutes default = 0 minutes</i>					
BADOCOPT BAD OP Conn Alarm Option	Off On	Bad Output Connection Alarm is not initially permitted Bad Output Connection Alarm is initially permitted					
BADOCPR BAD OP Conn Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint
OPHITP Output High Alarm Trip Point	_ _ _ _	<i>range = OPLOTP TO OPHILM and NaN default = NaN</i>					
OPHIPR Output High Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint
OPLOTP Output Low Alarm Trip Point	_ _ _ _	<i>range = OPLOLM TO OPHITP and NaN default = NaN</i>					
OPLOPR Output Low Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint
OPALDB Output Alarm Deadband	_ _ _ _	<i>range = 0 - 25.0 but ≤ (OPHITP- OPLOTP)/2 default = 5.0 Output Units</i>					
BADCTLPR Bad-Control Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint
ALENBST Point Alarm Enable Status	Enable Disable Inhibit	<u>Displayed</u> Yes No No		<u>Logged</u> Yes Yes No		<u>Reported To EIP</u> Yes Yes No	

PVALGID	Null	No affect
PV Algorithm Identifier	Calcultr	Calculator

APM Calculator PV Algorithm

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PV Configuration Display

PVEUHI PV High Range In Engr Units	_ _ _ _ _ _ _	<i>range = PVEULO to PVEXEUHI</i> <i>default = 100.0</i>
PVEULO PV Low Range In Engr Units	_ _ _ _ _ _ _	<i>range = PVEXEULO to PVEUHI</i> <i>default = 0.0</i>
PVFORMAT PV Decimal Point Format	DO D1	D2 D3 D0=9999. D1=999.9 D2=99.99 D3 = 9.999
PVEXEUHI PV Extended Hi Range in EUs	_ _ _ _ _ _ _ _ _	<i>range = >=PVEUHI</i> <i>default = 102.9</i>
PVEXEULO PV Extended Lo Range in EUs	_ _ _ _ _ _ _ _ _	<i>range = <=PVEULO</i> <i>default = -2.9</i>
PVCLAMP PV Clamping Option	NoClamp Clamp	No clamping of the PV value Clamp PV value at range extension limit

N/A if PNTFORM = Componnt.

PVSRCOPT PV Source Option	Only Auto All	PV Source selection is not available Selection from PVSOURCE is available
PVSOURCE PV Source	Auto Man Sub	Field wiring/memory fetch supplies PV PV is supplied by Operator or program A value is substituted by a CL program

PV Configuration Display

TF PV Filter Lag Time in Minutes	_ _ _ _	<i>range = 0.0 to 60.0 (minutes)</i> <i>default = 0.0 (PV will not be delayed)</i>
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PVTV PV Target Value in EUs	_ _ _ _ _ _ _ _ _	<i>range = PVEXEULO-PVEXEUHI, & NaN</i> <i>default = NaN</i>
OVERVAL Overview Display Value	_ _ _	<i>range = 0 to 100</i> <i>default = 25</i>

Reg PV Calculator Algo Display

N # Inputs To Calcultr Algo	_	<i>range = 1 to 6</i> <i>default = 2</i>
C1 Calculator Constant # 1	_ _ _ _ _ _ _ _ _	<i>range = N/A</i> <i>default = 0.0</i>
C2 Calculator Constant # 2	_ _ _ _ _ _ _ _ _	<i>range = N/A</i> <i>default = 0.0</i>
C3 Calculator Constant # 3	_ _ _ _ _ _ _ _ _	<i>range = N/A</i> <i>default = 0.0</i>

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Alarming Display - N/A if PNTFORM = Component		
\$AUXUNIT Auxiliary Unit	_.	<i>range</i> = Any configured LCN Unit ID <i>default</i> = NaN
\$ALDLOPT (R620 and later) Alarm Delay Option	None OffDlyLo OffDlyHi OffDlyEm OnDlyLo OnDlyHi OnDlyEm	Off Delay Lo Off Delay Hi Off Delay Emergency On Delay Lo On Delay Hi On Delay Emergency
\$DLBADAL (R681 and later) Delay for BADPV	None Offdelay	<i>range</i> = Offdelay, None <i>default</i> = None
\$DLTIME (R620 and later) Delay Time	_._._.	<i>range</i> = 0 to 9999 seconds <i>default</i> = 0 seconds
\$REALARM (R660 and later) Re-alarms Interval	_._._.	<i>range</i> = 0 to 9999 minutes <i>default</i> = 0 minutes

APM Calculator PV Algorithm

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Alarming Display (Continued)

	Half	One	Two	Three	Four	Five	EU	
PVALDB PV Alarm Deadband In %								
PVALDBEU PV Alarm Deadband In EU	_ _ _ _							range > = 0.0 default = 1.0
PVHITP PV High Alarm Trip Point	_ _ _ _ _ _ _							range = PVLOTP - PVHHTP, & NaN default = NaN
PVHIPR PV High Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint	
\$HISELCT (R681 and later) PV High Selective Cutout	Cutout Allow							range = Cutout, Allow default = Cutout
PVLOTP PV Low Alarm Trip Point	_ _ _ _ _ _ _							range = PVLLTP - PVHITP, & NaN default = NaN
PVLOPR PV Low Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint	
\$LOSELCT (R681 and later) PV Low Selective Cutout	Cutout Allow							range = Cutout, Allow default = Cutout
PVHHTP (RegCtl) PV Hi Hi Alarm Trip Point	_ _ _ _ _ _ _							range = PVHITP to PVEUHI, & NaN default = NaN
PVHHPR PV High High Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint	
\$HHSELCT (R681 and later) PV High High Selective Cutout	Cutout Allow							range = Cutout, Allow default = Cutout
PVLLTP (RegCtl) PV Low Low Alarm Trip Point	_ _ _ _ _ _ _							range = PVEULO - PVLOTP, & NaN default = NaN
PVLLPR PV Low Low Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint	
\$LLSELCT (R681 and later) PV Low Low Selective Cutout	Cutout Allow							range = Cutout, Allow default = Cutout
PVSGCHTP PV Significant Change Alarm Trip Point	_ _ _ _ _ _ _							range = >=0.0, and NaN default = NaN
PVROCTP PV Pos Rate Of Change Trip Point	_ _ _ _ _ _ _							range = >=0.0, and NaN default = NaN
PVROCPPR PV Pos Rate Of Chg Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint	
\$RPSELCT (R681 and later) PV Positive Rate of Change Selective Cutout	Cutout Allow							range = Cutout, Allow default = Cutout

APM Calculator PV Algorithm

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Alarming Display (Continued)

PVROCNTP PV Neg Rate Of Change Trip Point	_ _ _ _ _ _ _	<i>range = >=0.0, and NaN default = NaN</i>					
PVROCNPR PV Neg Rate Of Chg Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint
\$RNSELCT (R681 and later) PV Negative Rate of Change Selective Cutout	Cutout Allow	<i>range = Cutout, Allow default = Cutout</i>					
BADPVPR Bad-PV Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint
ALENBST Point Alarm Enable Status				<u>Displayed</u>	<u>Logged</u>	<u>Reported To EIP</u>	
	Enable			Yes	Yes	Yes	
	Disable			No	Yes	Yes	
	Inhibit			No	No	No	

PVALGID PV Algorithm Identifier	NULL  DataAcq	No affect Data Acquisition
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APM Data Acquisition PV Algorithm

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PV Configuration Display

PVEUHI PV High Range In Engr Units	_ _ _ _ _ _ _	<i>range</i> = PVEULO to PVEXEUHI <i>default</i> = 100.0
PVEULO PV Low Range In Engr Units	_ _ _ _ _ _ _	<i>range</i> = PVEXEULO to PVEUHI <i>default</i> = 0.0
PVFORMAT PV Decimal Point Format	DO D2 D1 D3	D0 = 9999. D2 = 99.99 D1 = 999.9 D3 = 9.999
PVEXEUHI PV Extended Hi Range in EUs	_ _ _ _ _ _ _ _ _ _ _ _ _	<i>range</i> = >=PVEUHI <i>default</i> = 102.9
PVEXEULO PV Extended Lo Range in EUs	_ _ _ _ _ _ _ _ _ _ _ _ _	<i>range</i> = <=PVEULO <i>default</i> = -2.9
PV CLAMP PV Clamping Option	NoClamp Clamp	No clamping of the PV value Clamp PV value at range extension limit

PVSRCOPT PV Source Option	Only Auto All	PV source selection is not available Selection from PVSOURCE is available
PVSOURCE PV Source	Auto Man Sub	Field wiring/memory fetch supplies PV PV is supplied by Operator or program A value is substituted by a CL program

TF PV Filter Lag Time in Minutes	0.0000	<i>range</i> = 0.0 to 60.0 (minutes) <i>default</i> = 0.0 (PV will not be delayed)
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PVTV PV Target Value in EUs		<i>range</i> = PVEXEULO-PVEXEUHI, & NaN <i>default</i> = NaN
OVERVAL Overview Display Value		<i>range</i> = 0 to 100 <i>default</i> = 25

Reg PV Data Acq Algo Display

PISRC (PV-Input-Connection Source)		Destination Parameter (PIDSTN)
Input#	PISRC (point.parameter)	
1	_____	P1

APM Data Acquisition PV Algorithm

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Alarming Display - N/A if PNTFORM = Componnt

As you enter alarm trip points and alarm priorities in the next section please refer to the following alarm priority definitions and the note.

NoAction	Alarm is not reported to the system and not annunciated (R670 & later)
Journal	Alarm is historized, not reported to US, not annunciated
Low High	Alarm is historized, reported to US, and annunciated
Emergency	Alarm is historized, reported to US, and annunciated
Printer	Alarm is historized, reported to US, and annunciated
JnlPrint	Not historized, reported to the printer, not annunciated
	Alarm historized, reported to the printer, not annunciated

Note: If you do not enter a trip point value for a given pair of trip-points and alarm-priorities, do not circle an alarm priority option for that pair (i.e., no trip point, no alarm priority).

Alarming Display - N/A if PNTFORM = Componnt

\$AUXUNIT Auxiliary Unit		<i>range = Any configured LCN Unit ID default = NaN</i>
\$ALDLOPT (R620 and later) Alarm Delay Option	None OffDlyLo Off Delay Lo OffDlyHi Off Delay Hi OffDlyEm Off Delay Emergency OnDlyLo On Delay Lo OnDlyHi On Delay Hi OnDlyEm On Delay Emergency	
\$DLBADAL (R681 and later) Delay for BADPV	None Offdelay	<i>range = Offdelay, None default = None</i>
DLYTIME (R620 and later) Delay Time		<i>range = 0 to 9999 seconds default = 0 seconds</i>
\$REALARM Re-alarming Interval (R660 and later)		<i>range = 0 to 9999 minutes default = 0 minutes</i>
PVALDB PV Alarm Deadband In %	Half One Two Three Four Five EU	
PVALDBEU PV Alarm Deadband In EU		<i>range > = 0.0 default = 1.0</i>
PVHITP PV High Alarm Trip Point		<i>range = PVLOTP - PVHITP, & NaN default = NaN</i>
PVHIPR PV High Alarm Priority	NoAction Journal Low High Emergency Printer JnlPrint (R670 and later)	
\$HISELCT (R681 and later) PV High Selective Cutout	Cutout Allow	<i>range = Cutout, Allow default = Cutout</i>
PVLOTP PV Low Alarm Trip Point		<i>range = PVLLTP - PVHITP, & NaN default = NaN</i>
PVLOPR PV Low Alarm Priority	NoAction Journal Low High Emergency Printer JnlPrint (R670 and later)	
\$LOSELCT (R681 and later) PV Low Selective Cutout	Cutout Allow	<i>range = Cutout, Allow default = Cutout</i>

APM Data Acquisition PV Algorithm

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Alarming Display – N/A if PNTFORM=Compnt

PVHHTP(RegCtl) PV Hi Hi Alarm Trip Point	_ _ _ _ _ _ _									<i>range</i> = PVHITP to PVEUHI, & NaN <i>default</i> = NaN
PVHHPR PV High High Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint			
\$HHSELCT (R681 and later) PV High High Selective Cutout	Cutout Allow									<i>range</i> = Cutout, Allow <i>default</i> = Cutout
PVLLTP(RegCtl) PV Low Low Alarm Trip Point	_ _ _ _ _ _ _									<i>range</i> = PVEULO-PVLOTP,& NaN <i>default</i> = NaN
PVLLPR PV Low Low Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint			
\$LLSELCT (R681 and later) PV Low Low Selective Cutout	Cutout Allow									<i>range</i> = Cutout, Allow <i>default</i> = Cutout
PVSGCHTP PV Significant Change Alarm Trip Point	_ _ _ _ _ _ _									<i>range</i> = >=0.0, and NaN <i>default</i> = NaN
PVROCPTP PV Pos Rate Of Change Trip Point	_ _ _ _ _ _ _									<i>range</i> = >=0.0, and NaN <i>default</i> = NaN
PVROCPPR PV Pos Rate Of Chg Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint			
\$RPSELCT (R681 and later) PV Positive Rate of Change Selective Cutout	Cutout Allow									<i>range</i> = Cutout, Allow <i>default</i> = Cutout
PVROCNTP PV Neg Rate Of Change Trip Point	_ _ _ _ _ _ _									<i>range</i> = >=0.0, and NaN <i>default</i> = NaN
PVROCNPR PV Neg Rate Of Chg Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint			
\$RNSELCT (R681 and later) PV Negative Rate of Change Selective Cutout	Cutout Allow									<i>range</i> = Cutout, Allow <i>default</i> = Cutout
BADPVPR Bad-PV Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint			
ALENBST Point Alarm Enable Status										
		Enable		<u>Displayed</u>	<u>Logged</u>	<u>Reported To EIP</u>				
		Disable		Yes	Yes	Yes				
		Inhibit		No	Yes No	Yes				
				No		No				

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USER ID User ID Reservation	_ _ _ _ _ _ _ _ _ _ _ _ _ _ _	<i>range</i> = Sixteen Character String <i>default</i> = Dashes
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APM Digital Composite Point

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Point Assignment Display

NOSTATES	2	<i>default</i> = Not applicable
Number Of Digital States	3	2 digital switch states can be configured 3 digital switch states can be configured
NODINPTS	0	0 digin connections can be configured
# Of Digital Input Connections	1	1 digin connections can be configured
	2	2 digin connections can be configured
NODOPTS	0	0 digout connections can be configured
# Of Digital Output Connections	1	1 digout connections can be configured
	2	2 digout connections can be configured
	3	3 digout connections can be configured

State Configuration Display

N/A if PNTFORM = Componnt.

PVTXTTOPT PV Text Option	Off On	<i>range</i> = Not applicable <i>default</i> = "Off"		
STATETXT () State - Descriptor Text	(1) (0) (2) 	<i>range</i> = Not applicable <i>default</i> = "On" <i>default</i> = "Off" <i>default</i> = "State2"		
BOXCLR (1) State 1 Box Color	Red Cyan	Green (Top box default) Yellow	White Blue	Black Magenta
BOXCLR (0) State 0 Box Color	Red Cyan	Green Yellow (Middle box default)	White Blue	Black Magenta
BOXCLR (2) State 2 Box Color	Red (Lower box default) Cyan	Green Yellow	White Blue	Black Magenta
MOVPTXT Moving PV Text		<i>range</i> = Eight character string <i>default</i> = "Moving"		
BADPVTXT Bad PV Text		<i>range</i> = Eight character string <i>default</i> = "Bad"		

APM Digital Composite Point

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State Configuration Display (Continued)

MOMSTATE (N/A if NODOPTS = 0) Momentary Output States	None Mom_0 Mom_1 Mom_2 Mom_1_2	No momentary output states State 0 is valid if (NOSTATES = 2) State 1 is valid if (NOSTATES = 2 or 3) State 2 is valid if (NOSTATES = 3) State 1 or 2 is valid if (NOSTATES = 3)
LOGICSRC Logic Source	Logic Point driving the interlock to this Digital Composite Point	
DISRC (Digital-Input-Connection Source) Input # (N/A if NODINPTS = 0)	DISRC (point.parameter)	Destination
1		D1
2		D2

PV / Input Configuration Display -Only if NODINPTS = 1

D1_1 D1_One PV State	PVState0 PVState1	Use & display contents of STATETXT0 Use & display contents of STATETXT1
D2D1_00 D2_D1 Zero_Zero PV State D2 = Off, D1 = Off	PVState1 PVState0 PVState2 MovPV BadPV	Use & display contents of STATETXT1 Use & display contents of STATETXT0 Use & display contents of STATETXT2* Use & display contents of MOVPTXT Use & display contents of BADPTXT
D2D1_01 D2D1 Zero_One PV State D2 = Off, D1 = On	PVState1 PVState0 PVState2 MovPV BadPV	Use & display contents of STATETXT1 Use & display contents of STATETXT0 Use & display contents of STATETXT2* Use & display contents of MOVPTXT Use & display contents of BADPTXT
D2D1_10 D2D1 One_Zero PV State D2 = On, D1 = Off	PVState1 PVState0 PVState2 MovPV BadPV	Use & display contents of STATETXT1 Use & display contents of STATETXT0 Use & display contents of STATETXT2* Use & display contents of MOVPTXT Use & display contents of BADPTXT
D2D1_11 D2D1 One_One PV State D2 = On, D1 = On	PVState1 PVState0 PVState2 MovPV BadPV	Use & display contents of STATETXT1 Use & display contents of STATETXT0 Use & display contents of STATETXT2* Use & display contents of MOVPTXT Use & display contents of BADPTXT

*Only applies if NOSTATES = 3

PV / Input Configuration Display (Continued) - N/A if NODINPTS = 0

PVSRCOPT PV Source Option	Only Auto All	PV Source selection is not available and field wiring or memory fetch supplies PV. Selection from PVSOURCE is available & Operator/CL-program may supply PV.
PVSOURCE PV Source	Auto Man Sub Track	Field wiring or memory -fetch supplies PV PV is supplied by Operator or program A value is substituted by a CL program PV tracks the output

APM Digital Composite Point

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Output Configuration Display – N/A if NODOPTS = 0

		<u>NOSTATES</u>	<u>NODOPTS</u>
ST1_OP1	Off	2 - 3	1 - 3
State 1 - Output 1	On		
ST1_OP2	Off	2 - 3	2 - 3
State 1 - Output 2	On		
ST1_OP3	Off	2 - 3	3
State 1 - Output 3	On		
ST0_OP1	Off	1 - 3	1 - 3
State 0 - Output 1	On		
ST0_OP2	Off	1 - 3	2 - 3
State 0 - Output 2	On		
ST0_OP3	Off	1 - 3	3
State 0 - Output3	On		
ST2_OP1	Off	3	1 - 3
State 2 - Output 1	On		
ST2_OP2	Off	3	2 - 3
State 2 - Output 2	On		
ST2_OP3	Off	3	3
State 2 - Output 3	On		
DODSTN (Digital Output Connection Destination)			
<u>Output #</u>	<u>DODSTN (point.parameter)</u>		<u>NODOPTS</u>
1			1
2			2
3			3
PULSEWTH		range = 0.0 to 60.0	
Pulse Width For DigComp Point Outputs		default = 1.0	

Applies if NOSTATES = 3, and MOMSTATE = None

STCHGOPT	None	range = Not Applicable
State Change Option	State 0	default = None
PAUSETIM		range = 0 to 1000 seconds
State 0 Pause Time		default = 0

SEALOPT	None	range = Not Applicable
Seal-in Circuit Option	Sealin	default = None

APM Digital Composite Point

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Maintenance Display

MAINTOPT Maintenance Support	Off On	range = Not Applicable default = Off
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Maintenance Statistics

<u>Maximum Hours Allowed in Each State</u>		<u>Maximum Number of Transitions into State</u>	
STATE 1 MAXTIM1H	_ _ _ _ _ _ _	MAXTRAN1	_ _ _ _ _ _ _
STATE 0 MAXTIM0H	_ _ _ _ _ _ _	MAXTRAN0	_ _ _ _ _ _ _
STATE 2 MAXTIM2H	_ _ _ _ _ _ _	MAXTRAN2	_ _ _ _ _ _ _
range = N/A default = 0		range = 0 (No Limit) default = 0.0	

Mode Configuration Display - N/A if NODOPTS = 0

NMODATTR Normal Mode Attribute	None Operator Program	Parameter NMODE sets "normal" mode Operator sets "normal" operating mode Program sets "normal" operating mode
MODEPERM Mode Permissive	Permit NotPerm	Operator can change this point's mode Optr. cannot change this point's mode
OROPT Interlock Bypass Option	Off On	Optr. cannot bypass this point's interlocks Operator can bypass this point's interlocks If BYPASS = ON
BYPASS Interlock Bypass State	Off On	Outputs not allowed to ignore interlocks Stores to OP & SO are allowed to ignore the interlocks
\$MODESEL (R630 and later) Mode Selection List	Man Prog	Man mode and Program mode attribute are allowed for this data point.

APM Digital Composite Point

Alarming Display - N/A if PNTFORM = Componnt or if NODINPTS = 0

\$AUXUNIT Auxiliary Unit	_ _	<i>range = Any configured LCN Unit ID default = NaN</i>												
\$REALARM Re-alarming Interval (R660 and later)	_ _ _ _	<i>range = 0 to 9999 minutes default = 0 minutes</i>												
PVNORMAL PV Value's Normal State Text	_ _ _ _ _ _ _	<i>range = STATETXT(0) – STATETXT(2) default = None</i>												
FBTIME Feedback Time	_ _ _ _	<i>range = 1 to 1000 default = 0</i>												
N/A if FBTIME = 0.														
CMDFALTM Command Failure Timeout	_ _ _ _	<i>range = < Feedback Time default = 0</i>												
ALENBST Alarm Enable Status	Enable Disable Inhibit	<table> <thead> <tr> <th><u>Displayed</u></th> <th><u>Logged</u></th> <th><u>Reported to EIP</u></th> </tr> </thead> <tbody> <tr> <td>Yes</td> <td>Yes</td> <td>Yes</td> </tr> <tr> <td>No</td> <td>Yes No</td> <td>Yes</td> </tr> <tr> <td>No</td> <td></td> <td>No</td> </tr> </tbody> </table>	<u>Displayed</u>	<u>Logged</u>	<u>Reported to EIP</u>	Yes	Yes	Yes	No	Yes No	Yes	No		No
<u>Displayed</u>	<u>Logged</u>	<u>Reported to EIP</u>												
Yes	Yes	Yes												
No	Yes No	Yes												
No		No												

Alarming Display- N/A if PNTFORM = Componnt. or if NODINPTS = 0

SI0ALOPT Safety Override Alarm	None Auto_Rtn Cnfm_Rqd	No alarms are generated Alarms generated, return to normal when override clears Alarms generated, confirmed before returning to normal If override clears. SIOALPPR must ≠ NoAction or Journal.
SI0DESC SI0 Alarm Descriptor	_ _ _ _ _ _ _	<i>range = 8 Alphanumeric Characters default = blank</i>
OVRDALOP Override Alarm Option	None Auto_Rtn Cnfm_Rqd	No alarms are generated Alarms generated, return to normal when override clears Alarms generated, confirmed before returning to normal If override clears. OVRDALPR must ≠ NoAction or Journal.
I1DESC I1 Alarm Descriptor	_ _ _ _ _ _ _	<i>range = 8 Alphanumeric Characters default = blank</i>
I0DESC I0 Alarm Descriptor	_ _ _ _ _ _ _	<i>range = 8 Alphanumeric Characters default = blank</i>
I2DESC I2 Alarm Descriptor	_ _ _ _ _ _ _	<i>range = 8 Alphanumeric Characters default = blank</i>

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BADPVPR Bad PV Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint
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N/A if FBTIME = 0.

CMDDISPR Command Disagree Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint
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OFFNRMPR Offnormal Alarm Priority	NoAction (R670 and later)	Alarm not reported to the system and not annunciated					
	Journal	Alarm is historized, not reported to US not annunciated					
	Low	Alarm is historized, reported to US, and annunciated					
	High	Alarm is historized, reported to US, and annunciated					
	Emergency	Not historized, reported to the printer, not annunciated					
	Printer	Alarm historized, reported to the printer, not annunciated					
	JnlPrint						
\$OFSELECT (R681 and later) Off Normal Selective Cutout	Cutout Allow	<i>range</i> = Cutout, Allow <i>default</i> = Cutout					

N/A if SI0ALOPT = None.

SI0ALPR Override SI0 Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint
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N/A if OVRDALOP = None.

OVRDALPR Override Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint
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EVTOPT Event Reporting Option	None Eip	No Reporting Digital PV changes invoke Event-Initiated Processing					
EIPPCODE Event Init Proc Destination							<i>range</i> = Not applicable <i>default</i> = Null

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Point Assignment Display – Begin DI Status Type configuration

Point Configuration		
NAME Tag Name _____		<i>range</i> = Not applicable <i>default</i> = Blank
NODETYP Node Type APM		An APM Point Is Being Configured If a PM, HPM, LM or SM is being Configured, go to the appropriate forms for those devices.
PNTFORM Point Form Full	Component	Point is fully displayed and alarmed Pt. is partially displayed but not alarmed

[illegible]

PRIMMOD		<i>range = Not Applicable</i>
Primary Module ID	_ _ _ _ _ _ _ _ _ _ _ _ _ _ _	<i>default = Null</i>

PNTMODTY APM Module Type	DI DISOE	Digital Input Module High Resolution DI Module
DITYPE Digital Input Type	Status Latched Accum	Status input (open/closed contacts) Steady-state input Pulsed input

APM Digital Input, Status Type Point

Unit:	Tagname:	Engineer:	Date:
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Operating Configuration Display - N/A if PNTFORM = Componnt.

STATETXT State 1-Description Text	1) _____	<i>range = Not applicable</i> <i>default => "On"</i>		
STATETXT State 0-Description Text	0) _____	<i>default = Not applicable</i> <i>default = >"Off"</i>		
BOXCLR (1) State 1 Box Color	Red Cyan	Green (Upper box default) Yellow	White Blue	Black Magenta
BOXCLR (0) State 0 Box Color	Red Cyan	Green Yellow (Lower box default)	White Blue	Black Magenta

Operating Configuration Display - N/A if PNTFORM = Componnt (Continued)

INPTDIR Digital Input Direction	Direct Reverse	Open=>pvraw=off; Closed=>pvraw=on Closed=>pvraw=off; Open=>pvraw=on
PVSRCOPT PV Source Option	Only Auto All	PV source selection is not available and field wiring or memory fetch supplies PV. Selection from PVSOURCE is available & Operator/CL-program may supply PV.
PVSOURCE PV Source	Auto Man Sub	Field wiring or memory fetch supplies PV PV is supplied by Operator or program A value is substituted by a CL program

Applies if PNTMODTY = DISOE

DEBOUNCE Debounce Time in Seconds	_ _	<i>range = 0-50 Milliseconds</i> <i>default = 10 Milliseconds</i>
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Alarming Display – N/A if PNTFORM = Componnt.

As you enter alarm priorities in the next section please refer to the following.

NoAction	Alarm is not reported to the system and not annunciated
Journal	Alarm is historized, not reported to US, not annunciated
Low	Alarm is historized, reported to US, and annunciated
High	Alarm is historized, reported to US, and annunciated
Emergency	Alarm is historized, reported to US, and annunciated
Printer	Not historized, reported to the printer, not annunciated
JnlPrint	Alarm historized, reported to the printer, not annunciated

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Unit:	Tagname:	Engineer:	Date:
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NAME Tag Name		<i>range</i> = Not applicable <i>default</i> = Blank
NODETYP Node Type	APM	An APM Point Is Being Configured If a PM, HPM, LM or SM is being configured, go to the appropriate forms for those devices.
PNTFORM Point Form	Full Component	Point is fully displayed and alarmed Pt. is partially displayed but not alarmed

PTDESC Point Descriptor	_____	<i>range</i> = Not applicable <i>default</i> = Blank
EUDESC Engineering Units Descriptor	_____	<i>range</i> = Not applicable <i>default</i> = Blank
KEYWORD Keyword Descriptor	_____	<i>range</i> = Not applicable <i>default</i> = Blank
ASSOCDSP Associated Display	_____	<i>range</i> = Not applicable <i>default</i> = Blank
\$CDETAIL (R620 and later) Point Custom Display	_____	<i>range</i> = Not applicable <i>default</i> = Blank
UNIT Process Unit Point Is Assigned To	____	<i>range</i> =Two Alphanumeric Characters <i>default</i> = Not applicable
NTWKNUM NIM's UCN That This Point Is In	____	<i>range</i> = 1 to 20 <i>default</i> = Not applicable
NODENUM APMM's Address On The UCN	____	<i>range</i> = 1 to 64 <i>default</i> = Not applicable
MODNUM I/O Module Number	____	<i>range</i> = 1 to 40 <i>default</i> =Not applicable
SLOTNUM Slot Number	____	<i>range</i> = 1-32 <i>default</i> = Not applicable

PRIMMOD		<i>range = Not Applicable</i>
Primary Module ID		<i>default = Null</i>

PNTMODTY APM Module Type	DI DISOE	Digital Input Module High Resolution DI Module
DITYPE Digital Input Type	Status Latched Accum	Status input (open/closed contacts) Steady-state input Pulsed input

APM Digital Input, Latched Type Point

Unit:	Tagname:	Engineer:	Date:
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Operating Configuration Display – N/A if PNTFORM = Componnt.

STATETXT (1) State 1-Description Text			range = Not applicable default =>"On"	
STATETXT (0) State 0-Description Text			range = Not applicable default =>"Off"	
BOXCLR (1) State 1 Box Color	Red Cyan	Green (Upper box default) Yellow	White Blue	Black Magenta
BOXCLR (0) State 0 Box Color	Red Cyan	Green Yellow (Lower box default)	White Blue	Black Magenta

Operating Configuration Display - N/A if PNTFORM = Component

Operating Configuration Display - PV and PV-CLM Components		
INPTDIR Digital Input Direction	Direct Reverse	Open=>pvraw=off; Closed=>pvraw=on Closed=>pvraw=off; Open=>pvraw=on
PVSRCOPT PV Source Option	Only Auto All	PV Source selection is not available and Field wiring or memory fetch supplies PV. Selection from PVSOURCE is available & Operator/CL-program may supply PV.
PVSOURCE PV Source	Auto Man Sub	Field wiring or memory fetch supplies PV. PV is supplied by Operator or program A value is substituted by a CL program

Applies if PNTMODTY = DISOE

DEBOUNCE Debounce Time in Secondss	100	<i>range = 0 - 50 Milliseconds</i> <i>default = 10 Milliseconds</i>
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Alarming Display - N/A if PNTFORM = Component

Alarming Display - N/A if PNTFORM = Component	
EVTOPT Event Reporting Option	None Neither Eip or Soe is allowed Process special is triggered in AM/CG
EIPPCODE Event Init Proc Destination	range = Not applicable default = Null

Applies if PNTMODTY = DISOE

PVCHGDLY		<i>range = 0 – 60 Seconds</i>
PV Change Delay in Seconds	<u> </u> . <u> </u>	<i>default = 0</i>

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[illegible]

APM Digital Input, Accum Type Point

Unit:	Tagname:	Engineer:	Date:
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N/A if PNTFORM = Componnt.

PRIMMOD		<i>range = Not Applicable</i>
Primary Module ID		<i>default = Null</i>

PNTMODTY	DI	Digital Input Module
PM Module Type	DISOE	High Res. DI Module (N/A for Accum Point)
DITYPE	Status	Status input (open/closed contacts)
Digital Input Type	Latched	Steady-state input
	Accum	Pulsed input

Operating Configuration Display - N/A if PNTFORM = Componnt

INPTDIR	Direct	Open=>pvraw=off; Closed=>pvraw=on
Digital Input Direction	Reverse	Closed=>pvraw=off; Open=>pvraw=on
COUNTDWN	Off	<i>range = Off-On</i>
Accumulator Count Down Flag	On	<i>default = Off</i>
AVTV		<i>range = > = 0.0 and NaN</i>
Accumulated Value Target Value		<i>default = 0</i>
RESETVAL		<i>range = > = 0</i>
Accumulator Reset Value		<i>default = 0</i>
OVERVAL	. . .	<i>range = 0 to 100</i>
Overview Display Value		<i>default = 25</i>

OPTDIR DO Direct/Reverse Action	<i>Direct</i> Reverse	For OP pulse = "On", OP = "On" signal For OP pulse = "Off", OP = "Off" signal
PERIOD Pulse Width Modulation Cycle Time	_._._._._	<i>range</i> = 1.0 to 120.0 <i>default</i> = 10.0 (Seconds)

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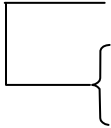
PV Algorithm Display		
PVALGID PV Algorithm Identifier	Null FlowComp	No affect Flow Compensation

APM Flow Compensation PV Algorithm

Unit:	Tagname:	Engineer:	Date:
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PV Configuration Display

PVEUHI PV High Range In Engr Units	_ _ _ _ _ _ _	range =PVEULO to PVEXEUHI default = 100.0
PVEULO PV Low Range In Engr Units	_ _ _ _ _ _ _	range = PVEXEULO to PVEUHI default = 0.0
PVFORMAT PV Decimal Point Format	DO D2 D1 D3	D0=9999. D2=99.99 D1=999.9 D3 = 9.999
PVEXEUHI PV Extended Hi Range in EUs	_ _ _ _ _ _ _ _ _ _ _ _ _	range = >=PVEUHI default = 102.9
PVEXEULO PV Extended Lo Range in EUs	_ _ _ _ _ _ _ _ _ _ _ _ _	range = <=PVEULO default = -2.9
PVCLAMP PV Clamping Option	NoClamp Clamp	No clamping of the PV value Clamp PV value at range extension limit

PVSRCOPT PV Source Option		Only Auto All	PV Source selection is not available Selection from PVSOURCE is available
PVSOURCE PV Source		Auto Man Sub	Field wiring /memory -fetch supplies PV PV is supplied by Operator or program A value is substituted by a CL program

PV Configuration Display (Continued)

TF PV Filter Lag Time in Minutes	_ _ _ _ _ _ _	range = 0.0 to 60.0 (minutes) default = 0.0 (PV will not be delayed)
PVTV PV Target Value in EUs	_ _ _ _ _ _ _ _ _ _ _ _ _	range = PVEXEULO-PVEXEUHI, & NaN default = NaN
OVERVAL Overview Display Value	_ _ _ _ _ _ _	range = 0 to 100 default = 25

APM Flow Compensation PV Algorithm

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Unit:	Tagname:	Engineer:	Date:
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Reg PV Flow Comp Algo Display

		Inputs	Type of compensation
PVEQN(FlowComp)	EqA	F and G	Mass/vol. Flow of liquid
PV Equation Type	EqB	F, P, and T	Mass flow of gas
	EqC	F, G, P, and T	Mass flow of gas
	EqD	F, G, P, and T	Volumetric flow of gas
	EqE	F, P, T, X, and Q	Mass flow of steam
PVCHAR	Linear		Linear PV Characterization
PV Characterization	SqrRoot		Square Root PV Characterization
C	_ _ _ _ _ _ _		range = N/A default = 1.0
Scaling Constant			
C1(FlowComp)	_ _ _ _ _ _ _		range = N/A default = 1.0
Correction Factor			
C2(FlowComp)	_ _ _ _ _ _ _		range => 0.1 default = 1.0
Correction Factor			
COMPHILM	_ _ _ _		range = COMPLOLM to 10.0, & NaN default = 1.25
Compensation Term High Limit			
COMPLOLM	_ _ _ _		range = 0.0 to COMPHILM, and NaN default = 0.8
Compensation Term Low Limit			

<u>Reference Constants</u>			<u>PV Equation =</u>				
			A	B	C	D	E
RG	_ _ _ _ _ _ _	range = Real # default = 1.0	9		9	9	
Reference Specific Gravity							
RP	_ _ _ _ _ _ _	range = Real # default = 1.0		9	9	9	9
Reference Pressure							
RT	_ _ _ _ _ _ _	range = Real # default = 1.0		9	9	9	9
Reference Temperature							
RX	_ _ _ _ _ _ _	range = Real # default = 1.0					9
Reference Steam Compressability							
RQ	_ _ _ _ _ _ _	range = Real # default = 1.0					9
Reference Steam Quality Factor							
P0 (Flow Comp)	_ _ _ _ _ _ _	range = Real # default = 0.0		9	9	9	9
Zero Reference For Pressure							
T0	_ _ _ _ _ _ _	range = Real # default = 0.0		9	9	9	9
Zero Reference for Temperature							

APM Flow Compensation PV Algorithm

Unit:	Tagname:	Engineer:	Date:
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Reg PV Flow Comp Algo Display (Continued)

PISRC(n) Input#	(PV-Input-Con- nection Source)					PISRC (point.parameter)	Destination (PIDSTN)
Equation =	A	B	C	D	E		
n =	1	1	1	1	1		F
n =	2		2	2			G
n =		2	3	3	2		P
n =		3	4	4	3		T
n =					4		X
n =					5		Q

Alarming Display – N/A if PNTFORM = Comppnt

As you enter alarm trip points and alarm priorities in the next section please refer to the following alarm Priority definitions and the note.

NoAction	Alarm is not reported to the system and not annunciated (R670 & later)
Journal	Alarm is historized, not reported to US, not annunciated
Low	Alarm is historized, reported to US, and annunciated
High	Alarm is historized, reported to US, and annunciated
Emergency	Alarm is historized, reported to US, and annunciated
Printer	Not historized, reported to the printer, not annunciated
JnlPrint	Alarm historized, reported to the printer, not annunciated

Note: If you do not enter a trip point value for a given pair of trip-points and alarm-priorities, do not Circle an alarm priority option for that pair (i.e., no trip point, no alarm priority).

Alarming Display - N/A if PNTFORM = Comppnt

\$AUXUNIT Auxiliary Unit	..	range = Any configured LCN Unit ID default = NaN
\$ALDLOPT (R620 and later) Alarm Delay Option	None OffDlyLo OffDlyHi OffDlyEm OnDlyLo OnDlyHi OnDlyEm	Off Delay Lo Off Delay Hi Off Delay Emergency On Delay Lo On Delay Hi On Delay Emergency
\$DLBADAL (R681 and later) Delay for BADPV	None Offdelay	range = Offdelay, None default = None
DLTIME (R620 and later) Delay Time	..	range = 0 to 9999 seconds default = 0 seconds
\$REALARM Re-alarming Interval (R660 and later)	..	range = 0 to 9999 minutes default = 0 minutes

APM Flow Compensation PV Algorithm

Unit:	Tagname:	Engineer:	Date:
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Alarming Display - N/A if PNTFORM = Componnt

PVALDB PV Alarm Deadband In %	Half	One	Two	Three	Four	Five	EU
PVALDBEU PV Alarm Deadband In EU							} range > = 0.0 default = 1.0
PVHITP PV High Alarm Trip Point							range = PVLOTP - PVHHTP, & NaN default = NaN
PVHIPR PV High Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergncy	Printer	JnlPrint
\$HISELCT (R681 and later) PV High Selective Cutout	Cutout Allow				range = Cutout, Allow default = Cutout		
PVLOTP PV Low Alarm Trip Point					range = PVLLTP - PVHITP, & NaN default = NaN		
PVLOPR PV Low Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergncy	Printer	JnlPrint
\$LOSELCT (R681 and later) PV Low Selective Cutout	Cutout Allow				range = Cutout, Allow default = Cutout		
PVHHTP PV Hi Hi Alarm Trip Point					range = PVHITP to PVEUHI, & NaN default = NaN		
PVHHPR PV High High Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergncy	Printer	JnlPrint
\$HHSELCT (R681 and later) PV High High Selective Cutout	Cutout Allow				range = Cutout, Allow default = Cutout		

PVLLTP PV Low Low Alarm Trip Point	_ _ _ _ _ _ _	<i>range = PVEULO - PVLOTP, & NaN default = NaN</i>						
PVLLPR PV Low Low Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint	
\$LLSELCT (R681 and later) PV Low Low Selective Cutout	Cutout Allow	<i>range = Cutout, Allow default = Cutout</i>						
PVSGCHTP PV Significant Change Alarm Trip Point	_ _ _ _ _ _ _	<i>range = >=0.0, and NaN default = NaN</i>						
PVROCPTP PV Pos Rate Of Change Trip Point	_ _ _ _ _ _ _	<i>range = >=0.0, and NaN default = NaN</i>						
PVROCPPR PV Pos Rate Of Chg Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint	
\$RPSELCT (R681 and later) PV Positive Rate of Change Selective Cutout	Cutout Allow	<i>range = Cutout, Allow default = Cutout</i>						
PVROCNTP PV Neg Rate Of Change Trip Point	_ _ _ _ _ _ _	<i>range = >=0.0, and NaN default = NaN</i>						
PVROCNP PV Neg Rate Of Chg Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint	
\$RNSELCT (R681 and later) PV Negative Rate of Change Selective Cutout	Cutout Allow	<i>range = Cutout, Allow default = Cutout</i>						
BADPVPR Bad-PV Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint	
ALENBST Point Alarm Enable Status	Enable Disable Inhibit	<u>Displayed</u> Yes No No	<u>Logged</u> Yes Yes No	<u>Reported To EIP</u> Yes Yes No				

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		N/A if PNTFORM = Component.
PRIMMOD		<i>range</i> = Not Applicable
Primary Module ID	_.	<i>default</i> = Null
USERID		Sixteen Character String
User ID Reservation	_.	<i>default</i> = dashes

APM General Linearization PV Algorithm

Unit:	Tagname:	Engineer:	Date:
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PV Algorithm Display

PVALGID PV Algorithm Identifier	NULL GenLin	No affect General Linearity
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PV Configuration Display

PVEUHI PV High Range In Engr Units	_ _ _ _ _ _ _	<i>range</i> = PVEULO to PVEXEUHI <i>default</i> = 100.0
PVEULO PV Low Range In Engr Units	_ _ _ _ _ _ _	<i>range</i> = PVEXEULO to PVEUHI <i>default</i> = 0.0
PVFORMAT PV Decimal Point Format	DO D2 D1 D3	D0 = 9999. D2 = 99.99 D1 = 999.9 D3 = 9.999
PVEXEUHI PV Extended Hi Range in EUs	_ _ _ _ _ _ _ _ _ _ _ _ _	<i>range</i> = >=PVEUHI <i>default</i> = 102.9
PVEXEULO PV Extended Lo Range in EUs	_ _ _ _ _ _ _ _ _ _ _ _ _	<i>range</i> = <=PVEULO <i>default</i> = -2.9
PV CLAMP PV Clamping Option	NoClamp Clamp	No clamping of the PV value Clamp PV value at range extension limit

N/A if PNTFORM = Componnt

PVSRCOPT PV Source Option	Only Auto All	PV Source selection is not available Selection from PVSOURCE is available
PVSOURCE PV Source	Auto Man Sub	Field wiring /memory -fetch supplies PV PV is supplied by Operator or program A value is substituted by a CL program

PV Configuration Display (continued)

TF PV Filter Lag Time in Minutes	_ _ _ _ _ _ _	<i>range</i> = 0.0 to 60.0 (minutes) <i>default</i> = 0.0 (PV will not be delayed)
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N/A if PNTFORM = Componnt

PVTV PV Target Value in EUs	_ _ _ _ _ _ _ _ _ _ _ _ _	<i>range</i> = PVEXEULO-PVEXEUHI, & NaN <i>default</i> = NaN
OVERVAL Overview Display Value	_ _ _ _ _ _ _	<i>range</i> = 0 to 100 <i>default</i> = 25

APM General Linearization PV Algorithm

Unit:	Tagname:	Engineer:	Date:
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Reg PV General/Linear Algo Display

SEGTOT		range = 1 to 12	
# Of Characteristic Curve Segments		default = 1	
<u>Characteristic Curve Coordinates</u>			
IN0-IN12 range = >=prev. coord.&<=next coord. default = NaN		OUT0-OUT12 range = >=prev. coord.&<=next coord. default = NaN	
<u>Input</u> <u>Coordinate</u>	<u>Value</u>	<u>Output</u> <u>Coordinate</u>	<u>Value</u>
IN0		OUT0	
IN1		OUT1	
IN2		OUT2	
IN3		OUT3	
IN4		OUT4	
IN5		OUT5	
IN6		OUT6	
IN7		OUT7	
IN8		OUT8	
IN9		OUT9	
IN10		OUT10	
IN11		OUT11	
IN12		OUT12	

PISRC (PV-Input-Connection Source)	<u>PISRC</u> (point.parameter)	<u>Destination Para</u> <u>-meter (PIDSTN)</u>
<u>Input#</u>		
1		P1

Alarming Display – N/A if PNTFORM = Component

As you enter alarm trip points and alarm priorities in the next section please refer to the following alarm priority definitions and the note.

NoAction	Alarm is not reported to the system and not annunciated (R670 & later)
Journal	Alarm is historized, not reported to US, not annunciated
Low	Alarm is historized, reported to US, and annunciated
High	Alarm is historized, reported to US, and annunciated
Emergency	Alarm is historized, reported to US, and annunciated
Printer	Not historized, reported to the printer, not annunciated
JnlPrint	Alarm historized, reported to the printer, not annunciated

Note: If you do not enter a trip point value for a given pair of trip-points and alarm-priorities, do not circle an alarm priority option for that pair (i.e., no trip point, no alarm priority).

APM General Linearization PV Algorithm

Unit:	Tagname:	Engineer:	Date:
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Alarming Display - N/A if PNTFORM = Componnt

\$AUXUNIT Auxiliary Unit	_ _	range = Any configured LCN Unit ID default = NaN
\$ALDLOPT (R620 and later) Alarm Delay Option	None OffDlyLo OffDlyHi OffDlyEm OnDlyLo OnDlyHi OnDlyEm	Off Delay Lo Off Delay Hi Off Delay Emergency On Delay Lo On Delay Hi On Delay Emergency
\$DLBADAL (R681 and later) Delay for BADPV	None Offdelay	range = Offdelay, None default = None
\$DLTIME (R620 and later) Delay Time	_ _ _ _	range = 0 to 9999 seconds default = 0 seconds
\$REALARM (R660 and later) Re-alarming Interval	_ _ _ _	range = 0 to 9999 minutes default = 0 minutes
PVALDB PV Alarm Deadband In %	Half One Two Three Four Five EU	
PVALDBEU PV Alarm Deadband In EU	_ _ _ _	range > = 0.0 default = 1.0
PVHITP PV High Alarm Trip Point	_ _ _ _ _ _ _	range = PVLOTP - PVHHTP, & NaN default = NaN
PVHIPR PV High Alarm Priority	NoAction Journal Low High Emergency Printer JnlPrint (R670 and later)	
\$HISELCT (R681 and later) PV High Selective Cutout	Cutout Allow	range = Cutout, Allow default = Cutout
PVLOTP PV Low Alarm Trip Point	_ _ _ _ _ _ _	range = PVLLTP - PVHITP, & NaN default = NaN
PVLOPR PV Low Alarm Priority	NoAction Journal Low High Emergency Printer JnlPrint (R670 and later)	
\$LOSELCT (R681 and later) PV Low Selective Cutout	Cutout Allow	range = Cutout, Allow default = Cutout
PVHHTP(RegCtl) PV Hi Hi Alarm Trip Point	_ _ _ _ _ _ _	range = PVHITP to PVEUHI, & NaN default = NaN
PVHHPR PV High High Alarm Priority	NoAction Journal Low High Emergency Printer JnlPrint (R670 and later)	
\$HHSELCT (R681 and later) PV High High Selective Cutout	Cutout Allow	range = Cutout, Allow default = Cutout
PVLLTP(RegCtl) PV Low Low Alarm Trip Point	_ _ _ _ _ _ _	range = PVEULO – PVLOTP, & NaN default = NaN

PVLLPR	NoAction	Journal	Low	High	Emergency	Printer	JnlPrint
PV Low Low Alarm Priority	(R670 and later)						
\$ LSELCT (R681 and later)	Cutout				range = Cutout, Allow		
PV Low Low Selective Cutout	Allow				default = Cutout		
PVSGCHTP	_ . _ .				range = >=0.0, and NaN		
PV Significant Change Alarm Trip Point					default = NaN		
PVROCPTP	_ .				range = >=0.0, and NaN		
PV Pos Rate Of Change Trip Point					default = NaN		
PVROCPPRR	NoAction	Journal	Low	High	Emergency	Printer	JnlPrint
PV Pos Rate Of Chg Alarm Priority	(R670 and later)						
\$ RPSLCT (R681 and later)	Cutout				range = Cutout, Allow		
PV Positive Rate of Change Selective Cutout	Allow				default = Cutout		
PVROCNTP	_ .				range = >=0.0, and NaN		
PV Neg Rate Of Change Trip Point					default = NaN		
PVROCNPR	NoAction	Journal	Low	High	Emergency	Printer	JnlPrint
PV Neg Rate Of Chg Alarm Priority	(R670 and later)						
\$ RNSLCT (R681 and later)	Cutout				range = Cutout, Allow		
PV Negative Rate of Change Selective Cutout	Allow				default = Cutout		
BADPVPR	NoAction	Journal	Low	High	Emergency	Printer	JnlPrint
Bad-PV Alarm Priority	(R670 and later)						
ALENBST							
Point Alarm Enable Status							
	Enable			Displayed	Logged	Reported To EIP	
	Disable			Yes	Yes	Yes	
	Inhibit			No	Yes No	Yes	
				No		No	

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PVALGID	PV Algorithm Display	Effect
PV Algorithm Identifier	Null HiLoAvg	No affect High Low Average Selector

APM High Low Average PV Algorithm

Unit:	Tagname:	Engineer:	Date:
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PV Configuration Display

PVEUHI PV High Range In Engr Units	_ _ _ _	<i>range</i> = PVEULO to PVEXEUHI <i>default</i> = 100.0
PVEULO PV Low Range In Engr Units	_ _ _ _	<i>range</i> = PVEXEULO to PVEUHI <i>default</i> = 0.0
PVFORMAT PV Decimal Point Format	DO D1	D2 D3 D0 = 9999. D1 = 999.9 D2 = 99.99 D3 = 9.999
PVEXEUHI PV Extended Hi Range in EUs	_ _ _ _ _ _ _	<i>range</i> = >=PVEUHI <i>default</i> = 102.9
PVEXEULO PV Extended Lo Range in EUs	_ _ _ _ _ _ _	<i>range</i> = <=PVEULO <i>default</i> = -2.9
PVCLAMP PV Clamping Option	NoClamp Clamp	No clamping of the PV value Clamp PV value at range extension limit

N/A if PNTFORM = Componnt.

PVSRCOPT PV Source Option		Only Auto All	PV Source selection is not available Selection from PVSOURCE is available
PVSOURCE PV Source		Auto Man Sub	Field wiring or memory fetch supplies PV PV is supplied by Operator or program A value is substituted by a CL program

PV Configuration Display (Continued)

TF PV Filter Lag Time in Minutes	_ _ _ _	<i>range</i> = 0.0 to 60.0 (minutes) <i>default</i> = 0.0 (PV will not be delayed)
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N/A if PNTFORM = Componnt.

PVTV PV Target Value in EUs	_ _ _ _ _ _ _	<i>range</i> = PVEXEULO-PVEXEUHI, & NaN <i>default</i> = NaN
OVERVAL Overview Display Value	_ _ _	<i>range</i> = 0 to 100 <i>default</i> = 25

APM High Low Average PV Algorithm

Unit:	Tagname:	Engineer:	Date:
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Reg PV High/Low Avg Algo Display

PVEQN (HiLoAvg) PV Equation Type	EqA EqB EqC	Select & identify highest of up to 6 inputs Select & identify lowest of up to 6 inputs Calculate the average of up to 6 inputs
N (HiLoAvg) # Inputs To HiLoAvg Algo		range = 2 to 6 default = 2
NMIN Minimum No. Of Good Inputs		range = 1 to N (# Inputs parameter) default = 1
FRCPERM Forced Input Permissive	☐ Off ☐ On	Optr. cannot select input to HiLoAvg Operator can select input to HiLoAvg
FSELIN Input To Be Force Selected	☐ SelectP1 ☐ SelectP2 ☐ SelectP3 ☐ SelectP4 ☐ SelectP5 ☐ SelectP6	Input P1 is the forced input Input P2 is the forced input Input P3 is the forced input Input P4 is the forced input Input P5 is the forced input Input P6 is the forced input
PISRC (PV-Input-Connection Source)		<u>Destination Para- meter (PIDSTN)</u>
<u>Input#</u>	<u>PISRC (point.parameter)</u>	
1		P1
2		P2
3		P3
4		P4
5		P5
6		P6

Alarming Display – N/A if PNTFORM = Componnt

As you enter alarm trip points and alarm priorities in the next section please refer to the following alarm Priority definitions and the note.

NoAction	Alarm is not reported to the system and not annunciated (R670 & later)
Journal	Alarm is historized, not reported to US, not annunciated
Low	Alarm is historized, reported to US, and annunciated
High	Alarm is historized, reported to US, and annunciated
Emergency	Alarm is historized, reported to US, and annunciated Not
Printer	historized, reported to the printer, not annunciated Alarm
JnlPrint	historized, reported to the printer, not annunciated

Note: If you do not enter a trip point value for a given pair of trip-points and alarm-priorities, do not Circle an alarm priority option for that pair (i.e., no trip point, no alarm priority).

Alarming Display – N/A if PNTFORM = Componnt

\$AUXUNIT Auxiliary Unit		range = Any configured LCN Unit ID default = NaN
\$ALDLOPT (R620 and later) Alarm Delay Option	None OffDlyLo OffDlyHi OffDlyEm OnDlyLo OnDlyHi OnDlyEm	Off Delay Lo Off Delay Hi Off Delay Emergency On Delay Lo On Delay Hi On Delay Emergency

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Alarming Display - N/A if PNTFORM = Component

\$DLBADAL (R681 and later) Delay for BADPV	None Offdelay	<i>range = Offdelay, None default = None</i>
DLTIME (R620 and later) Delay Time	_ _ _ _	<i>range = 0 to 9999 seconds default = 0 seconds</i>
REALARM Re-alarming Interval (R660 and later)	_ _ _ _	<i>range = 0 to 9999 minutes default = 0 minutes</i>
PVALDB PV Alarm Deadband In %	Half One Two Three Four Five EU	}
PVALDBEU PV Alarm Deadband In EU	_ _ _ _	
PVHITP PV High Alarm Trip Point	_ _ _ _ _ _ _	<i>range = PVLOTP - PVHITP, & NaN default = NaN</i>
PVHIPR PV High Alarm Priority	NoAction Journal Low	High Emergency Printer JnlPrint
\$HISELCT (681 and later) PV High Selective Cutout	Cutout Allow	<i>range = Cutout, Allow default = Cutout</i>
PVLOTP PV Low Alarm Trip Point	_ _ _ _ _ _ _	<i>range = PVLLTP - PVHITP, & NaN default = NaN</i>
PVLOPR PV Low Alarm Priority	NoAction Journal Low	High Emergency Printer JnlPrint
\$LOSELCT (681 and later) PV Low Selective Cutout	Cutout Allow	<i>range = Cutout, Allow default = Cutout</i>
PVHHTP PV Hi Hi Alarm Trip Point	_ _ _ _ _ _ _	<i>range = PVHITP to PVEUHI, & NaN default = NaN</i>
PVHHPR PV High High Alarm Priority	NoAction Journal Low	High Emergency Printer JnlPrint
\$HHSELCT (681 and later) PV High High Selective Cutout	Cutout Allow	<i>range = Cutout, Allow default = Cutout</i>
PVLLTP PV Low Low Alarm Trip Point	_ _ _ _ _ _ _	<i>range = PVEULO - PVLOTP, & NaN default = NaN</i>
PVLLPR PV Low Low Alarm Priority	NoAction Journal Low	High Emergency Printer JnlPrint
\$LLSELCT (681 and later) PV Low Low Selective Cutout	Cutout Allow	<i>range = Cutout, Allow default = Cutout</i>
PVSGCHTP PV Significant Change Alarm Trip Point	_ _ _ _ _ _ _	<i>range = >=0.0, and NaN default = NaN</i>
PVROCTP PV Pos Rate Of Change Trip Point	_ _ _ _ _ _ _	<i>range = >=0.0, and NaN default = NaN</i>

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Alarming Display - N/A if PNTFORM = Componnt

PVROCPPR PV Pos Rate Of Chg Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint
\$RPSELCT (R681 and later) PV Positive Rate of Change Selective Cutout	Cutout Allow				<i>range = Cutout, Allow</i> <i>default = Cutout</i>		
PVROCNTP PV Neg Rate Of Change Trip Point	_.:_.:_.:_.:_.:_.:_.				<i>range = >=0.0, and NaN</i> <i>default = NaN</i>		
PVROCNPR PV Neg Rate Of Chg Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint
\$RNSELCT (R681 and later) PV Negative Rate of Change Selective Cutout	Cutout Allow				<i>range = Cutout, Allow</i> <i>default = Cutout</i>		
BADPVPR Bad-PV Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint
ALENBST Point Alarm Enable Status			Enable Disable Inhibit	<u>Displayed</u> Yes No No	<u>Logged</u> Yes Yes No	<u>Reported To EIP</u> Yes Yes No	

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Point Assignment Display

Control Algorithm Display

PV Configuration Display

OVERVAL	_. .	<i>range = 0 to 100</i>
Overview Display Value		<i>default = 25</i>

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Mode Configuration Display

RCASOPT Remote Cascade Option	None Ddc	Local cascade is the only valid cascade AM strategy stores to OP. No OP limits.
SHEDTIME Remote Cascade Shed Time		range = 0 to 1000 (seconds) default = 0 (mode shed disabled)
SHEDMODE Shed Mode	Man } Bcas }	Supervisory point will shed to Manual Supervisory point will shed to BackupCas
NMODE (RegCtl) RegCtl Point's Normal Mode	None } Man } Cas } Bcas }	No configured "normal" operating mode Manual is the "normal" operating mode Cascade is "normal" operating mode Backup Cas is "normal" operating mode
NMODATTR Normal Mode Attribute	None Operator Program	Parameter NMODE sets "normal" mode Operator sets "normal" operating mode Program sets "normal" operating mode
MODEPERM Mode Permissive	Permit NotPerm	Operator can change this point's mode Optr. cannot change this point's mode
\$MODESEL (R630 and later) Mode Selection List	Man Prog Cas Bcas	Man, Cas, Bcas mode and Prog mode attribute are allowed for this data point.
EXTSWOPT External Mode Switching	None Ems	No external mode switching is allowed Ext. source can change point's mode

Reg Control Incr/Sum Algo Display

M # Of Initializable Control Algorithm Inputs		range = 2 to 4 default = 2
XEUHI X Input High Range In EUs		range = >= XEULO default = 100.0
XEULO X Input Low Range in Eus		range = <= XEUHI default = 0.0
K1 (IncrSum) Gain Constant For X1 Input		range = >=0.0 default = 1.0
K2 (IncrSum) Gain Constant For X2 Input		range = >=0.0 default = 1.0
K3 (IncrSum) Gain Constant For X3 Input		range = >=0.0 default = 1.0
K4 (IncrSum) Gain Constant For X4 Input		range = >=0.0 default = 1.0
NOCINPTS # Of Control Input Connections		range = 0 to 4 default = 0

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Reg Control Incr/Sum Algo Display (Continued)

CISRC (Control-Input-Connection Source)	CISRC (point.parameter)	Destination Parameter (CODSTN)
Input #		
1		
2		
3		
4		

Output Configuration Display

NOCOPTS		range = 0 to 4 default = 1
# Of Control Output Connections		
CODSTN (Control-Output-Connection Destination)		
Output #	CODSTN (point.parameter)	
1		
2		
3		
4		
OPHILM		range = OPLOLM – 106.9 & NaN default = 105.0
Output High Limit In Percent		
OPLOLM		range = -6.9 – OPHILM, & NaN default = -5.0
Output Low Limit In Percent		
SAFEOP		range = -6.9 to to 106.9 & NaN default = NaN
Safe Output Limit In Percent		
OPMCHLM		range = >=0.0, and NaN default = 0.0
Output Minimum Change In %		
OPROCLM		range = >=0.0, and NaN default = NaN
OP Rate Of Change Limit In %		
\$OPTOL		range = 0.0 to 106.9 default = 0.0
Output Tolerance in %		
OUTIND	Direct	4 - 20mA = 0 – 100%
Reverse Output Indicator	Reverse	20 - 4mA = 0 – 100%
Note: With R670 and later, when this parameter is set to Reverse, the values of the displayed output and the related output parameters are changed. Refer to the definition of the OUTIND parameter in the PM Family Parameter Dictionary, XPM09.		

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Alarming Display - N/A if PNTFORM = Componnt

As you enter alarm priorities in the next section please refer to the following.

NoAction	Alarm is not reported to the system and not annunciated (R670 & later)
Journal	Alarm is historized, not reported to US, not annunciated
Low	Alarm is historized, reported to US, and annunciated
High	Alarm is historized, reported to US, and annunciated
Emergency	Alarm is historized, reported to US, and annunciated Not
Printer	historized, reported to the printer, not annunciated Alarm
JnlPrint	historized, reported to the printer, not annunciated

Alarming Display - N/A if PNTFORM = Componnt

\$AUXUNIT Auxiliary Unit	_ _ _	<i>range = Any configured LCN Unit ID default = NaN</i>		
\$ REALARM (R660 and Later) Re-alarming Interval	_ _ _	<i>range = 0 to 9999 minutes default = 0 minutes</i>		
BADOCOPT BAD OP Conn Alarm Option	Off On	Bad Output Connection Alarm is not initially permitted Bad Output Connection Alarm is Initially permitted		
BADOCPR BAD OP Conn Alarm Priority	NoAction (R670 and later)	Journal	Low	High
		Emergency	Printer	JnlPrint
OPALDB Output Alarm Deadband	_ _ _ _	<i>range = 0 - 25.0 but $\leq$ (OPHITP-OPLOTP)/2 default = 5.0 Output Units</i>		
OPHITP Output High Alarm Trip Point	_ _ _ _	<i>range = OPLOTP to OPHILM and NaN default = NaN</i>		
OPHIPR Output High Alarm Priority	NoAction (R670 and later)	Journal	Low	High
		Emergency		JnlPrint
OPLOTP Output Low Alarm Trip Point	_ _ _ _	<i>range = OPLOLM to OPHITP and NaN default = NaN</i>		
OPLOPR Output Low Alarm Priority	NoAction (R670 and later)	Journal	Low	High
		Emergency		JnlPrint
ALENBST Point Alarm Enable Status	Enable Disable Inhibit	<u>Displayed</u> Yes No No	<u>Logged</u> Yes Yes No	<u>Reported To EIP</u> Yes Yes No

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LOGMIX	12_24_4	12 inputs	24 logic blocks	4 outputs
Logic Mix	12_16_8	12 inputs	16 logic blocks	8 outputs
	12_8_12	12 inputs	8 logic blocks	12 outputs

APM Logic Point

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Input Connections Display

LISRC (Logic Point Inputs)	
Input	Source Point.Parameter
L1	
L2	
L3	
L4	
L5	
L6	
L7	
L8	
L9	
L10	
L11	
L12	

LIBADOPT	Off	"Off" state is substituted for bad input
Logic Bad Input Handling Option	On	"On" state is substituted for bad input
	Hold	Last good value is subed. for bad input

Input Connections Display (Continued)

Logic Flags And Numerics - Initial Values
Numerics 1-8 are used as local variables. Entry is a real number. Default = NAN

NN1		NN2	
NN3		NN4	
NN5		NN6	
NN7		NN8	

Flags 7-12 are used as local variables

FL7	Off	On	FL8	Off	On
FL9	Off	On	FL10	Off	On
FL11	Off	On	FL12	Off	On

APM Logic Point

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Logic Output Connections Display

The following 2 parameters are not to be configured but are to be used for making choices when filling out the output-connections table below:

LOSRC Logic-Output-Conn Source	L1...L12 FL1...FL12 NN1...NN8 SO1..SO24	L(external input) = a logical or a real value FL(flag) = either a 1(On) or a 0(Off) NN(numeric) = data type of Real SO(Status Output) = Boolean (0 or 1)
LOENBL Logic Output Enable	L1...L12 FL1...FL12 SO1..SO16	L(external input) = a logical or a real value FL(flag) = either a 1(On) or a 0(Off) SO(Status Output) = Boolean (0 or 1)

Logic Output Destination Table LODSTN (Logic- Output-Connection Destination) Destination

<u>LOSRC</u> defaults = FL1	<u>Point.Parameter</u> defaults = blank.blank	<u>LOENBL</u> defaults = FL2
01		
02		
03		
04		
Enter data in lines 5 through 8 if LOGMIX = 12_16_8 or 12_8_12		
05		
06		
07		
08		
Enter data in lines 9 through 12 if LOGMIX = 12_8_12		
09		
10		
11		
12		

APM Logic Point

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Logic Descriptors Display – N/A if PNTFORM = Componnt.

NODESC

Number of Generic Descriptors

|_ |

range = 0 to 12

default = 0

The following parameter is not to be configured but is to be used for making choices when filling out the logic descriptors table below:

PRMDESC(1-12)

Parameter Descriptor Assignment

L1...L12

FL1..FL12

NN1..NN8

SO1..SO24

Logic point inputs

Flags

Numerics

Status Outputs

Logic Descriptors Table

<u>Descr#</u>	<u>Generic Descriptors</u> (defaults = blanks)	<u>Descr.Assign</u> (defaults = SO1)
1	GENDESC (1) _.	PRMDESC (1) _.
2	GENDESC (2) _.	PRMDESC (2) _.
3	GENDESC (3) _.	PRMDESC (3) _.
4	GENDESC (4) _.	PRMDESC (4) _.
5	GENDESC (5) _.	PRMDESC (5) _.
6	GENDESC (6) _.	PRMDESC (6) _.
7	GENDESC (7) _.	PRMDESC (7) _.
8	GENDESC (8) _.	PRMDESC (8) _.
9	GENDESC (9) _.	PRMDESC (9) _.
10	GENDESC (10) _.	PRMDESC (10) _.
11	GENDESC (11) _.	PRMDESC (11) _.
12	GENDESC (12) _.	PRMDESC (12) _.

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Logic Alarming Display – N/A if PNTFORM = Component.

\$AUXUNIT Auxiliary Unit	_ _	<i>range</i> = Any configured LCN Unit ID <i>default</i> = NaN					
\$REALARM (R660 and later) Re-alarming Interval	_ _ _ _	<i>range</i> = 0 to 9999 minutes <i>default</i> = 0 minutes					
ALENBST Alarm Enable State	Enable Disable Inhibit	Normal Alarm/Event Reporting Disable Alarms Reporting to US Inhibit Alarms not Reported to US					
C1PR Custom Alarm #1 Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint
C1DESC Custom Alarm #1 Descriptor	_ _ _ _ _ _ _	<i>range</i> = 8 Character String <i>default</i> = Blank					
C1SRC Custom Alarm #1 Source	_ _ _ _	<i>range</i> = None, FL1 to FL12, SO1 to SO24, L1 to L12 <i>default</i> = None					
C2PR Custom Alarm #2 Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint
C2DESC Custom Alarm #2 Descriptor	_ _ _ _ _ _ _	<i>range</i> = 8 Character String <i>default</i> = Blank					
C2SRC Custom Alarm #2 Source	_ _ _ _	<i>range</i> = None, FL1 to FL12, SO1 to SO24, L1 to L12 <i>default</i> = None					
C3PR Custom Alarm #3 Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint
C3DESC Custom Alarm #3 Descriptor	_ _ _ _ _ _ _	<i>range</i> = 8 Character String <i>default</i> = Blank					
C3SRC Custom Alarm #3 Source	_ _ _ _	<i>range</i> = None, FL1 to FL12, SO1 to SO24, L1 to L12 <i>default</i> = None					
C4PR Custom Alarm #4 Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint
C4DESC Custom Alarm #4 Descriptor	_ _ _ _ _ _ _	<i>range</i> = 8 Character String <i>default</i> = Blank					
C4SRC Custom Alarm #4 Source	_ _ _ _	<i>range</i> = None, FL1 to FL12, SO1 to SO24, L1 to L12 <i>default</i> = None					
EIPPCODE EIP Point ID	_ _ _ _ _ _ _	<i>range</i> = Not Applicable <i>default</i> = Null					

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Use the appropriate section of this form for each Logic Block (n) required. Circle the Algorithm type (e.g., AND, OR, etc.) and enter/select the parameters that apply for that algorithm.

In the following tables, valid entries for S1, S2, S3, or S4, are: L1 - L12, SO1 - SO24, or FL1 - FL12. Valid entries for R1 and R2 are L1 - L12 or NN1 - NN8. The Logic Block outputs are SO1 through SO16 or SO1 through SO24.

LOGALGID (n) Logic Block n = ____ Enter 1-24 for LOGMIX 12-24-4; Enter 1-16 for LOGMIX 12-16-8

NULL	AND	OR	NAND	NOR		
S1(n)		_ _ _ _		Default = L1	S1Rev(n)	OFF ON
S2(n)		_ _ _ _		Default = L1	S2Rev(n)	OFF ON
S3(n)		_ _ _ _		Default = L1	S3Rev(n)	OFF ON

LOGALGID (n) Logic Block n = ____ Enter 1-24 for LOGMIX 12-24-4; Enter 1-16 for LOGMIX 12-16-8

NULL	NOT	
S1(n)	_ _ _ _	Default = L1

LOGALGID (n) Logic Block n = ____ Enter 1-24 for LOGMIX 12-24-4; Enter 1-16 for LOGMIX 12-16-8

NULL	XOR	
S1(n)	_ _ _ _	Default = L1
S2(n)	_ _ _ _	Default = L1

APM Logic Point

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LOGALGID (n) Logic Block n = ____ Enter 1-24 for LOGMIX 12-24-4; Enter 1-16 for LOGMIX 12-16-8			
NULL	QOR2	QOR3	
S1(n)	_ . _ . _ . _		Default = L1
S2(n)	_ . _ . _ . _		Default = L1
S3(n)	_ . _ . _ . _		Default = L1
S4(n)	_ . _ . _ . _		Default = L1

LOGALGID (n) Logic Block n = ____ Enter 1-24 for LOGMIX 12-24-4; Enter 1-16 for LOGMIX 12-16-8			
NULL	SWITCH	FLIPFLOP	CHDETECT
S1(n)	_ . _ . _ . _		Default = L1
S2(n)	_ . _ . _ . _		Default = L1
S3(n)	_ . _ . _ . _		Default = L1

LOGALGID (n) Logic Block n = ____ Enter 1-24 for LOGMIX 12-24-4; Enter 1-16 for LOGMIX 12-16-8						
NULL	EQ	NE	GT	GE	LT	LE
R1(n)	_ . _ . _ . _					Default = L1
R2(n)	_ . _ . _ . _					Default = L1
Deadband (n)	_ . _ . _ . _					range = >=0 default = 1.0

LOGALGID (n) Logic Block n = ____ Enter 1-24 for LOGMIX 12-24-4; Enter 1-16 for LOGMIX 12-16-8		
NULL	CHECKBAD	
R1(n)	_ . _ . _ . _	Default = L1

APM Logic Point

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LOGALGID (n) Logic Block n = ____ Enter 1-24 for LOGMIX 12-24-4; Enter 1-16 for LOGMIX 12-16-8

NULL **PULSE** **MINPULSE** **MAXPULSE**

S1(n) |_. _ . _ . _| Default = L1

Dlytime(n) |_. _ . _ . _| range = 0-8000
default = 1.0

LOGALGID (n) Logic Block n = ____ Enter 1-24 for LOGMIX 12-24-4; Enter 1-16 for LOGMIX 12-16-8

NULL **DELAY**

S1(n) |_. _ . _ . _| Default = L1

LOGALGID (n) Logic Block n = ____ Enter 1-24 for LOGMIX 12-24-4; Enter 1-16 for LOGMIX 12-16-8

NULL **ONLY** **OFFDLY**

S1(n) |_. _ . _ . _| Default = L1

Dlytime(n) |_. _ . _ . _| range = 0-8000
default = 1.0

LOGALGID (n) Logic Block n = ____ Enter 1-24 for LOGMIX 12-24-4; Enter 1-16 for LOGMIX 12-16-8

NULL **WATCHDOG**

Dlytime(n) |_. _ . _ . _| range = 0-8000
default = 1.0

APM Middle of 3 PV Algorithm

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Unit:	Tagname:	Engineer:	Date:
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Point Assignment Display

NAME Tag Name	_____	range = Not applicable default = Blank
NODETYPE Node Type	PM APM HPM	A PM Point Is Being Configured An APM Point Is Being Configured An HPM Point Is Being Configured
PNTFORM Point Form	FuLL Componnt	Point is fully displayed and alarmed Pt. is partially displayed but not alarmed

PTDESC Point Descriptor	_____	range = Not applicable default = Blank
EUDESC Engineering Units Descriptor	_____	range = Not applicable default = Blank
KEYWORD Keyword Descriptor	_____	range = Not applicable default = Blank
ASSOCDSP Associated Display	_____	range = Not applicable default = Blank
\$CDETAIL (R620 and later) Point Custom Display	_____	range = Not applicable default = Blank
UNIT Process Unit Point Is Assigned To	____	range = Two Alphanumeric Characters default = Not applicable
NTWKNUM NIM's UCN That This Point Is In	____	range = 1 to 20 default = Not applicable
NODENUM APMM's Address On The UCN	____	range = 1 to 64 default = Not applicable
MODNUM I/O Module Number	____	range = (Future) default = 0
SLOTNUM Slot Number	____	range = 1-80 (but <=NPVSLOT) default = Not applicable

N/A if PNTFORM = Componnt.

PRIMMOD Primary Module ID	_____	range = Not Applicable default = Null
USERID User ID Reservation	_____	Sixteen Character String default = dashes

PV Algorithm Display

PVALGID PV Algorithm Identifier	NULL MidOf3	No affect Middle of Three
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APM Middle of 3 PV Algorithm

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PV Configuration Display (Continued)

PVEUHI PV High Range In Engr Units	_ _ _ _ _ _ _	<i>range</i> = PVEUHI to PVEXEUHI <i>default</i> = 100.0
PVEULO PV Low Range In Engr Units	_ _ _ _ _ _ _	<i>range</i> = PVEXEULO to PVEUHI <i>default</i> = 0.0
PVFORMAT PV Decimal Point Format	DO D2 D1 D3	D0=9999. D2=99.99 D1=999.9 D3 = 9.999
PVEXEUHI PV Extended Hi Range in EUs	_ _ _ _ _ _ _ _ _ _ _ _ _	<i>range</i> = >=PVEUHI <i>default</i> = 102.9
PVEXEULO PV Extended Lo Range in Eus	_ _ _ _ _ _ _ _ _ _ _ _ _	<i>range</i> = <=PVEULO <i>default</i> = -2.9
PV CLAMP PV Clamping Option	NoClamp Clamp	No clamping of the PV value Clamp PV value at range extension limit

PVSRCOPT PV Source Option		Only Auto All	PV Source selection is not available Selection from PVSOURCE is available
PVSOURCE PV Source		Auto Man Sub	Field wiring /memory-fetch supplies PV PV is supplied by Operator or program A value is substituted by a CL program
TF PV Filter Lag Time in Minutes			<i>range</i> = 0.0 to 60.0 (minutes) <i>default</i> = 0.0 (PV will not be delayed)
PVTV PV Target Value in EUs			<i>range</i> = PVEXEULO-PVEXEUHI, & NaN <i>default</i> = NaN
OVERVAL Overview Display Value			<i>range</i> = 0 to 100 <i>default</i> = 25

Req PV Middle Of 3 Algo Display

PVEQN(MidOf3)	EqA	Highest good input when 1 or 2 are bad
PV Equation Type	EqB	Lowest good input when 1 or 2 are bad
	EqC	Average of any and all good inputs

Reg PV Middle Of 3 Algo Display

PISRC (PV-Input-Connection Source) <u>Input#</u>	<u>PISRC</u> (point.parameter)	<u>Destination Para-</u> <u>-meter (PIDSTN)</u>
1		P1
2		P2
3		P3

APM Middle of 3 PV Algorithm

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Alarming Display – N/A if PNTFORM = Componnt
As you enter alarm trip points and alarm priorities in the next section please refer to the following alarm Priority definitions and the note.

NoAction	Alarm is not reported to the system and not annunciated (R670 & later)
Journal	Alarm is historized, not reported to US, not annunciated
Low	Alarm is historized, reported to US, and annunciated
High	Alarm is historized, reported to US, and annunciated
Emergency	Alarm is historized, reported to US, and annunciated Not
Printer	historized, reported to the printer, not annunciated Alarm
JnlPrint	historized, reported to the printer, not annunciated

Note: If you do not enter a trip point value for a given pair of trip-points and alarm-priorities, do not Circle an alarm priority option for that pair (that is, no trip point, no alarm priority).

Alarming Display - N/A if PNTFORM = Componnt

\$AUXUNIT Auxiliary Unit		<i>range = Any configured LCN Unit ID default = NaN</i>
\$ALDLOPT (R620 and later) Alarm Delay Option	None OffDlyLo OffDlyHi OffDlyEm OnDlyLo OnDlyHi OnDlyEm	Off Delay Lo Off Delay Hi Off Delay Emergency On Delay Lo On Delay Hi On Delay Emergency
\$DLBADAL (R681 and later) Delay for BADPV	None Offdelay	<i>range = Offdelay, None default = None</i>
DLTIME (R620 and later) Delay Time		<i>range = 0 to 9999 seconds default = 0 seconds</i>
\$REALARM Re-alarming Interval (R660 and later)		<i>range = 0 to 9999 minutes default = 0 minutes</i>
PVALDB PV Alarm Deadband In %	Half One Two Three Four Five EU	
PVALDBEU PV Alarm Deadband In EU		<i>range > = 0.0 default = 1.0</i>
PVHITP PV High Alarm Trip Point		<i>range = PVLOTP - PVHHTP, & NaN default = NaN</i>
PVHIPR PV High Alarm Priority	NoAction Journal Low (R670 and later)	High Emergency Printer JnlPrint
\$HISELCT (R681 and later) PV High Selective Cutout	Cutout Allow	<i>range = Cutout, Allow default = Cutout</i>
PVLOTP PV Low Alarm Trip Point		<i>range = PVLLTP - PVHITP, & NaN default = NaN</i>
PVLOPR PV Low Alarm Priority	NoAction Journal Low (R670 and later)	High Emergency Printer JnlPrint

Alarm Naming Display - N/A if PNTFORM = Component								
\$LOSELCT (R681 and later) PV Low Selective Cutout	Cutout Allow							<i>range</i> = Cutout, Allow <i>default</i> = Cutout
PVHHTP PV Hi Hi Alarm Trip Point	_ . _ . _ . _ .							<i>range</i> = PVHITP to PVEUHI, & NaN <i>default</i> = NaN
PVHHPR PV High High Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint	
\$HHSELCT (R681 and later) PV High High Selective Cutout	Cutout Allow							<i>range</i> = Cutout, Allow <i>default</i> = Cutout
PVLLTP PV Low Low Alarm Trip Point	_ . _ . _ . _ .							<i>range</i> = PVEULO - PVLOTP, & NaN <i>default</i> = NaN
PVLLPR PV Low Low Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint	
\$LLSELCT (R681 and later) PV Low Low Selective Cutout	Cutout Allow							<i>range</i> = Cutout, Allow <i>default</i> = Cutout
PVSGCHTP PV Significant Change Alarm Trip Point	_ . _ . _ . _ .							<i>range</i> = >=0.0, and NaN <i>default</i> = NaN
PVROCTPT PV Pos Rate Of Change Trip Point	_ . _ . _ . _ .							<i>range</i> = >=0.0, and NaN <i>default</i> = NaN
PVROCPR PV Pos Rate Of Chg Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint	
\$RPSELCT (R681 and later) PV Positive Rate of Change Selective Cutout	Cutout Allow							<i>range</i> = Cutout, Allow <i>default</i> = Cutout
PVROCNP PV Neg Rate Of Change Trip Point	_ . _ . _ . _ .							<i>range</i> = >=0.0, and NaN <i>default</i> = NaN
PVROCNP PV Neg Rate Of Chg Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint	
\$RNSELCT (R681 and later) PV Negative Rate of Change Selective Cutout	Cutout Allow							<i>range</i> = Cutout, Allow <i>default</i> = Cutout
BADPVPR Bad-PV Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint	
ALENBST Point Alarm Enable Status	Enable Disable Inhibit			<u>Displayed</u> Yes No	<u>Logged</u> Yes Yes No	<u>Reported To EIP</u> Yes Yes No		

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OVERVAL		range = 0 to 100
Overview Display Value	. . .	default = 25

APM Override Selector CTL Algorithm

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Mode Configuration Display

BADCTLOP Bad PV Mode Shed Option	No Shed Shed Hold Shed Low Shed High Shed Safe	See APM Parameter Reference Dictionary
RCASOPT Remote Cascade Option	None Ddc	Local cascade is the only valid cascade AM strategy stores to OP. No OP limits.
SHEDTIME Remote Cascade Shed Time		range = 0 to 1000 (seconds) default = 0(mode shed disabled)
SHEDMODE Shed Mode	Man BCas	Supervisory point will shed to Manual Supervisory point will shed to BackupCas
NMODE (RegCtl) RegCtl Point's Normal Mode	None Man Cas Bcas	No configured "normal" operating mode Manual is the "normal" operating mode Cascade is "normal" operating mode Backup Cas is "normal" operating mode
NMODATTR Normal Mode Attribute	None Operator Program	Parameter NMODE sets "normal" mode Operator sets "normal" operating mode Program sets "normal" operating mode
MODEPERM Mode Permissive	Permit NotPerm	Operator can change this point's mode Optr. cannot change this point's mode
\$MODESEL (R630 and later) Mode Selection List	Man Prog Cas Bcas	Man, Cas, Bcas mode and Prog mode attribute are allowed for this data point.
EXTSWOPT External Mode Switching	None Ems	No external mode switching is allowed Ext. source can change point's mode

Reg Control OrSel Algo Display

CTLEQN Control Equation Type	EqA EqB	Selects the highest input Selects the lowest input
OROPT Override Feedback Option	Off On	Hi Lo selector without override feedback Hi Lo selector with override feedback
OROFFSET Override Offset	Off On	No override offset is applied Offset of Gain times Error is applied
M # Of Initializable Control Algorithm Inputs		range = 2 to 4 default = 2
XEUHI X Input High Range in EUs		range = >= XEULO default = 100.0
XEULO X Input Low Range In EUs		range = <= XEUHI default = 0.0
NOCINPTS #Of Control Input Connections		range = 0 to 4 default = 0
CISRC (Control-Input-Connection Source)	CISRC (point.parameter)	Destination Parameter (CIDSTN)
Input#		
1		
2		
3		
4		

OPHILM Output High Limit In Percent	_. . . ._	<i>range</i> = OPLOLM – 106.9 & NaN <i>default</i> = 105.0
OPLOLM Output Low Limit In Percent	_. . . ._	<i>range</i> = -6.9 – OPHILM, & NaN <i>default</i> = -5.0
SAFEOP Safe Output Limit In Percent	_. . . ._	<i>range</i> = -6.9 to to 106.9 & NaN <i>default</i> = NaN
OPMCHLM Output Minimum Change In %	_. . . ._	<i>range</i> = >=0.0, and NaN <i>default</i> = 0.0
OPROCLM OP Rate Of Change Limit In %	_. . . ._	<i>range</i> = >=0.0, and NaN <i>default</i> = NaN
\$OPTOL Output Tolerance in %	_. . . ._	<i>range</i> = 0.0 to 106.9 <i>default</i> = 0.0
OUTIND (R620 and later) Reverse Output Indicator	Direct Reverse	4 - 20Ma = 0 – 100% 20 - 4mA = 0 – 100% Note: With R670 and later, when this parameter is set to Reverse, the values of the displayed output and the related output parameters are changed. Refer to the definition of the OUTIND parameter in the PM Family Parameter Dictionary, XPM09.

APM Override Selector CTL Algorithm

Unit:	Tagname:	Engineer:	Date:
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Alarming Display - N/A if PNTFORM = Component

As you enter alarm priorities in the next section please refer to the following.

NoAction	Alarm is not reported to the system and not annunciated (R670 & later)
Journal	Alarm is historized, not reported to US, not annunciated
Low	Alarm is historized, reported to US, and annunciated
High	Alarm is historized, reported to US, and annunciated
Emergency	Alarm is historized, reported to US, and annunciated
Printer	Not historized, reported to the printer, not annunciated
JnlPrint	Alarm historized, reported to the printer, not annunciated

\$AUXUNIT Auxiliary Unit		<i>range = Any configured LCN Unit ID default = NaN</i>
\$DLBADAL (R681 and later) Delay for BADCTL	None Offdelay	<i>range = Offdelay, None default = None</i>
DLTIME (R620 and later) Delay Time		<i>range = 0 to 9999 seconds default = 0 seconds</i>
\$REALARM (R660 and later) Re-alarming Interval		<i>range = 0 to 9999 minutes default = 0 minutes</i>
BADOCOPT BAD OP Conn Alarm Option	Off On	Bad Output Connection Alarm is not initially permitted Bad Output Connection Alarm is Initially permitted
BADOCPR BAD OP Conn Alarm Priority	NoAction Journal Low High Emergency JnlPrint Printer (R670 and later)	
OPALDB Output Alarm Deadband		<i>range = 0 - 25.0 but $\leq$ (OPHITP-OPLOTP)/2 default = 5.0 Output Units</i>
OPHITP Output High Alarm Trip Point		<i>range = OPLOTP TO OPHILM and NaN default = NaN</i>
OPHIPR Output High Alarm Priority	NoAction Journal Low High Emergency Printer JnlPrint (R670 and later)	
OPLOTP Output Low Alarm Trip Point		<i>range = OPLOLM TO OPHITP and NaN default = NaN</i>
OPLOPR Output Low Alarm Alarm Priority	NoAction Journal Low High Emergency Printer JnlPrint (R670 and later)	
BADCTLPR Bad-control Alarm Priority	NoAction Journal Low High Emergency JnlPrint Printer (R670 and later)	
ALENBST Point Alarm Enable Status	Enable Disable Inhibit	<u>Displayed</u> Yes No No <u>Logged</u> Yes Yes No <u>Reported To EIP</u> Yes Yes No

CTLALGID Control Algorithm Identifier	<i>Null</i> Pid	No affect Proportional, Integral and Derivative
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APM PID CTL Algorithm

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PV Configuration Display

PVEUHI PV High Range In Engr Units	_ _ _ _	<i>range =</i> >=PVEULO <i>default =</i> 100.0
PVEULO PV Low Range In Engr Units	_ _ _ _	<i>range =</i> <= PVEUHI <i>default =</i> 0.0
PVFORMAT PV Decimal Point Format	DO D1	D2 D3 D0=9999. D1=999.9 D2=99.99 D3 = 9.999

N/A if PNTFORM = Componnt.

PVSRCOPT PV Source Option	Only Auto All	PV Source selection is not available Selection from PVSOURCE is available
PVSOURCE PV Source	Auto Man Sub	Field-wiring/memory -fetch supplies PV PV is supplied by Operator or program A value is substituted by a CL program
OVERVAL Overview Display Value	_ _ _	<i>range =</i> 0 to 100 <i>default =</i> 25

Mode Configuration Display

BADCTLOP Bad PV Mode Shed Option	No_Shed Shed Hold Shed Low Shed High Shed Safe	See APM Parameter Reference Dictionary
RCASOPT Remote Cascade Option	None Spc Ddc DdcRsp Rsp	Local cascade is the only valid cascade AM strategy stores to SP within SP limits AM Strategy stores to OP. No OP limits. AM strategy does Ddc and Rsp functions AM strategy stores to SP when this point is in Auto and is being initialized by its secondary. SP limits are applied.
SHEDTIME Remote Cascade Shed Time	_ _ _ _	<i>range =</i> 0 to 1000 (seconds) <i>default =</i> 0 (mode shed disabled)
SHEDMODE (If Shedtime ≥ 1) Shed Mode	Man Auto BCas	Supervisory point will shed to Manual, Automatic, or Backup Cascade mode.
NMODE (RegCtl) RegCtl Point's Normal Mode	None Man Auto Cas Bcas	No configured "normal" operating mode Manual is the "normal" operating mode Automatic is "normal" operating mode Cascade is "normal" operating mode Backup Cascade is "normal" operating mode

N/A if NMODE = None

NMODATTR Normal Mode Attribute	None Operator Program	Parameter NMODE sets "normal" mode Operator sets "normal" operating mode Program sets "normal" operating mode
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APM PID CTL Algorithm

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Mode Configuration Display (continued)

MODEPERM Mode Permissive	Permit NotPerm	Operator can change this point's mode Optr. Cannot change this point's mode
\$MODESEL (R630 and later) Mode Selection List	Man Prog Auto Cas Bcas	Man, Auto, Cas and Bcas mode, and Prog mode attribute are allowed for this data point.
EXTSWOPT External Mode Switching	None Ems	No external mode switching is allowed Ext. source can change point's mode

Setpoint Display

SPHILM SP High Limit		range = SPLOLM – SPEUHI, and NaN default = NaN
SPLOLM SP Low Limit		range = SPEULO – SPHILM, and NaN default = NaN
SP Set Point In Engineering Units Of The PV		range = SPLOLM to SPHILM default = 0.0
SPOPT Set Point Option Asp is invalid if PNTFORM=Componnt.	None Tv Asp	No specialized options are available Target value processing Advisory deviation alarming
RBOPT Ratio Bias Option		No ratio/bias is used to calculate the SP Fixed ratio (R) and fixed bias(B) are used R is back calculated during initialization B is back calculated during initialization
RTHILM Ratio High Limit		range = RTLOLM – 100.0, & NaN default = 100.0
RTLOLM Ratio Low Limit		range = 0.0001 - RTHILM, & NaN default = 0.0001
BSHILM Bias High Limit		range = >=BSLOLM, & NaN default = 50.0
BSLOLM Bias Low Limit		range = <=BSHILM, & NaN default = -50.0
RATIO Ratio		range = RTLOLM to RTHILM default = 1.0
BIAS Bias Parameter		range = BSLOLM to BSHILM default = 0.0

APM PID CTL Algorithm

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Reg Control Pid Algo Display

PIDFORM PID Controller Form	Interact Ideal	(Proportional + Integral) x (Derivative) (Proportional + Integral) + (Derivative)
CTLEQN (Pid) Control Equation Type	EqA EqB EqC EqD	P, I & D act on Error P & I act on Error, D acts on PV I acts on Error, P & D act on PV Integral-only control
PVTRACK PV Tracking Option	NoTrack Track	SP will never be set equal to PV Man mode or init.causes SP to track PV
CTLACTN Control Action	Reverse Direct	As PV increases, Output decreases As PV increases, Output increases

Reg Control Pid Algo Display (continued)

N/A if CTLEQN = EqD.

GAINOPT Gain (K) Option	Ext NonLin Gap Lin	K = KLIN*KEXT (gain from external source) K = KLIN*((NLFM)+(NLGAIN*[PVP-SPP]/[100]) K = KLIN*KGAP if PV>(SP-GAPLO) or if PV<(SP+GAPHI), else K = KLIN. K = the value entered by the user	
K (Pid) Overall Gain In % Per %	_ _ _ _ _ _ _	range = 0.0 – 240.0 default = Not applicable	
KLIN Linear Gain Factor	{ _ _ _ _ _ _ _ }	range = 0.0 – 240.0 default = 1.0	
KGAP Gap Gain Factor	_ _ _ _	range = 0.0 to 1.0 default = 1.0	
GAPHI Gap High Limit	_ _ _ _ _ _ _	range = >=0.0 default = 0.0	
GAPLO Gap Low Limit	_ _ _ _ _ _ _	range = >=0.0 default = 0.0	
NLGAIN Non Linear Gain	_ _ _ _ _ _ _	range = 0.0 – 240.0 default = 0.0	
NLFM Non Linearity Form	NLFM = 0 NLFM = 1	K can go to zero K cannot go to < 1	
KEXT External Gain Modifier	_ _ _ _ _ _ _	range = 0.0 to 240.0 default = 1.0	
T1 Integral Time In Minutes	T1 _ _ _ _ _ _ _	range = 0.0 – 1440.0 default = 0.0 (minutes)	
T2 Derivative Time In Minutes	T2 _ _ _ _ _ _ _	range = 0.0 – 1440.0 default = 0.0 (minutes)	

APM PID CTL Algorithm

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NOCINPTS # Of Control Input Connections		range = 0 to 2 default = 1
CISRC (Control Input Connection Source)		
<u>Input #</u>	<u>CISRC(point.parameter)</u>	<u>Destination Parameter (CIDSTN)</u>
1		PVAUTO
2		SP

Output Configuration Display

NOCOPTS # Of Control Output Connections		range = 0 to 4 default = 1
CODSTN (Control-Output-Connection Destination)		
<u>Output #</u> (N/A if NOCOPTS = 0)	<u>CODSTN (point.parameter)</u>	
1		
2		
3		
4		

(N/A if NOCOPTS > 0)

CVEUHI Output Engineering Unit High Range		range = > = CVEULO default = 100.0
CVEULO Output Engineering Unit Low Range		range = > = CVEUHI default = 0.0

OPHILM Output High Limit In Percent		range = OPLOLM - 106.9 & NaN default = 105.0
OPLOLM Output Low Limit In Percent		range = -6.9 - OPHILM,& NaN default = -5.0
SAFEOP Safe Output Limit In Percent		range = -6.9 to 106.9, & NaN default = NaN
OPMCHLM Output Minimum Change In %		range = > = 0.0, and NaN default = 0.0
OPROCLM OP Rate Of Change Limit In %		range = > = 0.0, and NaN default = NaN
\$OPTOL Output Tolerance in %		range = -6.9 to 106.9, & NaN default = NaN
OUTIND (R620 and later) Reverse Output Indicator	Direct Reverse	4 - 20mA = 0 – 100% 20 – 4mA = 0 – 100% Note: With R670 and later, when this parameter is set to Reverse, the values of the displayed output and the related output parameters are changed. Refer to the definition of the OUTIND parameter in the PM Family Parameter Dictionary, XPM09.

APM PID CTL Algorithm

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Alarming Display - N/A if PNTFORM = Componnt

As you enter alarm trip points and alarm priorities in the next section please refer to the following alarm priority definitions and the note.

NoAction	Alarm is not reported to the system and not annunciated (R670 & later)
Journal	Alarm is historized, not reported to US, not annunciated
Low	Alarm is historized, reported to US, and annunciated
High	Alarm is historized, reported to US, and annunciated
Emergency	Alarm is historized, reported to US, and annunciated Not
Printer	historized, reported to the printer, not annunciated Alarm
JnlPrint	historized, reported to the printer, not annunciated

Note: If you do not enter a trip point value for a given pair of trip-points and alarm-priorities, do not circle an alarm priority option for that pair (i.e., no trip point, no alarm priority).

Alarming Display - N/A if PNTFORM = Componnt

\$AUXUNIT Auxiliary Unit	_ _ _	<i>range</i> = Any configured LCN Unit ID <i>default</i> = NaN
\$ALDLOPT (R620 and later) Alarm Delay Option	None OffDlyLo OffDlyHi OffDlyEm OnDlyLo OnDlyHi OnDlyEm	Off Delay Lo Off Delay Hi Off Delay Emergency On Delay Lo On Delay Hi On Delay Emergency
\$DLBADAL (R681 and later) Delay for BADPV	None Offdelay	<i>range</i> = Offdelay, None <i>default</i> = None
\$DLTIME (R620 and later) Delay Time	_ _ _ _	<i>range</i> = 0 to 9999 seconds <i>default</i> = 0 seconds
\$REALARM (R660 and later) Re-alarming Interval	_ _ _ _	<i>range</i> = 0 to 9999 minutes <i>default</i> = 0 minutes
BADOCOPT BAD OP Conn Alarm Option	Off On	Bad Output Connection Alarm is not initially permitted Bad Output Connection Alarm is Initially permitted
BADOCPR BAD OP Conn Alarm Priority	NoAction Journal Low High Emergency Printer JnlPrint (R670 and later)	
OPALDB Output Alarm Deadband	_ _ _ _	<i>range</i> = 0 – 25.0 but $\leq$ (OPHITP-OPLOTP)/2 <i>default</i> = 5.0 Output Units
OPHITP Output High Alarm Trip Point	_ _ _ _	<i>range</i> = OPLOTP TO OPHILM and NaN <i>default</i> = NaN
OPHIPR Output High Alarm Alarm Priority	NoAction Journal Low High Emergency Printer JnlPrint (R670 and later)	
OPLOTP Output Low Alarm Trip Point	_ _ _ _	<i>range</i> = OPLOLM TO OPHITP and NaN <i>default</i> = NaN
OPLOPR Output Low Alarm Alarm Priority	NoAction Journal Low High Emergency Printer JnlPrint (R670 and later)	

APM PID CTL Algorithm

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PVALDB PV Alarm Deadband In %	Half	One	Two	Three	Four	Five	EU
PVALDBEU PV Alarm Deadband In EU	_ _ _ _ _ _ _				range >= 0.0 default = 1.0		
PVHITP PV High Alarm Trip Point	_ _ _ _ _ _ _				range = PVLOTP - PVHHTP, & NaN default = NaN		
PVHIPR PV High Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint
\$HISELCT (R681 and later) PV High Selective Cutout	Cutout Allow			range = Cutout, Allow default = Cutout			
PVLOTP PV Low Alarm Trip Point	_ _ _ _ _ _ _				range = PVLLTP - PVHITP, & NaN default = NaN		
PVLOPR PV Low Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint
\$LOSELCT (R681 and later) PV Low Selective Cutout	Cutout Allow			range = Cutout, Allow default = Cutout			

Alarming Display – N/A if PNTFORM = Component

PVHHTP PV Hi Hi Alarm Trip Point	_ _ _ _ _ _ _				range = PVHITP to PVEUHI, & NaN default = NaN		
PVHHPR PV High High Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint
\$HHSELCT (R681 and later) PV High High Selective Cutout	Cutout Allow			range = Cutout, Allow default = Cutout			
PVLLTP(RegCtl) PV Low Low Alarm Trip Point	_ _ _ _ _ _ _				range = PVEULO – PVLOTP, & NaN default = NaN		
PVLLPR PV Low Low Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint
\$LLSELCT (R681 and later) PV Low Low Selective Cutout	Cutout Allow			range = Cutout, Allow default = Cutout			
PVSGCHTP PV Signif Chg Alarm Trip Point	_ _ _ _ _ _ _				range = >=0.0, and NaN default = NaN		
PVROCTP PV Pos Rate Of Chg Trip Point	_ _ _ _ _ _ _				range = >=0.0, and NaN default = NaN		
PVROCPPR PV Pos Rate Of Chg Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint

APM **PID** CTL Algorithm

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Alarming Display – N/A if PNTFORM = Component

\$RPSLECT (R681 and later) PV Positive Rate Of Change Selective Cutout	Cutout Allow							range = Cutout, Allow default = Cutout
PVROCNTP PV Neg Rate Of Chg Trip Point	_ _ _ _ _ _ _ _ _							range = >=0.0, and NaN default = NaN
PVROCNPBR PV Neg Rate Of Chg Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint	
\$RNSELECT (R681 and later) PV Negative Rate of Change Selective Cutout	Cutout Allow							range = Cutout, Allow default = Cutout
BADPVPR Bad-PV Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint	
DEVHITP Deviation High Alarm Trip Point	_ _ _ _ _ _ _ _ _							range = >=0.0, and NaN default = NaN
DEWHIPR Deviation High Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint	
\$DHSELECT (R681 and later) Deviation High Selective Cutout	Cutout Allow							range = Cutout, Allow default = Cutout
DEVLOTP Deviation High Alarm Trip Point	_ _ _ _ _ _ _ _ _							range = >=0.0, and NaN default = NaN
DEVLOPR Deviation Low Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint	
\$DLSELECT (R681 and later) Deviation Low Selective Cutout	Cutout Allow							range = Cutout, Allow default = Cutout
ADVDEVTP Advisory Dev. Alarm Trip Point (When SPOPT = ASP)	_ _ _ _ _ _ _ _ _							range = >=0.0, and NaN default = NaN
ADVDEVPR Advisory Deviation Alarm Priority	NoAction (R670 and later) (When SPOPT = ASP)	Journal	Low	High	Emergency	Printer	JnlPrint	
ALENBST Point Alarm Enable Status	Enable Disable Inhibit			Displayed	Logged	Reported To EIP		Yes No No

APM PID with External Reset Feedback CTL Algorithm

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Point Assignment Display

NAME Tag Name	_____	<i>range = Not applicable</i> <i>default = Blank</i>
NODETYP Node Type	PM APM HPM	A PM Point Is Being Configured An APM Point Is Being Configured An HPM Point Is Being Configured
PNTFORM Point Form	Full Componnt	Point is fully displayed and alarmed <i>Pt. is partially displayed but not alarmed</i>

PTDESC Point Descriptor	_____	<i>range = Not applicable</i> <i>default = Blank</i>
EUDESC Engineering Units Descriptor	_____	<i>range = Not applicable</i> <i>default = Blank</i>
KEYWORD Keyword Descriptor	_____	<i>range = Not applicable</i> <i>default = Blank</i>
ASSOCDSP Associated Display	_____	<i>range = Not applicable</i> <i>default = Blank</i>
\$CDETAIL (R620 and later) Point Custom Display	_____	<i>range = Not applicable</i> <i>default = Blank</i>
UNIT Process Unit Point Is Assigned To	_____	<i>range = Two Alphanumeric Characters</i> <i>default = Not applicable</i>
NTWKNUM NIM's UCN That This Point Is In	_____	<i>range = 1 to 20</i> <i>default = Not applicable</i>
NODENUM APMM's Address On The UCN	_____	<i>range = 1 to 64</i> <i>default = Not applicable</i>
MODNUM I/O Module Number	_____	<i>range = 0</i> <i>default = 0 (Control Processor Point)</i>
SLOTNUM Slot Number	_____	<i>range = 1-160 (But <=NCTLSLOT)</i> <i>default = Not applicable</i>

N/A if PNTFORM = Componnt.

PRIMMOD Primary Module ID	_____	<i>range = Not Applicable</i> <i>default = Null</i>
USERID User ID Reservation	_____	<i>Sixteen Character String</i> <i>default = dashes</i>

Control Algorithm Display

CTLALGID Control Algorithm Identifier	Null PidErfb	No affect PID with External Reset Feedback
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PV Configuration Display

PVEUHI PV High Range In Engr Units	_____	<i>range = >=PVEULO</i> <i>default = 100.0</i>
PVEULO PV Low Range In Engr Units	_____	<i>range = <= PVEUHI</i> <i>default = 0.0</i>
PVFORMAT PV Decimal Point Format	DO D2 D1 D3	D0=9999. D2=99.99 D1=999.9 D3 = 9.999

APM PID with External Reset Feedback CTL Algorithm

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PVSRCOPT PV Source Option	Only Auto All	PV Source selection is not available Selection from PVSOURCE is available
PVSOURCE PV Source	Auto Man Sub	Field-wiring /memory -fetch supplies PV PV is supplied by Operator or program A value is substituted by a CL program
OVERVAL Overview Display Value	. . .	range = 0 to 100 default = 25

Mode Configuration Display

BADCTLOP Bad PV Mode Shed Option	No Shed Shed Hold Shed Low Shed High Shed Safe	See APM Parameter Reference Dictionary
RCASOPT Remote Cascade Option	None Spc Ddc DdcRsp Rsp	Local cascade is the only valid cascade AM strategy stores to SP within SP limits AM Strategy stores to OP. No OP limits. AM strategy does Ddc and Rsp functions AM strategy stores to SP when this point is in Auto and is being initialized by its secondary. SP limits are applied.
SHEDTIME Remote Cascade Shed Time	. . .	range = 0 to 1000 (seconds) default = 0 (mode shed disabled)
SHEDMODE (If Shedtime ≥ 1) Shed Mode	Man Auto BCas	Supervisory point will shed to Manual, Automatic, or Backup Cascade mode.
NMODE (RegCtl) RegCtl Point's Normal Mode	None Man Auto Cas Bcas	No configured "normal" operating mode Manual is the "normal" operating mode Automatic is "normal" operating mode Cascade is "normal" operating mode Backup Cascade is "normal" operating mode

N/A if NMODE =None

NMODATTR Normal Mode Attribute	None Operator Program	Parameter NMODE sets "normal" mode Operator sets "normal" operating mode Program sets "normal" operating mode
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MODEPERM Mode Permissive	Permit NotPerm	Operator can change this point's mode Optr. cannot change this point's mode
\$MODESEL (R630 and later) Mode selection list	Man Prog Auto Cas Bcas	Man, Auto, Cas and Bcas mode, and Prog mode attribute are allowed for this data point.
EXTSWOPT External Mode Switching	None Ems	No external mode switching is allowed Ext. source can change point's mode

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SPHILM SP High Limit		<i>range</i> = SPLOLM – SPEUHI, and NaN <i>default</i> = NaN
SPLOLM SP Low Limit		<i>range</i> = SPEULO – SPHILM, and NaN <i>default</i> = NaN
\$SPTOL Set Point Tolerance		<i>range</i> = ≥ 0.0 <i>default</i> = 0.0
SP Set Point In Engineering Units Of The PV		<i>range</i> = SPLOLM to SPHILM <i>default</i> = 0.0
SPOPT Set Point Option Asp is invalid if PNTFORM=Componnt.	None Tv Asp	No specialized options are available Target value processing Advisory deviation alarming
RBOPT Ratio Bias Option	NoRatBi FixRatBi AutoRat AutoBi	No ratio/bias is used to calculate the SP Fixed ratio (R) and fixed bias(B) are used R is back calculated during initialization B is back calculated during initialization
RTHILM Ratio High Limit		<i>range</i> = RTLOLM – 100.0, & NaN <i>default</i> = 100.0
RTLOLM Ratio Low Limit		<i>range</i> = 0.0001 - RTHILM, & NaN <i>default</i> = 0.0001
BSHILM Bias High Limit		<i>range</i> = $\geq$ BSLOLM, & NaN <i>default</i> = 50.0
BSLOLM Bias Low Limit		<i>range</i> = $\leq$ BSHILM, & NaN <i>default</i> = -50.0
RATIO Ratio		<i>range</i> = RTLOLM to RTHILM <i>default</i> = 1.0
BIAS Bias Parameter		<i>range</i> = BSLOLM to BSHILM <i>default</i> = 0.0

PIDFORM PID Controller Form	Interact Ideal	(Proportional + Integral) x (Derivative) (Proportional) + (Integral) + (Derivative)
CTLEQN(Pid) Control Equation Type	EqA EqB EqC EqD	P, I & D act on Error P & I act on Error, D acts on PV I acts on Error, P & D act on PV Integral-only control
PVTRACK PV Tracking Option	No Track Track	SP will never be set equal to PV Man mode or init. Causes SP to track PV
CTLACTN Control Action	Reverse Direct	As PV increases, Output decreases As PV increases, Output increases

APM PID with External Reset Feedback CTL Algorithm

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Reg Control PidErfb Algo Display (continued)

N/A if CTLEQN = EqD.

GAINOPT		Ext		K = K1*KEXT (gain from external source)	
Gain (K) Option	NonLin	Gap	K = K1*((NLFM)+(NLGAIN*[PVP-SPP]/[100])		
			K = KLIN*KGAP if PV>(SP-GAPLO) or if		
			PV<(SP+GAPHI), else K = KLIN		
	Lin		K = the value entered by the user		
K (Pid)	_ _ _ _		range = 0.0 – 240.0		
Overall Gain In % Per %			default = Not applicable		
KLIN	_ _ _ _		range = 0.0 – 240.0		
Linear Gain Factor			default = 1.0		
KGAP	_ _		range = 0.0 to 1.0		
Gap Gain Factor			default = 1.0		
GAPHI	_ _ _ _ _ _		range = >=0.0		
Gap High Limit			default = 0.0		
GAPLO	_ _ _ _ _ _		range = >=0.0		
Gap Low Limit			default = 0.0		
NLGAIN	_ _ _ _		range = 0.0 – 240.0		
Non Linear Gain			default = 0.0		
NLFM		NLFM = 0	K can go to zero		
Non Linearity Form		NLFM = 1	K cannot go to < 1		
KEXT	_ _ _ _		range = 0.0 to 240.0		
External Gain Modifier			default = 1.0		
T1	T1 _ _ _ _		range = 0.0 to 1440.0		
Integral Time In Minutes			default = 0.0 (minutes)		
T2	T2 _ _ _ _		range = 0.0 to 1440.0		
Derivative Time In Minutes			default = 0.0 (minutes)		
K1 (PidErfb)		_ _	range = 0.0 to 1.0		
External Reset Feedback Gain Factor			default = 1.0		

NOCINPTS		range = 3 to 4 default = 3
#Of Control Input Connections		
CISRC (Control-Input-Connection Source)		Destination Parameter (CIDSTN)
Input#	CISRC (point.parameter)	
1		PVAUTO
2		RFB
3		TRFB SP
4		

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NOCOPTS			<i>range = 0 to 4</i>
# Of Control Output Connections	_		<i>default = 1</i>
CODSTN (Control-Output-Connection Destination)			
<u>Output #</u>		<u>CODSTN (point.parameter)</u>	
1	_._._._._.		
2	_._._._._.		
3	_._._._._.		
4	_._._._._.		
CVEUHI	_._._._._.		<i>range = >=0.0</i>
CV High Limit In EUs			<i>default = 100.00</i>
CVEULO	_._._._._.		<i>range = >=0.0</i>
CV Low Limit In EUs			<i>default = 0.0</i>
OPHILM	_._._._.		<i>range = OPLOLM – 106.9 & NaN</i>
Output High Limit In Percent			<i>default = 105.0</i>
OPLOLM	_._._._.		<i>range = -6.9 – OPHILM, & NaN</i>
Output Low Limit In Percent			<i>default = -5.0</i>
SAFEOP	_._._._.		<i>range = -6.9 to to 106.9 & NaN</i>
Safe Output Limit In Percent			<i>default = NaN</i>
OPMCHLM	_._._._.		<i>range = >=0.0, and NaN</i>
Output Minimum Change In %			<i>default = 0.0</i>
OPROCLM	_._._._.		<i>range = >=0.0, and NaN</i>
OP Rate Of Change Limit In %			<i>default = NaN</i>
\$OPTOL	_._._._.		<i>range = 0.0 to 106.9</i>
Output Low Limit in Percent			<i>default = 105.0</i>
OUTIND (R620 and later)	Direct		4 - 20mA = 0 – 100%
Reverse Output Indicator	Reverse		20 – 4mA = 0 – 100%
			Note: With R670 and later, when this parameter is set to Reverse, the values of the displayed output and the related output parameters are changed. Refer to the definition of the OUTIND parameter in the PM Family Parameter Dictionary, XPM09.

APM PID with External Reset Feedback CTL Algorithm

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Alarming Display - N/A if PNTFORM = Componnt

As you enter alarm trip points and alarm priorities in the next section please refer to the following alarm priority definitions and the note.

NoAction	Alarm is not reported to the system and not annunciated (R670 & later)
Journal	Alarm is historized, not reported to US, not annunciated
Low	Alarm is historized, reported to US, and annunciated
High	Alarm is historized, reported to US, and annunciated
Emergency	Alarm is historized, reported to US, and annunciated Not
Printer	historized, reported to the printer, not annunciated Alarm
JnlPrint	historized, reported to the printer, not annunciated

Note: If you do not enter a trip point value for a given pair of trip-points and alarm-priorities, do not circle an alarm priority option for that pair (i.e., no trip point, no alarm priority).

Alarming Display - N/A if PNTFORM = Componnt

\$AUXUNIT			<i>range = Any configured LCN Unit ID</i>
Auxiliary Unit	_. _		<i>default = NaN</i>
\$ALDLOPT (R620 and later)		None	
Alarm Delay Option		OffDlyLo	Off Delay Lo
		OffDlyHi	Off Delay Hi
		OffDlyEm	Off Delay Emergency
		OnDlyLo	On Delay Lo
		OnDlyHi	On Delay Hi
		OnDlyEm	On Delay Emergency
\$DLBADAL (R681 and later)		None	<i>range = Offdelay, None</i>
Delay for BADPV		Offdelay	<i>default = None</i>
\$DLTIME (R620 and later)		_. _ . _ . _	<i>range = 0 to 9999 seconds</i>
Delay Time			<i>default = 0 seconds</i>
\$REALARM (R660 and later)		_. _ . _	<i>range = 0 to 9999 minutes</i>
Re-alarming Interval			<i>default = 0 minutes</i>
BADOCOPT		Off	Bad Output Connection Alarm is not initially permitted
BADOC Alarm Option		On	Bad Output Connection Alarm is Initially permitted
BADOCPR		NoAction	Journal Low High Emergency Printer JnlPrint
BADOC Alarm Priority		(R670 and later)	

OPALDB		_. _ . _ . _ . _	<i>range = 0 - 25.0 but $\leq$ (OPHITP-OPLOTP)/2</i>
Output Alarm Deadband			<i>default = 5.0 Output Units</i>
OPHITP		_. _ . _ . _ . _	<i>range = OPLOTP TO OPHILM and NaN</i>
Output High Alarm Trip Point			<i>default = NaN</i>
OPHIPR		NoAction	Journal Low High Emergency Printer JnlPrint
Output High Alarm Priority		(R670 and later)	
OPLOTP		_. _ . _ . _ . _	<i>range = OPLOLM TO OPHITP and NaN</i>
Output Low Alarm Trip Point			<i>default = NaN</i>
OPLOPR		NoAction	Journal Low High Emergency Printer JnlPrint
Output Low Alarm Priority		(R670 and later)	

APM PID with External Reset

Feedback CTL Algorithm

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Alarming Display – N/A if PNTFORM = Componnt

PVALDB PV Alarm Deadband In %	Half	One	Two	Three	Four	Five	EU
PVALDBEU PV Alarm Deadband In EU	_ _ _ _					range > = 0.0 default = 1.0	

PVHITP PV High Alarm Trip Point	_ _ _ _	range = PVLOTP - PVHHTP, & NaN default = NaN					
PVHIPR PV High Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint
\$HISELCT (R681 and later) PV High Selective Cutout	Cutout Allow	range = Cutout, Allow default = Cutout					
PVLOTP PV Low Alarm Trip Point	_ _ _ _	range = PVLLTP - PVHITP, & NaN default = NaN					
PVLOPR PV Low Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint
\$LOSELCT (R681 and later) PV Low Selective Cutout	Cutout Allow	range = Cutout, Allow default = Cutout					
PVHHTP PV Hi Hi Alarm Trip Point	_ _ _ _	range = PVHITP to PVEUHI, & NaN default = NaN					
PVHHPR PV High High Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint
\$HHSELCT (R681 and later) PV High High Selective Cutout	Cutout Allow	range = Cutout, Allow default = Cutout					
PVLLTP PV Low Low Alarm Trip Point	_ _ _ _	range = PVEULO – PVLOTP, & NaN default = NaN					
PVLLPR PV Low Low Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint
\$LLSELCT (R681 and later) PV Low Low Selective Cutout	Cutout Allow	range = Cutout, Allow default = Cutout					
PVSGCHTP PV Signif Chg Alarm Trip Point	_ _ _ _	range = >=0.0, and NaN default = NaN					
PVROCTP PV Pos Rate Of Chg Trip Point	_ _ _ _	range = >=0.0, and NaN default = NaN					
PVROCPPR PV Pos Rate Of Chg Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint
\$RPSELCT (R681 and later) PV Positive Rate of Change Selective Cutout	Cutout Allow	range = Cutout, Allow default = Cutout					

APM PID with External Reset Feedback CTL Algorithm

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PVROCNTP PV Neg Rate Of Chg Trip Point	_ . _ . _ . _ . _ . _ . _	<i>range = >=0.0, and NaN default = NaN</i>						
PVROCNPR PV Neg Rate Of Chg Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint	
\$RNSELCT (R681 and later) PV Negative Rate of Change Selective Cutout	Cutout Allow	<i>range = Cutout, Allow default = Cutout</i>						
BADPVPR Bad-PV Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint	
DEVHITP Deviation High Alarm Trip Point	_ . _ . _ . _ . _ . _ . _	<i>range = >=0.0, and NaN default = NaN</i>						
DEVHIPR Deviation High Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint	
\$DHSELCT (R681 and later) Deviation High Selective Cutout	Cutout Allow	<i>range = Cutout, Allow default = Cutout</i>						
DEVLOTP Deviation Low Alarm Trip Point	_ . _ . _ . _ . _ . _ . _	<i>range = >=0.0, and NaN default = NaN</i>						
DEVLOPR Deviation Low Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint	
\$DLSELCT (R681 and later) Deviation Low Selective Cutout	Cutout Allow	<i>range = Cutout, Allow default = Cutout</i>						
ADVDEVTP Advisory Deviation Alarm Trip Point (When SPOPT = ASP)	_ . _ . _ . _ . _ . _ . _	<i>range = >=0.0, and NaN default = NaN</i>						
ADVDEVPR Advisory Deviation Alarm Priority (When SPOPT = ASP)	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint	
ALENBST Point Alarm Enable Status		<u>Displayed</u>		<u>Logged</u>		<u>Reported To EIP</u>		
	Enable	Yes		Yes		Yes		
	Disable	No		Yes No		Yes		
	Inhibit	No				No		

APM PID with Feedforward CTL Algorithm

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Point Assignment Display

NAME Tag Name		<i>range</i> = Not applicable <i>default</i> = Blank
NODETYPE Node Type	PM APM HPM	A PM Point Is Being Configured An APM Point Is Being Configured An HPM Point Is Being Configured
PNTFORM Point Form	FuLL Componnt	Point is fully displayed and alarmed Pt. is partially displayed but not alarmed

PTDESC Point Descriptor		<i>range</i> = Not applicable <i>default</i> = Blank
EUDESC Engineering Units Descriptor		<i>range</i> = Not applicable <i>default</i> = Blank
KEYWORD Keyword Descriptor		<i>range</i> = Not applicable <i>default</i> = Blank
ASSOCDSP Associated Display		<i>range</i> = Not applicable <i>default</i> = Blank
\$CDETAIL (R620 and later) Point Custom Display		<i>range</i> = Not applicable <i>default</i> = Blank
UNIT Process Unit Point Is Assigned To		<i>range</i> = Two Alphanumeric Characters <i>default</i> = Not applicable
NTWKNUM NIM's UCN That This Point Is In		<i>range</i> = 1 to 20 <i>default</i> = Not applicable
NODENUM APMM's Address On The UCN		<i>range</i> = 1 to 64 <i>default</i> = Not applicable
MODNUM I/O Module Number		<i>range</i> = 0 <i>default</i> = 0 (Control Processor Point)
SLOTNUM Slot Number		<i>range</i> = 1-160 (but <= NCTLSLOT) <i>default</i> = Not applicable

N/A if PNTFORM = Componnt.

PRIMMOD Primary Module ID		<i>range</i> = Not Applicable <i>default</i> = Null
USERID User ID Reservation		Sixteen Character String <i>default</i> = dashes

Control Algorithm Display

CTLALGID Control Algorithm Identifier	Null PidFf	No affect PID with Feedforward
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PV Configuration Display

PVEUHI PV High Range In Engr Units	_ _ _ _	<i>range</i> = >= PVEULO <i>default</i> = 100.0
PVEULO PV Low Range In Engr Units	_ _ _ _	<i>range</i> = <= PVEUHI <i>default</i> = 0.0
PVFORMAT PV Decimal Point Format	D0 D2 D1 D3	D0 = 9999. D2 = 99.99 D1 = 999.9 D3 = 9.999
PVSRCOPT PV Source Option	OnlyAuto All	PV source selection is not available Selection from PVSOURCE is available
PVSOURCE PV Source	Auto Man Sub	Field-wiring/memory-fetch supplies PV PV is supplied by Operator or program A value is substituted by a CL program
OVERVAL Overview Display Value	_ _ _ _	<i>range</i> = 0 to 100 <i>default</i> = 25

Mode Configuration Display

BADCTLOP Bad PV Mode Shed Option	No_Shed Shed Hold Shed Low Shed High Shed Safe	See APM Parameter Reference Dictionary
RCASOPT Remote Cascade Option	None Spc Ddc DdcRsp Rsp	Local cascade is the only valid cascade AM strategy stores to SP within SP limits AM strategy stores to OP. No OP limits. AM strategy does Ddc and Rsp functions AM strategy stores to SP when this point is in Auto and is being initialized by its secondary. SP limits are applied.
SHEDTIME Remote Cascade Shed Time	_ _ _ _	<i>range</i> = 0 to 1000 (seconds) <i>default</i> = 0 (mode shed disabled)
SHEDMODE (If Shedtime ≥1) Shed Mode	Man Auto Bcas	Supervisory point will shed to Manual, Automatic, or Backup Cascade mode.
NMODE(RegCtl) RegCtl Point's Normal Mode	None Man Auto Cas Bcas	No configured "normal" operating mode Manual is the "normal" operating mode Automatic is "normal" operating mode Cascade is "normal" operating mode Backup Cascade is "normal" operating mode

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Unit:	Tagname:	Engineer:	Date:
N/A if NMODE = None			

NMODATTR Normal Mode Attribute	None Operator Program	Parameter NMODE sets "normal" mode Operator sets "normal" operating mode Program sets "normal" operating mode
MODEPERM Mode Permissive	Permit NotPerm	Operator can change this point's mode Optr. cannot change this point's mode
\$MODESEL (R630 and later) Mode Selection List	Man Prog Auto Cas Bcas	Man, Auto, Cas and Bcas mode, and Prog mode attribute are allowed for this data point.
EXTSWOPT External Mode Switching	None Ems	No external mode switching is allowed Ext. source can change point's mode

Setpoint Display

SPHILM SP High Limit		range = SPLOLM – SPEUHI, and NaN default = NaN
SPLOLM SP Low Limit		range = SPEULO – SPHILM, and NaN default = NaN
SP Set Point In Engineering Units Of The PV		range = SPLOLM to SPHILM default = 0.0
\$SPTOL Set Point Tolerance		range = >=0.0 default = 0.0
SPOPT Set Point Option Asp is invalid if PNTFORM=Componnt.	None Tv Asp	No specialized options are available Target value processing Advisory deviation alarming
RBOPT Ratio Bias Option	NoRatBi FixRatBi AutoRat AutoBi	No ratio/bias is used to calculate the SP Fixed ratio (R) and fixed bias (B) are used R is back calculated during initialization B is back calculated during initialization
RTHILM Ratio High Limit	{ }	range = RTLOLM – 100.0, & NaN default = 100.0
RTLOLM Ratio Low Limit	{ }	range = 0.0001 - RTHILM, & NaN default = 0.0001
BSHILM Bias High Limit	{ }	range = > = BSLOLM, & NaN default = 50.0
BSLOLM Bias Low Limit	{ }	range = < = BSHILM, & NaN default = 50.0
RATIO Ratio	{ }	range = RTLOLM to RTHILM default = 1.0
BIAS Bias Parameter	{ }	range = BSLOLM to BSHILM default = 0.0

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Reg Control PidFf Algo Display

PIDFORM PID Controller Form	Interact Ideal	(Proportional + Integral) x (Derivative) (Proportional) + (Integral) + (Derivative)
CTLEQN(Pid) Control Equation Type	EqA EqB EqC EqD	P, I & D act on Error P & I act on Error, D acts on PV I acts on Error, P & D act on PV Integral-only control
PVTRACK PV Tracking Option	No Track Track	SP will never be set equal to PV Man mode or init. causes SP to track PV
CTLACTN Control Action	Reverse Direct	As PV increases, Output decreases As PV increases, Output increases

Reg Control Pid Ff Algo Display (continued) N/A if CTLEQN = EqD.

GAINOPT Gain (K) Option	Ext NonLin Gap Lin	K = KFF*KEXT (gain from external source) K = KFF*((NLFM)+(NLGAIN*[PVP-SPP]/[100])) K = KLIN*KGAP if PV>(SP-GAPLO) or if PV<(SP+GAPHI), else K = KLIN. K = the value entered by the user	
K(Pid) Overall Gain In % Per %	_ _ _ _ _ _ _	range = 0.0 - 240.0 default = Not applicable	
KLIN Linear Gain Factor	_ _ _ _ _ _ _	range = 0.0 - 240.0 default = 1.0	
KGAP Gap Gain Factor	_ _ _ _	range = 0.0 to 1.0 default = 1.0	
GAPHI Gap High Limit	_ _ _ _ _ _ _ _ _ _ _	range = >=0.0 default = 0.0	
GAPLO Gap Low Limit	_ _ _ _ _ _ _ _ _ _ _	range = >=0.0 default = 0.0	
NLGAIN Non Linear Gain	_ _ _ _ _ _ _	range = 0.0 - 240.0 default = 0.0	
NLFM Non Linearity Form	0 1	K can go to zero K cannot go to < 1	
KEXT External Gain Modifier	_ _ _ _ _ _ _	range = 0.0 to 240.0 default = 1.0	
T1 Integral Time In Minutes	_ _ _ _ _ _ _	range = 0.0 to 1440.0 default = 0.0 (minutes)	
T2 Derivative Time In Minutes	_ _ _ _ _ _ _	range = 0.0 to 1440.0 default = 0.0 (minutes)	

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Reg Control PidFf Algo Display (continued)

FFOPT	Add	(Scaled Feedforward) + (Feedback)
Feed Forward Type	Multiply	(Feedback)x(Scaled, Biased Feedfwd)
BFF		range = >=0.0
Feed Forward Input Bias	_.	default = 0.0
KFF		range = >=0.0 to <=1.0
Gain For Feed Forward Input	_. . .	default = 1.0
NOCINPTS		range = 0 to 3
#Of Control Input Connections	_	default = 2
CISRC (Control-Input-Connection Source)		<u>Destination Parameter (CIDSTN)</u>
<u>Input#</u>	<u>CISRC (point.parameter)</u>	
1	_.	PVAUTO
2	_.	FF
3	_.	SP

Output Configuration Display

NOCOPTS range = 0 to 4
 #Of Control Output Connections default = 1

CODSTN (Control-Output-Connection Destination)

<u>Output #</u>	(N/A if NOCOPTS = 0)	<u>CODSTN</u> (point.parameter)
1		
2		
3		
4		

(N/A if NOCOPTS > 0)

CVEUHI		<i>range</i> = >=CVEULO
Output Engineering Unit High Range	_ _ _ _	<i>default</i> = 100.0
CVEULO		<i>range</i> = >=CVEUHI
Output Engineering Unit Low Range	_ _ _ _	<i>default</i> = 0.0

OPHILM Output High Limit In Percent		<i>range</i> = OPLOLM – 106.9 & NaN <i>default</i> = 105.0
OPLOLM Output Low Limit In Percent		<i>range</i> = -6.9 – OPHILM,& NaN <i>default</i> = -5.0
SAFEOP Safe Output Limit In Percent		<i>range</i> = -6.9 to 106.9, & NaN <i>default</i> = NaN
OPMCHLM Output Minimum Change In %		<i>range</i> = >=0.0, and NaN <i>default</i> = 0.0
OPROCLM OP Rate Of Change Limit In %		<i>range</i> = >=0.0, and NaN <i>default</i> = 0.0

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\$OPTOL Output Tolerance in %	_ . _ . _ . _ . _ .	<i>range</i> = -6.9 to 106.9 , & NaN <i>default</i> = NaN
OUTIND (R620 and later) Reverse Output Indicator	Direct Reverse	4 - 20mA = 0 - 100% 20 - 4mA = 0 - 100% Note: With R670 and later, when this parameter is set to Reverse, the values of the displayed output and the related output parameters are changed. Refer to the definition of the OUTIND parameter in the PM Family Parameter Dictionary, XPM09.

<i>Alarming Display – N/A if PNTFORM = Componnt</i>	
As you enter alarm trip points and alarm priorities in the next section please refer to the following alarm Priority definitions and the note.	
NoAction	Alarm is not reported to the system and not annunciated (R670 & later)
Journal	Alarm is historized, not reported to US, not annunciated
Low	Alarm is historized, reported to US, and annunciated
High	Alarm is historized, reported to US, and annunciated
Emergency	Alarm is historized, reported to US, and annunciated Not
Printer	historized, reported to the printer, not annunciated Alarm
JnlPrint	historized, reported to the printer, not annunciated
Note: If you do not enter a trip point value for a given pair of trip-points and alarm-priorities, do not Circle an alarm priority option for that pair (that is, no trip point, no alarm priority).	

\$AUXUNIT Auxiliary Unit	_ . _ .	<i>range</i> = Any configured LCN Unit ID <i>default</i> = NaN
\$ALDLOPT (R620 and later) Alarm Delay Option	None OffDlyLo OffDlyHi OffDlyEm OnDlyLo OnDlyHi OnDlyEm	Off Delay Lo Off Delay Hi Off Delay Emergency On Delay Lo On Delay Hi On Delay Emergency
\$DLBADAL (R681 and later) Delay for BADPV	None Offdelay	<i>range</i> = Offdelay, None <i>default</i> = None
DLTIME (R620 and later) Delay Time	_ . _ . _ . _ .	<i>range</i> = 0 to 9999 seconds <i>default</i> = 0 seconds
\$REALARM (R660 and later) Re-alarming Interval	_ _ _ . _ .	<i>range</i> = 0 to 9999 minutes <i>default</i> = 0 minutes

Alarming Display – N/A if PNTFORM = Componnt.

BADOCPT BAD Op Conn Alarm Option	Off On	Bad Output Connection Alarm is not initially permitted Bad Output Connection Alarm is initially permitted					
BADOCPR BAD Op Conn Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint

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CTL Algorithm

Unit:	Tagname:	Engineer:	Date:
OPALDB Output Alarm Deadband	_ _ _ _ _ _ _	<i>range</i> = 0 - 25.0 but $\leq$ (OPHITP-OPLOTP)/2 <i>default</i> = 5.0 Output Units	
OPHITP Output High Alarm Trip Point	_ _ _ _ _ _ _	<i>range</i> = OPLOTP TO OPHILM and NaN <i>default</i> = NaN	
OPHIPR Output High Alarm Priority	NoAction (R670 and later)	Journal	Low High Emergency Printer JnlPrint
OPLOTP Output Low Alarm Trip Point	_ _ _ _ _ _ _	<i>range</i> = OPLOLM TO OPHITP and NaN <i>default</i> = NaN	
OPLOPR Output Low Alarm Priority	NoAction (R670 and later)	Journal	Low High Emergency Printer JnlPrint
PVALDB PV Alarm Deadband In %	Half One Two Three Four Five EU		
PVALDBEU PV Alarm Deadband in EU	_ _ _ _ _		
PVHITP PV High Alarm Trip Point	_ _ _ _ _ _ _	<i>range</i> = PVLOTP - PVHHTP, & NaN <i>default</i> = NaN	
PVHIPR PV High Alarm Priority	NoAction (R670 and later)	Journal	Low High Emergency Printer JnlPrint
\$HISELCT (R681 and later) PV High Selective Cutout	Cutout Allow	<i>range</i> = Cutout, Allow <i>default</i> = Cutout	
PVLOTP PV Low Alarm Trip Point	_ _ _ _ _ _ _	<i>range</i> = PVLLTP - PVHITP, & NaN <i>default</i> = NaN	
PVLOPR PV Low Alarm Priority	NoAction (R670 and later)	Journal	Low High Emergency Printer JnlPrint
\$LOSELCT (R681 and later) PV Low Selective Cutout	Cutout Allow	<i>range</i> = Cutout, Allow <i>default</i> = Cutout	
PVHHTP(RegCtl) PV Hi Hi Alarm Trip Point	_ _ _ _ _ _ _	<i>range</i> = PVHITP to PVEUHI, & NaN <i>default</i> = NaN	
PVHHPR PV High High Alarm Priority	NoAction (R670 and later)	Journal	Low High Emergency Printer JnlPrint
\$HHSELCT (R681 and later) PV High High Selective Cutout	Cutout Allow	<i>range</i> = Cutout, Allow <i>default</i> = Cutout	
PVLLTP(RegCtl) PV Low Low Alarm Trip Point	_ _ _ _ _ _ _	<i>range</i> = PVEULO – PVLOTP, & NaN <i>default</i> = NaN	

APM PID with Feedforward CTL Algorithm

Unit:	Tagname:	Engineer:	Date:
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Alarming Display – N/A if PNTFORM = Compnnt. (Continued)

PVLLPR								
PV Low Low Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint	
\$LLSELCT (R681 and later)	Cutout							
PV Low Low Selective Cutout	Allow				range = Cutout, Allow default = Cutout			
PVSGCHTP	_ . _ .							
PV Significant Change Alarm Trip Point					range = >=0.0, and NaN default = NaN			
PVROCPTP	_ .							
PV Pos Rate Of Change Trip Point					range = >=0.0, and NaN default = NaN			
PVROCPPR	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint	
PV Pos Rate Of Change Alarm Priority								
\$RPSELCT (R681 and later)	Cutout							
PV Positive Rate of Change Selective Cutout	Allow				range = Cutout, Allow default = Cutout			
PVROCNTP	_ .							
PV Neg Rate Of Change Trip Point					range = >=0.0, and NaN default = NaN			
PVROCNPR	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint	
PV Neg Rate Of Change Alarm Priority								
\$RNSELCT (R681 and later)	Cutout							
PV Negative Rate of Change Selective Cutout	Allow				range = Cutout, Allow default = Cutout			
BADPVPR	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint	
Bad-PV Alarm Priority								
DEVHITP	_ .							
Deviation High Alarm Trip Point					range = >=0.0, and NaN default = NaN			
DEVHIPR	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint	
Deviation High Alarm Priority								
\$DHSELCT (R681 and later)	Cutout							
Deviation High Selective Cutout	Allow				range = Cutout, Allow default = Cutout			
DEVLOTP	_ .							
Deviation Low Alarm Trip Point					range = >=0.0, and NaN default = NaN			

APM PID with Feedforward CTL Algorithm

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Alarming Display – N/A if PNTFORM = Compnnt. (Continued)

DEVLOPR Deviation Low Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint
\$DLSELCT (R681 and later) Deviation Low Selective Cutout	Cutout Allow				<i>range = Cutout, Allow default = Cutout</i>		
ADVDEVTP (When SPOPT = ASP) Advisory Deviation Alarm Trip Point	_.				<i>range = >=0.0, and NaN default = NaN</i>		
ADVDEVPR (When SPOPT = ASP) Advisory Deviation Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint
ALENBST Point Alarm Enable Status	Enable Disable Inhibit			<u>Displayed</u> Yes No No	<u>Logged</u> Yes Yes No	<u>Reported to EIP</u> Yes Yes No	

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Control Algorithm Display		
CTLALGID	NULL	No affect
Control Algorithm Identifier	PosProp	Position Proportional

APM Position Proportional CTL Algorithm

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PV Configuration Display

PVEUHI PV High Range In Engr Units	_ _ _ _	<i>range = > = PVEULO</i> <i>default = 100.0</i>
PVEULO PV Low Range In Engr Units	_ _ _ _	<i>range = < = to PVEUHI</i> <i>default = 0.0</i>
PVFORMAT PV Decimal Point Format	D0 D2 D1 D3	D0 = 9999. D2 = 99.99 D1 = 999.9 D3 = 9.999

PVSRCOPT PV Source Option	OnlyAuto All	PV source selection is not available Selection from PVSOURCE is available
PVSOURCE PV Source	Auto Man Sub	Field-wiring/memory-fetch supplies PV PV is supplied by Operator or program A value is substituted by a CL program
OVERVAL Overview Display Value	_ _ _	<i>range = 0 to 100</i> <i>default = 25</i>

Mode Configuration Display

BADCLTOP Bad PV Mode Shed Option	No_Shed Shed Hold Shed Low Shed High Shed Safe	<i>range = Not applicable</i> <i>default = No_Shed</i>
RCASOPT Remote Cascade Option	None Spc	Local cascade is the only valid cascade AM strategy stores to SP within SP limits
SHEDTIME Remote Cascade Shed Time	_ _ _	<i>range = 0 to 1000 (seconds)</i> <i>default = 0 (mode shed disabled)</i>
SHEDMODE (If Shedtime $\geq$ 1) Shed Mode	Man Auto Bcas	Supervisory point will shed to Manual, Automatic or Backup Cascade mode.
NMODE (RegCtl) RegCtl Point's Normal Mode	None Man Auto Cas Bcas	No configured "normal" operating mode Manual is the "normal" operating mode Automatic is "normal" operating mode Cascade is "normal" operating mode Backup Cascade is "normal" operating mode

APM Position Proportional CTL Algorithm

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N/A if NMODE = None

NMODATTR Normal Mode Attribute	None Operator Program	Parameter NMODE sets "normal" mode Operator sets "normal" operating mode Program sets "normal" operating mode
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MODEPERM Mode Permissive	Permit NotPerm	Operator can change this point's mode Optr. cannot change this point's mode
\$MODESEL (R630 and later) Mode Selection List	Man Prog Auto Cas Bcas	Man, Auto, Cas and Bcas mode, and Prog mode attribute are allowed for this data point.
EXTSWOPT External Mode Switching	None Ems	No external mode switching is allowed Ext. source can change point's mode

Setpoint Display

SPHILM SP High Limit	_ _ _ _ _ _ _	<i>range</i> = SPLOLM - SPEUHI, and NaN <i>default</i> = NaN
SPLOLM SP Low Limit	_ _ _ _ _ _ _	<i>range</i> = SPEULO -SPHILM, and NaN <i>default</i> = NaN
\$SPTOL Set Point Tolerance	_ _ _ _ _ _	<i>range</i> = >=0.0 <i>default</i> = 0.0
SP Set Point In Engineering Units Of The PV	_ _ _ _ _ _ _	<i>range</i> = SPLOLM to SPHILM <i>default</i> = 0.0
SPOPT Set Point Option	None Tv Asp	No specialized options are available Target value processing Advisory deviation alarming

Reg Control Posn/Prop Algo Display

DEADBAND(PosProp) Dead Band In % Of Full Scale	_ _ _ _ _	<i>range</i> = 0.0 to 100.0 <i>default</i> = 5.0%
CYCLETIM PosProp Output Cycle Time In Seconds	_ _ _ _ _ _	<i>range</i> = 0.25 to 1000.0 <i>default</i> = 10.0 (seconds)
K Gain Value	_ _ _ _	<i>range</i> = 0.0 to 10.0 <i>default</i> = 1.0
RAISRATE Raise OP Stroke Rate In %/Sec	_ _ _ _ _	<i>range</i> = >=0.0 <i>default</i> = 100.0

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Alarming Display - N/A if PNTFORM = Componnt

As you enter alarm trip points and alarm priorities in the next section please refer to the following alarm priority definitions and the note.

NoAction	Alarm is not reported to the system and not annunciated (R670 & later)
Journal	Alarm is historized, not reported to US, not annunciated
Low	Alarm is historized, reported to US, and annunciated
High	Alarm is historized, reported to US, and annunciated
Emergncy	Alarm is historized, reported to US, and annunciated
Printer	Not historized, reported to the printer, not annunciated
JnlPrint	Alarm historized, reported to the printer, not annunciated

Note: If you do not enter a trip point value for a given pair of trip-points and alarm-priorities, do not circle an alarm priority option for that pair (that is, no trip point, no alarm priority).

Alarming Display - N/A if PNTFORM = Componnt

\$AUXUNIT Auxiliary Unit	_ _	range = Any configured LCN Unit ID default = NaN
\$ALDLOPT (R620 and later) Alarm Delay Option	None OffDlyLo OffDlyHi OffDlyEm OnDlyLo OnDlyHi OnDlyEm	Off Delay Lo Off Delay Hi Off Delay Emergency On Delay Lo On Delay Hi On Delay Emergency
\$DLBADAL (R681 and later) Delay for BADPV	None Offdelay	range = Offdelay, None default = None
DLTIME (R620 and later) Delay Time	_ _ _	range = 0 to 9999 seconds default = 0 seconds
\$REALARM Re-arming Interval (R660 and later)	_ _ _	range = 0 to 9999 minutes default = 0 minutes
BADOCPT BAD Op Conn Alarm Option	Off On	Bad Output Connection Alarm is not initially permitted Bad Output Connection Alarm is initially permitted
BADOCPR BAD Op Conn Alarm Priority	NoAction Journal Low High Emergncy Printer JnlPrint (R670 and later)	
PVALDB PV Alarm Deadband In %	Half One Two Three Four Five EU	
PVALDBEU PV Alarm Deadband In EU	_ _ _	range > = 0.0 default = 1.0
PVHITP PV High Alarm Trip Point	_ _ _ _	range = PVLOTP - PVHHTP, & NaN default = NaN
PVHIPR PV High Alarm Priority	NoAction Journal Low High Emergncy Printer JnlPrint (R670 and later)	
\$HISELCT (R681 and later) PV High Selective Cutout	Cutout Allow	range = Cutout, Allow default = Cutout

Unit:	Tagname:	Engineer:	Date:
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PVLOTP PV Low Alarm Trip Point	PVLLTP PV Low Low Alarm Trip Point	PVHHTP PV Hi Hi Alarm Trip Point	PVHHPR PV High High Alarm Priority	\$LLOSELECT (R681 and later) PV Low Selective Cutout	\$HHSSELECT (R681 and later) PV High High Selective Cutout	PVSGCHTP PV Significant Change Alarm Trip Point	PVROCPTP PV Pos Rate Of Change Trip Point	PVROCPPR PV Pos Rate Of Change Alarm Priority	\$RPSSELECT (R681 and later) PV Positive Rate of Change Selective Cutout	PVROCNTP PV Neg Rate Of Change Trip Point	PVROCNP PV Neg Rate Of Change Alarm Priority	\$RNSSELECT (R681 and later) PV Negative Rate of Change Selective Cutout
NoAction (R670 and later)	NoAction (R670 and later)	NoAction (R670 and later)	NoAction (R670 and later)	Cutout Allow	Cutout Allow	NoAction (R670 and later)	NoAction (R670 and later)	NoAction (R670 and later)	Cutout Allow	NoAction (R670 and later)	NoAction (R670 and later)	Cutout Allow
Journal	Journal	Journal	Journal				Journal	Journal			Journal	
Low	Low	Low	Low				Low	Low			Low	
High	High	High	High				High	High			High	
Emergency	Emergency	Emergency	Emergency				Emergency	Emergency			Emergency	
Printer	Printer	Printer	Printer				Printer	Printer			Printer	
JnlPrint	JnlPrint	JnlPrint	JnlPrint				JnlPrint	JnlPrint			JnlPrint	
<i>range = PVLLTP - PVHHTP, & NaN</i> <i>default = NaN</i>	<i>range = PVEULO - PVLOTP, & NaN</i> <i>default = NaN</i>	<i>range = PVHHTP to PVEUHI, & NaN</i> <i>default = NaN</i>		<i>range = Cutout, Allow</i> <i>default = Cutout</i>	<i>range = Cutout, Allow</i> <i>default = Cutout</i>	<i>range = >=0.0, and NaN</i> <i>default = NaN</i>	<i>range = >=0.0, and NaN</i> <i>default = NaN</i>		<i>range = Cutout, Allow</i> <i>default = Cutout</i>	<i>range = >=0.0, and NaN</i> <i>default = NaN</i>		<i>range = Cutout, Allow</i> <i>default = Cutout</i>

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Alarming Display – N/A if PNTFORM = Componnt. (Continued)

BADPVPR	NoAction	Journal	Low	High	Emergency	Printer	JnlPrint
Bad-PV Alarm Priority	(R670 and later)						
DEVHITP	_ _ _ _ _ _ _ _ _						
Deviation High Alarm Trip Point					<i>range = >=0.0, and NaN</i>		<i>default = NaN</i>
DEVHIPR	NoAction	Journal	Low	High	Emergency	Printer	JnlPrint
Deviation High Alarm Priority	(R670 and later)						
\$DHSELCT (R681 and later)	Cutout						
Deviation High Selective Cutout	Allow				<i>range = Cutout, Allow</i>		<i>default = Cutout</i>
DEVLOTP	_ _ _ _ _ _ _ _ _						
Deviation Low Alarm Trip Point					<i>range = >=0.0, and NaN</i>		<i>default = NaN</i>
DEVLOPR	NoAction	Journal	Low	High	Emergency	Printer	JnlPrint
Deviation Low Alarm Priority	(R670 and later)						
\$DLSELCT (R681 and later)	Cutout						
Deviation Low Selective Cutout	Allow				<i>range = Cutout, Allow</i>		<i>default = Cutout</i>
ADVDEVTP (When SPOPT = ASP)	_ _ _ _ _ _ _ _ _						
Advisory Deviation Alarm Trip Point					<i>range = >=0.0, and NaN</i>		<i>default = NaN</i>
ADVDEVPR (When SPOPT = ASP)	NoAction	Journal	Low	High	Emergency	Printer	JnlPrint
Advisory Deviation Alarm Priority	(R670 and later)						
ALENBST							
Point Alarm Enable Status					<u>Displayed</u>	<u>Logged</u>	<u>Reported to EIP</u>
	Enable				Yes	Yes	Yes
	Disable				No	Yes No	Yes
	Inhibit				No		No

CTLALGID Control Algorithm Identifier	NULL RampSoak	No affect Ramp Soak Algorithm
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NOCINPTS range = 0 to 1 # Of Control Input Connections default = 1		
CISRC (Control-Input-Connection Source)		
<u>Input #</u>	<u>CISRC (point.parameter)</u>	<u>Destination Parameter (CIDSTN)</u>
1		PV

Note: If you do not enter a trip point value for a given pair of trip-points and alarm-priorities, do not circle an alarm priority option for that pair (i.e., no trip point, no alarm priority).

APM Ramp Soak CTL Algorithm

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Alarming Display – N/A if PNTFORM = Componnt (Continued)

\$AUXUNIT Auxiliary Unit	_ _	<i>range = Any configured LCN Unit ID default = NaN</i>					
\$ALDLOPT (R620 or later) Alarm Delay Option	None OffDlyLo OffDlyHi OffDlyEm OnDlyLo OnDlyHi OnDlyEm	Off Delay Lo Off Delay Hi Off Delay Emergency On Delay Lo On Delay Hi On Delay Emergency					
DLYTIME (R620 and later) Delay Time	_ _ _ _	<i>range = 0 to 9999 seconds default = 0 seconds</i>					
\$REALARM (R660 or later) Re-alarming Interval	_ _ _ _	<i>range = 0 to 9999 minutes default = 0 minutes</i>					
BADOCPT BAD Op Conn Alarm Option	Off On	Bad Output Connection Alarm is not initially permitted Bad Output Connection Alarm is initially permitted					
BADOCPR BAD Op Conn Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint
DEVVHITP Deviation High Alarm Trip Point	_ _ _ _ _ _ _	<i>range = >=0.0, or NaN default = NaN</i>					
DEVHIPR Deviation High Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint
\$DHSELCT (R681 and later) Deviation High Selective Cutout	Cutout Allow	<i>range = Cutout, Allow default = Cutout</i>					
DEVLOTP Deviation Low Alarm Trip Point	_ _ _ _ _ _ _	<i>range = >=0.0, or NaN default = NaN</i>					
DEVLOPR Deviation Low Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint
\$DLSELCT (R681 and later) Deviation Low Selective Cutout	Cutout Allow	<i>range = Cutout, Allow default = Cutout</i>					
ALENBST Point Alarm Enable Status	Enable Disable Inhibit	<u>Displayed</u> Yes No No		<u>Logged</u> Yes Yes No		<u>Reported to EIP</u> Yes Yes No	

APM Ramp Soak CTL Algorithm

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Reg Control Ramp/Soak Algo Display (continued from Page 2)

<u>Ramp Rates and Soak Times/Values</u>			
Pair Number	Ramp Rate (RATEX) <i>range = >0.0(EUs); default = NaN</i>	Soak Time (SOAKTX) <i>range = 0.0 – 120.0; default = 0.0</i>	Soak Value (SOAKVX) <i>range = > = 00 (EUs); default = NaN</i>
1	_ _ _ _ _ _ _ _ _	_ _ _ _	_ _ _ _ _ _ _ _ _
2	_ _ _ _ _ _ _ _ _	_ _ _ _	_ _ _ _ _ _ _ _ _
3	_ _ _ _ _ _ _ _ _	_ _ _ _	_ _ _ _ _ _ _ _ _
4	_ _ _ _ _ _ _ _ _	_ _ _ _	_ _ _ _ _ _ _ _ _
5	_ _ _ _ _ _ _ _ _	_ _ _ _	_ _ _ _ _ _ _ _ _
6	_ _ _ _ _ _ _ _ _	_ _ _ _	_ _ _ _ _ _ _ _ _
7	_ _ _ _ _ _ _ _ _	_ _ _ _	_ _ _ _ _ _ _ _ _
8	_ _ _ _ _ _ _ _ _	_ _ _ _	_ _ _ _ _ _ _ _ _
9	_ _ _ _ _ _ _ _ _	_ _ _ _	_ _ _ _ _ _ _ _ _
10	_ _ _ _ _ _ _ _ _	_ _ _ _	_ _ _ _ _ _ _ _ _
11	_ _ _ _ _ _ _ _ _	_ _ _ _	_ _ _ _ _ _ _ _ _
12	_ _ _ _ _ _ _ _ _	_ _ _ _	_ _ _ _ _ _ _ _ _

<u>Mark Functions</u>			
<u>Segment Number 1</u>		<u>Segment Number 2</u>	
Begin Time (S1BGNTM) <i>range = 0.0 – 120 (Mins) default = 0.0</i>	End Time (S1ENDTM) <i>range = 0.0 – 120 (Mins) default = 0.0</i>	Begin Time (S2BGNTM) <i>range = 0.0 – 120 (Mins) default = 0.0</i>	End Time (S2ENDTM) <i>range = 0.0 – 120 (Mins) default = 0.0</i>
_ _ _ _ _ _ _ _ _	_ _ _ _ _ _ _ _ _	_ _ _ _ _ _ _ _ _	_ _ _ _ _ _ _ _ _
<u>Segment ID (S1SEGID)</u>		<u>Segment ID (S2SEGID)</u>	
RAMP1	SOAK1	RAMP 1	SOAK1
RAMP2	SOAK2	RAMP2	SOAK2
RAMP3	SOAK3	RAMP3	SOAK3
RAMP4	SOAK4	RAMP4	SOAK4
RAMP5	SOAK5	RAMP5	SOAK5
RAMP6	SOAK6	RAMP6	SOAK6
RAMP7	SOAK7	RAMP7	SOAK7
RAMP8	SOAK8	RAMP8	SOAK8
RAMP9	SOAK9	RAMP9	SOAK9
RAMP10	SOAK10	RAMP10	SOAK10
RAMP11	SOAK11	RAMP11	SOAK11
RAMP12	SOAK12	RAMP12	SOAK12

NOTE: Return to page 2 of this form and continue configuration at parameter **NOCINPTS**.

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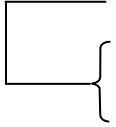
Unit:	Tagname:	Engineer:	Date:
<i>Point Assignment Display</i>			
NAME Tag Name	_ . _ . _ . _ . _ . _ . _ . _ .	range = Not applicable default = Blank	
NODETYP Node Type	PM APM HPM	A PM Point is being configured An APM Point is being configured An HPM Point is being configured	
PNTFORM Point Form	FULL Componnt	Point is fully displayed and alarmed Pt. is partially displayed but not alarmed	
PTDESC Point Descriptor	_ . _ . _ . _ . _ . _ . _ . _ .	range = Not applicable default = Blank	
EUDESC Engineering Units Descriptor	_ . _ . _ . _ .	range = Not applicable default = Blank	
KEYWORD Keyword Descriptor	_ . _ . _ . _ .	range = Not applicable default = Blank	
ASSOCDSP Associated Display	_ . _ . _ . _ .	range = Not applicable default = Blank	
\$CDETAIL (R620 and later) Point Custom Display	_ . _ . _ . _ .	range = Not applicable default = Blank	
UNIT Process Unit Point Is Assigned To	_ .	range = Two Alphanumeric Characters default = Not applicable	
NTWKNUM NIM's UCN That This Point Is In	_ .	range = 1 to 20 default = Not applicable	
NODENUM APMM's Address On The UCN	_ .	range = 1 to 64 default = Not applicable	
MODNUM I/O Module Number	_ .	range = 0 default = 0 (Control Processor Point)	
SLOTNUM Slot Number	_ . _ .	range = 1-160 (but <=NCTLSLOT) default = Not applicable	
N/A if PNTFORM = Componnt.			
PRIMMOD Primary Module ID	_ . _ . _ . _ . _ . _ .	range = Not Applicable default = Null	
USERID User ID Reservation	_ . _ . _ . _ . _ . _ .	range = Sixteen Character String default = Dashes	
<i>Control Algorithm Display</i>			
CTLALGID Control Algorithm Identifier	NULL RatioCtl	No affect Ratio Control	

APM Ratio Control CTL Algorithm

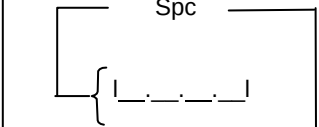
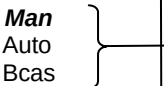
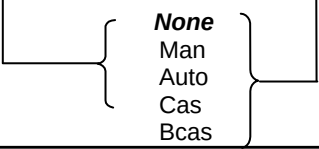
Unit:	Tagname:	Engineer:	Date:
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PV Configuration Display

PVEUHI PV High Range In Engr Units		<i>range = >=PVEULO</i> <i>default = 100.0</i>
PVEULO PV Low Range In Engr Units		<i>range = <=PVEUHI</i> <i>default = 0.0</i>
PVFORMAT PV Decimal Point Format	DO D1	D2 D3 D0=9999. D1=999.9 D2=99.99 D3 = 9.999

PVSRCOPT PV Source Option		Only Auto All PV Source selection is not available Selection from PVSOURCE is available
PVSOURCE PV Source		Auto Man Sub Field-wiring /memory-fetch supplies PV PV is supplied by Operator or program A value is substituted by a CL program
OVERVAL Overview Display Value		<i>range = 0 to 100</i> <i>default = 25</i>

Mode Configuration Display

RCASOPT Remote Cascade Option		None Spc Local cascade is the only valid cascade AM strategy stores to SP within SP limits
SHEDTIME Remote Cascade Shed Time		<i>range = 0 to 1000 (seconds)</i> <i>default = 0 (mode shed disabled)</i>
SHEDMODE Shed Mode		Man Auto Bcas Supervisory point will shed to Manual Supervisory point will shed to Automatic Supervisory point will shed to BackupCas
NMODE (RegCtl) RegCtl Point's Normal Mode		None Man Auto Cas Bcas No configured "normal" operating mode Manual is the "normal" operating mode Automatic is "normal" operating mode Cascade is "normal" operating mode Backup Cas is "normal" operating mode

N/A if NMODE =None

NMODATTR Normal Mode Attribute	None Operator Program	Parameter NMODE sets "normal" mode Operator sets "normal" operating mode Program sets "normal" operating mode
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MODEPERM Mode Permissive	Permit NotPerm	Operator can change this point's mode Optr. Cannot change this point's mode
\$MODESEL (R630 and later) Mode Selection List	Man Auto Cas Prog Bcas	Man, Auto, Cas and Bcas mode, and Prog mode attribute are allowed for this data point.
EXTSWOPT External Mode Switching	None Ems	No external mode switching is allowed Ext. source can change point's mode

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<i>Reg Control Ratio/Ctl Algo Display (Continued)</i>					
NOCINPTS					<i>range = 0 to 3</i>
#Of Control Input Connections		_			<i>default = 2</i>
CISRC (Control-Input-Connection Source)					<u>Destination Parameter (CIDSTN)</u>
<u>Input#</u>		<u>CISRC (point.parameter)</u>			
1	_.	.	.	.	PVAUTO
2	_.	.	.	.	X2
3	_.	.	.	.	SP

APM Ratio Control CTL Algorithm

Unit:	Tagname:	Engineer:	Date:
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Output Configuration Display

NOCOPTS range = 0 to 4
#Of Control Output Connections default = 1

CODSTN (Control-Output-Connection Destination)

Output# (N/A if NOCOPTS = 0)

CODSTN (point.parameter)

1	
2	
3	
4	

(N/A if NOCOPTS > 0)

CVEUHI Output Engineering Unit High Range		range = >=CVEULO default = 100.0
CVEULO Output Engineering Unit Low Range		range = >=CVEUHI default = 0.0
OPHILM Output High Limit In Percent		range = OPLOLM – 106.9 & NaN default = 105.0
OPLOLM Output Low Limit In Percent		range = -6.9 – OPHILM, & NaN default = -5.0
SAFEOP Safe Output Limit In Percent		range = -6.9 to 106.9, & NaN default = NaN
OPMCHLM Output Minimum Change In %		range = >=0.0, and NaN default = 0.0
OPROCLM OP Rate Of Change Limit In %		range = >=0.0, and NaN default = NaN
\$OPTOL Output Low Limit in %		range = 0.0 to 106.9 default = 105.0

OUTIND (R620 or later)
Reverse Output Indicator

Direct
Reverse

4 - 20mA = 0 – 100%
20 – 4mA = 0 – 100%

Note: With R670 and later, when this parameter is set to Reverse, the values of the displayed output and the related output parameters are changed. Refer to the definition of the OUTIND parameter in the PM Family Parameter Dictionary, XPM09.

APM Ratio Control CTL Algorithm

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Alarming Display – N/A if PNTFORM = Componnt

As you enter alarm trip points and alarm priorities in the next section please refer to the following alarm Priority definitions and the note.

NoAction	Alarm is not reported to the system and not annunciated (R670 & later)
Journal	Alarm is historized, not reported to US, not annunciated
Low	Alarm is historized, reported to US, and annunciated
High	Alarm is historized, reported to US, and annunciated
Emergency	Alarm is historized, reported to US, and annunciated
Printer	Not historized, reported to the printer, not annunciated
JnlPrint	Alarm historized, reported to the printer, not annunciated

Note: If you do not enter a trip point value for a given pair of trip-points and alarm-priorities, do not Circle an alarm priority option for that pair (i.e., no trip point, no alarm priority).

\$AUXUNIT Auxiliary Unit	_ _	<i>range = Any configured LCN Unit ID default = NaN</i>
\$ALDLOPT (R620 or later) Alarm Delay Option	None OffDlyLo OffDlyHi OffDlyEm OnDlyLo OnDlyHi OnDlyEm	Off Delay Lo Off Delay Hi Off Delay Emergency On Delay Lo On Delay Hi On Delay Emergency
\$DLBADAL (R681 and later) Delay for BADPV and BADCTL	None Offdelay	<i>range = Offdelay, None default = None</i>
DLTIME (R620 or later) Delay Time	_ _ _ _	<i>range = 0 to 9999 seconds default = 0 seconds</i>
\$REALARM (R660 or later) Re-alarming Interval	_ _ _ _	<i>range = 0 to 9999 minutes default = 0 minutes</i>
BADOCPT BAD Op Conn Alarm Option	Off On	Bad Output Connection Alarm is not initially permitted Bad Output Connection Alarm is initially permitted
BADOCPR BAD Op Conn Alarm Priority	NoAction (R670 and later) Journal Low High Emergency Printer JnlPrint	

OPALDB Output Alarm Deadband	_ _ _ _ _ _ _	<i>range = 0 – 25.0 but $\leq$ (OPHITP-OPLOTP)/2 default = 5.0 Output Units</i>
OPHITP Output High Alarm Trip Point	_ _ _ _ _ _ _	<i>range = OPLOTP TO OPHILM and NaN default = NaN</i>
OPHIPR Output High Alarm Priority	NoAction (R670 and later) Journal Low High Emergency Printer JnlPrint	
OPLOTP Output Low Alarm Trip Point	_ _ _ _ _ _ _	<i>range = OPLOLM TO OPHITP and NaN default = NaN</i>
OPLOPR Output Low Alarm Priority	NoAction (R670 and later) Journal Low High Emergency Printer JnlPrint	

APM Ratio Control CTL Algorithm

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Alarming Option – N/A if PNTFORM = Compnt.

PVALDB PV Alarm Deadband In %	Half One Two Three Four Five EU	
PVALDBEU PV Alarm Deadband in EU	_ _ _ _ _ _ _	range >= 0.0 default = 1.0
PVHITP PV High Alarm Trip Point	_ _ _ _ _ _ _	range = PVLOTP - PVHHTP, & NaN default = NaN
PVHIPR PV High Alarm Priority	NoAction Journal Low	High Emergency Printer JnlPrint
\$HISELCT (R681 and later) PV High Selective Cutout	Cutout Allow	range = Cutout, Allow default = Cutout
PVLOTP PV Low Alarm Trip Point	_ _ _ _ _ _ _	range = PVLLTP - PVHITP, & NaN default = NaN
PVLOPR PV Low Alarm Priority	NoAction Journal Low	High Emergency Printer JnlPrint
\$LOSELCT (R681 and later) PV Low Selective Cutout	Cutout Allow	range = Cutout, Allow default = Cutout
PVHHTP PV Hi Hi Alarm Trip Point	_ _ _ _ _ _ _	range = PVHITP to PVEUHI, & NaN default = NaN
PVHHPR PV High High Alarm Priority	NoAction Journal Low	High Emergency Printer JnlPrint
\$HHSELCT (R681 and later) PV High High Selective Cutout	Cutout Allow	range = Cutout, Allow default = Cutout
PVLLTP PV Low Low Alarm Trip Point	_ _ _ _ _ _ _	range = PVEULO – PVLOTP, & NaN default = NaN
PVLLPR PV Low Low Alarm Priority	NoAction Journal Low	High Emergency Printer JnlPrint
\$LLSELCT (R681 and later) PV Low Low Selective Cutout	Cutout Allow	range = Cutout, Allow default = Cutout

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Alarming Option – N/A if PNTFORM = Compnt.

PVSGCHTP PV Significant Change Alarm Trip Point	_ _ _ _ _ _ _ _ _									<i>range = >=0.0, and NaN default = NaN</i>
PVROCPTP PV Pos Rate Of Change Trip Point	_ _ _ _ _ _ _ _ _									<i>range = >=0.0, and NaN default = NaN</i>
PVROCPPR PV Pos Rate Of Change Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint			
\$RPNSELCT (R681 and later) PV Positive Rate of Change Selective Cutout	Cutout Allow									<i>range = Cutout, Allow default = Cutout</i>
PVROCNTP PV Neg Rate Of Change Trip Point	_ _ _ _ _ _ _ _ _									<i>range = >=0.0, and NaN default = NaN</i>
PVROCNPR PV Neg Rate Of Change Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint			
\$RNSELCT (R681 and later) PV Negative Rate of Change Selective Cutout	Cutout Allow									<i>range = Cutout, Allow default = Cutout</i>
BADPVPR Bad-PV Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint			
DEVHITP Deviation High Alarm Trip Point	_ _ _ _ _ _ _ _ _									<i>range = >=0.0, and NaN default = NaN</i>
DEVHIPR Deviation High Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint			
\$DHSELCT (R681 and later) Deviation High Selective Cutout	Cutout Allow									<i>range = Cutout, Allow default = Cutout</i>
DEVLOTP Deviation Low Alarm Trip Point	_ _ _ _ _ _ _ _ _									<i>range = >=0.0, and NaN default = NaN</i>
DEVLOPR Deviation Low Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint			
\$DLSELCT (R681 and later) Deviation Low Selective Cutout	Cutout Allow									<i>range = Cutout, Allow default = Cutout</i>
ADVDEVTP Advisory Deviation Alarm Trip Point (When SPOPT = ASP)	_ _ _ _ _ _ _ _ _									<i>range = >=0.0, and NaN default = NaN</i>

APM Ratio Control CTL Algorithm

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Alarming Option – N/A if PNTFORM = Compnnt.

ADVDEVPR Advisory Deviation Alarm Priority (When SPOPT = ASP)	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint
BADCTLPR Bad-control Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint
ALENBST Point Alarm Enable Status	Enable Disable Inhibit		<u>Displayed</u> Yes No No		<u>Logged</u> Yes Yes No		<u>Reported to EIP</u> Yes Yes No

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PV Algorithm Display		
PVALGID PV Algorithm Identifier	Null Summer	No affect Summer

APM Summer PV Algorithm

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PV Configuration Display

PVEUHI PV High Range In Engr Units	_ _ _ _ _ _ _	<i>range</i> = PVEULO to PVEXEUHI <i>default</i> = 100.0
PVEULO PV Low Range In Engr Units	_ _ _ _ _ _ _	<i>range</i> = PVEXEULO to PVEUHI <i>default</i> = 0.0
PVFORMAT PV Decimal Point Format	DO D2 D1 D3	D0 = 9999. D2 = 99.99 D1 = 999.9 D3 = 9.999
PVEXEUHI PV Extended Hi Range in EUs	_ _ _ _ _ _ _ _ _ _ _	<i>range</i> = >=PVEUHI <i>default</i> = 102.9
PVEXEULO PV Extended Lo Range in EUs	_ _ _ _ _ _ _ _ _ _ _	<i>range</i> = <=PVEULO <i>default</i> = -2.9
PV CLAMP PV Clamping Option	NoClamp Clamp	No clamping of the PV value Clamp PV value at range extension limit

PV Algorithm Display (Continued)

N/A if PNTFORM = Componnt.

PVSRCOPT PV Source Option		Only Auto All	PV Source selection is not available Selection from PVSOURCE is available
PVSOURCE PV Source		Auto Man Sub	Field wiring/ memory fetch supplies PV PV is supplied by Operator or program A value is substituted by a CL program

TF PV Filter Lag Time in Minutes	_ _ _ _ _	<i>range</i> = 0.0 to 60.0 (minutes) <i>default</i> = 0.0 (PV will not be delayed)
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N/A if PNTFORM = Componnt

PVTV PV Target Value in EUs	_ _ _ _ _ _ _ _ _ _ _	<i>range</i> = PVEXEULO-PVEXEUHI, & NaN <i>default</i> = NaN
OVERVAL Overview Display Value	_ _ _ _	<i>range</i> = 0 to 100 <i>default</i> = 25

APM Summer PV Algorithm

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Reg PV Summer Algo Display

PVEQN PV Equation Type	EqA EqB	P1 input is scaled and biased Up to 6 inputs are scaled and summed
C Scaling Constant	_ _ _ _ _ _ _	range = Not Applicable default = 1.0
D Overall Bias	_ _ _ _ _ _ _	range = Not Applicable default = 0.0
N (EqB only) # Inputs To Summer Algo		range = 2 to 6 default = 2
C1 (If $N \geq 2$) Scaling Constant for Input P1	_ _ _ _ _ _ _	range = Not Applicable default = 1.0
C2 (If $N \geq 2$) Scaling Constant for Input P2	_ _ _ _ _ _ _	range = Not Applicable default = 1.0
C3 (If $N \geq 3$) Scaling Constant for Input P3	_ _ _ _ _ _ _	range = Not Applicable default = 1.0
C4 (If $N \geq 4$) Scaling Constant for Input P4	_ _ _ _ _ _ _	range = Not Applicable default = 1.0
C5 (If $N \geq 5$) Scaling Constant for Input P5	_ _ _ _ _ _ _	range = Not Applicable default = 1.0
C6 (If $N = 6$) Scaling Constant for Input P6	_ _ _ _ _ _ _	range = Not Applicable default = 1.0
PISRC (PV-Input-Connection Source)	PISRC (point.parameter)	Destination Para -meter (PIDSTN)
Input#		
1	_ _ _ _ _ _ _	P1
2	_ _ _ _ _ _ _	P2
3	_ _ _ _ _ _ _	P3
4	_ _ _ _ _ _ _	P4
5	_ _ _ _ _ _ _	P5
6	_ _ _ _ _ _ _	P6

APM Summer PV Algorithm

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Alarming Display – N/A if PNTFORM = Component

As you enter alarm trip points and alarm priorities in the next section please refer to the following alarm Priority definitions and the note.

NoAction	Alarm is not reported to the system and not annunciated (R670 & later)
Journal	Alarm is historized, not reported to US, not annunciated
Low	Alarm is historized, reported to US, and annunciated
High	Alarm is historized, reported to US, and annunciated
Emergency	Alarm is historized, reported to US, and annunciated Not
Printer	historized, reported to the printer, not annunciated Alarm
JnlPrint	historized, reported to the printer, not annunciated

Note: If you do not enter a trip point value for a given pair of trip-points and alarm-priorities, do not Circle an alarm priority option for that pair (i.e., no trip point, no alarm priority).

\$AUXUNIT Auxiliary Unit	_. _	<i>range</i> = Any configured LCN Unit ID <i>default</i> = NaN
\$ALDLOPT (R620 and later) Alarm Delay Option	None OffDlyLo Off Delay Lo OffDlyHi Off Delay Hi OffDlyEm Off Delay Emergency OnDlyLo On Delay Lo OnDlyHi On Delay Hi OnDlyEm On Delay Emergency	
\$DLBADAL (R681 and later) Delay for BADPV	None Offdelay	<i>range</i> = Offdelay, None <i>default</i> = None
\$DLTIME (R620 and later) Delay Time	_. _ . _ . _	<i>range</i> = 0 to 9999 seconds <i>default</i> = 0 seconds
\$REALARM (R660 and later) Re-alarming Interval	_. _ . _ . _	<i>range</i> = 0 to 9999 minutes <i>default</i> = 0 minutes
PVALDB PV Alarm Deadband In %	Half One Two Three Four Five EU	
PVALDBEU PV Alarm Deadband In EU	_. _ . _ . _	<i>range</i> > = 0.0 } <i>default</i> = 1.0 }
PVHITP PV High Alarm Trip Point	_. _ . _ . _ . _ . _	<i>range</i> = PVLOTP - PVHHTP, & NaN <i>default</i> = NaN
PVHIPR PV High Alarm Priority	NoAction Journal Low High Emergency Printer JnlPrint (R670 and later)	
\$HISELCT (R681 and later) PV High Selective Cutout	Cutout Allow	<i>range</i> = Cutout, Allow <i>default</i> = Cutout
PVLOTP PV Low Alarm Trip Point	_. _ . _ . _ . _ . _	<i>range</i> = PVLLTP - PVHITP, & NaN <i>default</i> = NaN

APM Summer PV Algorithm

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Alarming Display - N/A if PNTFORM = Componnt (continued)

PVLOPR PV Low Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint
\$LOSELCT (R681 and later) PV Low Selective Cutout	Cutout Allow						<i>range = Cutout, Allow default = Cutout</i>
PVHHTP PV Hi Hi Alarm Trip Point							<i>range = PVHITP to PVEUHI, & NaN default = NaN</i>
PVHHPR PV High High Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint
\$HHSELCT (R681 and later) PV High High Selective Cutout	Cutout Allow						<i>range = Cutout, Allow default = Cutout</i>
PVLLTP PV Low Low Alarm Trip Point							<i>range = PVEULO - PVLOTP, & NaN default = NaN</i>
PVLLPR PV Low Low Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint
\$LLSELCT (R681 and later) PV Low Low Selective Cutout	Cutout Allow						<i>range = Cutout, Allow default = Cutout</i>
PVSGCHTP PV Significant Change Alarm Trip Point							<i>range = >=0.0, and NaN default = NaN</i>
PVROCTP PV Pos Rate Of Change Trip Point							<i>range = >=0.0, and NaN default = NaN</i>
PVROCPPR PV Pos Rate Of Chg Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint
\$RPSELCT (R681 and later) PV Positive Rate of Change Selective Cutout	Cutout Allow						<i>range = Cutout, Allow default = Cutout</i>
PVROCNTP PV Neg Rate Of Change Trip Point							<i>range = >=0.0, and NaN default = NaN</i>
PVROCNPR PV Neg Rate Of Chg Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint
\$RNSELCT (R681 and later) PV Negative Rate of Change Selective Cutout	Cutout Allow						<i>range = Cutout, Allow default = Cutout</i>
BADPVPR Bad-PV Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint
ALENBST Point Alarm Enable Status	Enable Disable Inhibit			<u>Displayed</u> Yes No No	<u>Logged</u> Yes Yes No	<u>Reported To EIP</u> Yes Yes No	

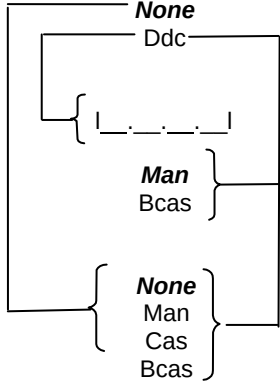
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OVERVAL	_. .	range = 0 to 100
Overview Display Value		default = 25

APM Switch CTL Algorithm

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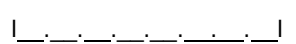
Mode Configuration Display

BADCTLOP Bad PV Mode Shed Option	No_Shed Shed Hold Shed Low Shed High Shed Safe	<i>range</i> = See APM Parameter Reference Dictionary <i>default</i> = No_Shed
RCASOPT Remote Cascade Option	None Ddc	Local cascade is the only valid cascade AM strategy stores to OP. No OP limits
SHEDTIME Remote Cascade Shed Time		<i>range</i> = 0 to 1000 (seconds) <i>default</i> = 0 (mode shed disabled)
SHEDMODE (If Shedtime ≥ 1) Shed Mode	Man Bcas	Supervisory point will shed to Manual, Backup Cascade mode. Supervisory point will shed to BackupCas
NMODE (RegCtl) RegCtl Point's Normal Mode	None Man Cas Bcas	No configured "normal" operating mode Manual is the "normal" operating mode Cascade is "normal" operating mode Backup Cas is "normal" operating mode

Mode Configuration Display (Continued) N/A if NMODE = None.

NMODATTR Normal Mode Attribute	None Operator Program	Parameter NMODE sets "normal" mode Operator sets "normal" operating mode Program sets "normal" operating mode
MODEPERM Mode Permissive	Permit NotPerm	Operator can change this point's mode Optr. cannot change this point's mode
\$MODESEL (R630 and later) Mode Selection List	Man Cas Prog Bcas	Man, Cas and Bcas mode, and Prog mode attribute are allowed for this data point.
EXTSWOPT External Mode Switching	None Ems	No external mode switching is allowed Ext. source can change point's mode

Reg Control Switch Algo Display

CTLEQN Control Equation Type	EqA EqB	Operator controls switch position Pgm. Or logic controls sw. position
TRACKING Selected Input Tracking	Off On	<i>range</i> = Off-On <i>default</i> = Off (Tracking disabled)
M # Of Initializable Control Algorithm Inputs		<i>range</i> = 2 to 4 <i>default</i> = 2
XEUHI X Input High Range in EUs		<i>range</i> = >=XEULO <i>default</i> = 100.0
XEULO X Input Low Range in EUs		<i>range</i> = <=XEUHI <i>default</i> = 0.0

N/A if CTLEQN = EqB, or if M < 2

SELXINP Selected X Input	SelectX1	SelectX2	SelectX3	SelectX4
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APM Switch CTL Algorithm

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NOCINPTS			range = 0 to 4 default = 0
#Of Control Input Connections			
CISRC (Control-Input-Connection Source)			<u>Destination Para-</u> <u>-meter (CIDSTN)</u>
<u>Input#</u>	<u>CISRC (point.parameter)</u>		
1			
2			
3			
4			

Output Configuration Display

NOCOPTS			range = 0 to 4 default = 1
# Of Control Output Connections			
CODSTN (Control-Output-Connection Destination)			
<u>Output #</u>	<u>CODSTN (point.parameter)</u>		
1			
2			
3			
4			
OPHILM			range = OPLOLM - 106.9 & NaN default = 105.0
Output High Limit In Percent			
OPLOLM			range = -6.9 - OPHILM, & NaN default = -5.0
Output Low Limit In Percent			
SAFEOP			range = -6.9 to 106.9 & NaN default = NaN
Safe Output Limit In Percent			
OPMCHLM			range = >=0.0, and NaN default = 0.0
Output Minimum Change In %			
OPROCLM			range = >=0.0, and NaN default = NaN
OP Rate Of Change Limit In %			
\$OPTOL			range = -6.9 to 106.9, & NaN default = NaN
Output Tolerance in %			
OUTIND (R620 and later)	Direct		4 - 20mA = 0 – 100%
Reverse Output Indicator	Reverse		20 - 4mA = 0 – 100%
Note: With R670 and later, when this parameter is set to Reverse, the values of the displayed output and the related output parameters are changed. Refer to the definition of the OUTIND parameter in the PM Family Parameter Dictionary, XPM09.			

APM Switch CTL Algorithm

Unit:	Tagname:	Engineer:	Date:
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Alarming Display - N/A if PNTFORM = Componnt

As you enter alarm trip points and alarm priorities in the next section please refer to the following alarm priority definitions and the note.

NoAction	Alarm is not reported to the system and not annunciated (R670 & later)
Journal	Alarm is historized, not reported to US, not annunciated
Low	Alarm is historized, reported to US, and annunciated
High	Alarm is historized, reported to US, and annunciated
Emergncy	Alarm is historized, reported to US, and annunciated Not
Printer	historized, reported to the printer, not annunciated Alarm
JnlPrint	historized, reported to the printer, not annunciated

Note: If you do not enter a trip point value for a given pair of trip-points and alarm-priorities, do not circle an alarm priority option for that pair (i.e., no trip point, no alarm priority).

Alarming Display - N/A if PNTFORM = Componnt

\$AUXUNIT Auxiliary Unit	_ _ _	<i>range = Any configured LCN Unit ID default = NaN</i>												
\$DLBADAL (R681 and later) Delay for BADCTL	None Offdelay	<i>range = Offdelay, None default = None</i>												
DLTIME (R620 and later) Delay Time	_ _ _ _	<i>range = 0 to 9999 seconds default = 0 seconds</i>												
\$REALARM Re-alarming Interval (R660 and later)	_ _ _ _	<i>range = 0 to 9999 minutes default = 0 minutes</i>												
BADOCOPT BAD Op Conn Alarm Option	Off On	Bad Output Connection Alarm is not initially permitted Bad Output Connection Alarm is Initially permitted												
BADOCPR BAD Op Conn Alarm Priority	NoAction Journal Low High Emergncy Printer JnlPrint (R670 and later)													
OPALDB Output Alarm Deadband	_ _ _ _ _ _	<i>range = 0 - 25.0 but ≤ (OPHITP-OPLOTP)/2 default = 5.0 Output Units</i>												
OPHITP Output High Alarm Trip Point	_ _ _ _ _ _	<i>range = OPLOTP to OPHILM and NaN default = NaN</i>												
OPHIPR Output High Alarm Priority	NoAction Journal Low High Emergncy Printer JnlPrint (R670 and later)													
OPLOTP Output Low Alarm Trip Point	_ _ _ _ _ _	<i>range = OPLOLM to OPHITP and NaN default = NaN</i>												
OPLOPR Output Low Alarm Priority	NoAction Journal Low High Emergncy Printer JnlPrint (R670 and later)													
BADCTLPR Bad-control Alarm Priority	NoAction Journal Low High Emergncy Printer JnlPrint (R670 and later)													
ALENBST Point Alarm Enable Status	Enable Disable Inhibit	<table> <tr> <td><u>Displayed</u></td> <td><u>Logged</u></td> <td><u>Reported To EIP</u></td> </tr> <tr> <td>Yes</td> <td>Yes</td> <td>Yes</td> </tr> <tr> <td>No</td> <td>Yes</td> <td>Yes</td> </tr> <tr> <td>No</td> <td>No</td> <td>No</td> </tr> </table>	<u>Displayed</u>	<u>Logged</u>	<u>Reported To EIP</u>	Yes	Yes	Yes	No	Yes	Yes	No	No	No
<u>Displayed</u>	<u>Logged</u>	<u>Reported To EIP</u>												
Yes	Yes	Yes												
No	Yes	Yes												
No	No	No												

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	N/A if PNTFORM = Componnt.
PRIMMOD Primary Module ID	_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ <i>range</i> = Not Applicable <i>default</i> = Null
USERID User ID Reservation	_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ Sixteen Character String <i>default</i> = dashes

APM Totalizer PV Algorithm

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PV Algorithm Display

PVALGID PV Algorithm Identifier	Null Totalizr	No affect Totalizer
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PV Configuration Display

PVEUHI PV High Range In Engr Units	_ _ _ _	<i>range</i> = PVEULO to PVEXEUHI <i>default</i> = 100.0
PVEULO PV Low Range In Engr Units	_ _ _ _	<i>range</i> = PVEXEULO to PVEUHI <i>default</i> = 0.0
PVFORMAT PV Decimal Point Format	D0 D2 D1 D3	D0 = 9999. D2 = 99.99 D1 = 999.9 D3 = 9.999
PVEXEUHI PV Extended Hi Range In EUs	_ _ _ _ _ _	<i>range</i> = >=PVEUHI <i>default</i> = 102.9
PVEXEULO PV Extended Lo Range In EUs	_ _ _ _ _ _	<i>range</i> = <=PVEULO <i>default</i> = -2.9
PVCLAMP PV Clamping Option	NoClamp Clamp	No clamping of the PV value Clamp PV value at range extension limit

N/A if PNTFORM = Compoint.

PVSRCOPT PV Source Option	OnlyAuto All	PV source selection is not available Selection from PVSOURCE is available
PVSOURCE PV Source	Auto Man Sub	Field-wiring/memory-fetch supplies PV PV is supplied by Operator or program A value is substituted by a CL program

TF PV Filter Lag Time In Minutes	_ _ _ _	<i>range</i> = 0.0 to 60.0 (minutes) <i>default</i> = 0.0 (PV will not be delayed)
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APM Totalizer PV Algorithm

Unit:	Tagname:	Engineer:	Date:
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Alarming Display - N/A if PNTFORM = Component

\$AUXUNIT Auxiliary Unit (R520 and later)	_ _	<i>range</i> = Any configured LCN Unit ID <i>default</i> = NaN
\$ALDLOPT (R620 and later) Alarm Delay Option	None OffDlyLo OffDlyHi OffDlyEm OnDlyLo OnDlyHi OnDlyEm	Off Delay Lo Off Delay Hi Off Delay Emergency On Delay Lo On Delay Hi On Delay Emergency
\$DLBADAL (R681 and later) Delay for BADPV	None Offdelay	<i>range</i> = Offdelay, None <i>default</i> = None
DLYTIME (R620 and later) Delay Time	_ _ _	<i>range</i> = 0 to 9999 seconds <i>default</i> = 0 seconds
\$REALARM (R660 and later) Re-alarming Interval	_ _ _	<i>range</i> = 0 to 9999 minutes <i>default</i> = 0 minutes
PVALDB PV Alarm Deadband In %	Half One Two Three Four Five EU	
PVALDBEU PV Alarm Deadband In EU	_ _ _	<i>range</i> > = 0.0 <i>default</i> = 1.0
PVHITP PV High Alarm Trip Point	_ _ _ _ _ _	<i>range</i> = PVLOTP - PVHHTP, & NaN <i>default</i> = NaN
PVHIPR PV High Alarm Priority	NoAction Journal Low High Emergency Printer JnlPrint (R670 and later)	
\$HISELCT (R681 and later) PV High Selective Cutout	Cutout Allow	<i>range</i> = Cutout, Allow <i>default</i> = Cutout
PVLOTP PV Low Alarm Trip Point	_ _ _ _ _ _	<i>range</i> = PVLLTP - PVHITP, & NaN <i>default</i> = NaN
PVLOPR PV Low Alarm Priority	NoAction Journal Low High Emergency Printer JnlPrint (R670 and later)	
\$LOSELCT (R681 and later) PV Low Selective Cutout	Cutout Allow	<i>range</i> = Cutout, Allow <i>default</i> = Cutout
PVHHTP(RegCtl) PV Hi Hi Alarm Trip Point	_ _ _ _ _ _	<i>range</i> = PVHITP to PVEUHI, & NaN <i>default</i> = NaN
PVHHPR PV High High Alarm Priority	NoAction Journal Low High Emergency Printer JnlPrint (R670 and later)	
\$HHSELCT (R681 and later) PV High High Selective Cutout	Cutout Allow	<i>range</i> = Cutout, Allow <i>default</i> = Cutout
PVLLTP(RegCtl) PV Low Low Alarm Trip Point	_ _ _ _ _ _	<i>range</i> = PVEULO - PVLOTP, & NaN <i>default</i> = NaN

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Alarming Display – N/A if PNTFORM = Compnnt. (Continued)

PVLLPR PV Low Low Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint
\$LLSELECT (R681 and later) PV Low Low Selective Cutout	Cutout Allow				<i>range</i> = Cutout, Allow <i>default</i> = Cutout		
PVSGCHTP PV Signif Change Alarm Trip Point	_____				<i>range</i> = >=0.0, and NaN <i>default</i> = NaN		
PVROCPTP PV Pos Rate Of Change Trip Point	_____				<i>range</i> = >=0.0, and NaN <i>default</i> = NaN		
PVROCPPR PV Pos Rate Of Chg Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint
\$RPSLECT (R681 and later) PV Positive Rate of Change Selective Cutout	Cutout Allow				<i>range</i> = Cutout, Allow <i>default</i> = Cutout		
PVROCNTP PV Neg Rate Of Change Trip Point	_____				<i>range</i> = >=0.0, and NaN <i>default</i> = NaN		
PVROCNP PV Neg Rate Of Chg Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint
\$RNSELECT (R681 and later) PV Negative Rate of Change Selective Cutout	Cutout Allow				<i>range</i> = Cutout, Allow <i>default</i> = Cutout		
BADPVPR Bad-PV Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint
ALENBST Point Alarm Enable Status	Enable Disable Inhibit			<u>Displayed</u> Yes No No	<u>Logged</u> Yes Yes No	<u>Reported To EIP</u> Yes Yes No	

APM Variable Dead Time with Lead Lag PV Algorithm

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Point Assignment Display

NAME Tag Name		<i>range</i> = Not applicable <i>default</i> = Blank
NODETYP Node Type	PM APM HPM	A PM Point Is Being Configured An APM Point Is Being Configured An HPM Point Is Being Configured
PNTFORM Point Form	FuLL Componnt	Point is fully displayed and alarmed Pt. is partially displayed but not alarmed

PTDESC Point Descriptor		<i>range</i> = Not applicable <i>default</i> = Blank
EUDESC Engineering Units Descriptor		<i>range</i> = Not applicable <i>default</i> = Blank
KEYWORD Keyword Descriptor		<i>range</i> = Not applicable <i>default</i> = Blank
ASSOCDSP (R510 and later) Associated Display		<i>range</i> = Not applicable <i>default</i> = Blank
\$CDETAIL (R620 and later) Point Custom Display		<i>range</i> = Not applicable <i>default</i> = Blank
UNIT Process Unit Point Is Assigned To		<i>range</i> = Two Alphanumeric Characters <i>default</i> = Not applicable
NTWKNUM NIM's UCN That This Point Is In		<i>range</i> = 1 to 20 <i>default</i> = Not applicable
NODENUM APMM's Address On The UCN		<i>range</i> = 1 to 64 <i>default</i> = Not applicable
MODNUM I/O Module Number		<i>range</i> = 0 <i>default</i> = 0 (Control Processor Point)
SLOTNUM Slot Number		<i>range</i> = 1-80 (but <= NPVSLOT) <i>default</i> = Not applicable

N/A if PNTFORM = Componnt.

PRIMMOD Primary Module ID		<i>range</i> = Not Applicable <i>default</i> = Null
USERID User ID Reservation		Sixteen Character String <i>default</i> = dashes

APM Variable Dead Time with Lead Lag PV Algorithm

Unit:	Tagname:	Engineer:	Date:
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PV Algorithm Display

PVALGID PV Algorithm Identifier	NULL VdtLdLag	No affect Variable Dead Time with Lead Lag
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PV Configuration Display

PVEUHI PV High Range In Engr Units	_ _ _ _	<i>range</i> = PVEULO to PVEXEUHI <i>default</i> = 100.0
PVEULO PV Low Range In Engr Units	_ _ _ _	<i>range</i> = PVEXEULO to PVEUHI <i>default</i> = 0.0
PVFORMAT PV Decimal Point Format	D0 D2 D1 D3	D0 = 9999. D2 = 99.99 D1 = 999.9 D3 = 9.999
PVEXEUHI PV Extended Hi Range In EUs	_ _ _ _ _ _ _	<i>range</i> = >=PVEUHI <i>default</i> = 102.9
PVEXEULO PV Extended Lo Range In EUs	_ _ _ _ _ _ _	<i>range</i> = <=PVEULO <i>default</i> = -2.9
PVCLAMP PV Clamping Option	NoClamp Clamp	No clamping of the PV value Clamp PV value at range extension limit

PV Configuration Display (Continued)

N/A if PNTFORM = Componnt.

PVSRCOPT PV Source Option	OnlyAuto All	PV source selection is not available Selection from PVSOURCE is available
PVSOURCE PV Source	Auto Man Sub	Field-wiring/memory-fetch supplies PV PV is supplied by Operator or program A value is substituted by a CL program

TF PV Filter Lag Time In Minutes	_ _ _ _	<i>range</i> = 0.0 to 60.0 (minutes) <i>default</i> = 0.0 (PV will not be delayed)
--	---------	---

N/A if PNTFORM = Componnt.

PVTV PV Target Value in EUs	_ _ _ _ _ _ _	<i>range</i> = PVEXEULO-PVEXEUHI, & NaN <i>default</i> = NaN
OVERVAL Overview Display Value	_ _ _ _	<i>range</i> = 0 to 100 <i>default</i> = 25

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[illegible]

APM Variable Dead Time with Lead Lag PV Algorithm

Unit:	Tagname:	Engineer:	Date:
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Alarming Display – N/A if PNTFORM = Component

As you enter alarm trip points and alarm priorities in the next section please refer to the following alarm Priority definitions.

NoAction	Alarm is not reported to the system and not annunciated (R670 & later)
Journal	Alarm is historized, not reported to US, not annunciated
Low	Alarm is historized, reported to US, and annunciated
High	Alarm is historized, reported to US, and annunciated
Emergency	Alarm is historized, reported to US, and annunciated Not
Printer	historized, reported to the printer, not annunciated Alarm
JnlPrint	historized, reported to the printer, not annunciated

\$AUXUNIT (R520 and later)

Auxiliary Unit

|_.|

range = Any configured LCN Unit ID

default = NaN

\$ALDLOPT (R620 and later)

Alarm Delay Option

None

OffDlyLo

Off Delay Lo

OffDlyHi

Off Delay Hi

OffDlyEm

Off Delay Emergency

OnDlyLo

On Delay Lo

OnDlyHi

On Delay Hi

OnDlyEm

On Delay Emergency

\$DLBADAL (R681 and later)

Delay for BADPV

None

Offdelay

range = Offdelay, None

default = None

DLTIME (R620 and later)

Delay Time

|_.|

range = 0 to 9999 seconds

default = 0 seconds

\$REALARM (R660 and later)

Re-alarming Interval

|_.|

range = 0 to 9999 minutes

default = 0 minutes

PVALDB

PV Alarm Deadband In %

Half

One

Two

Three

Four

Five

EU

PVALDBEU

PV Alarm Deadband in EU

|_.|

range >= 0.0

default = 1.0

PVHITP

PV High Alarm Trip Point

|_.|

range = PVLOTP - PVHHTP, & NaN

default = NaN

PVHIPR

PV High Alarm Priority

NoAction

(R670 and later)

Journal

Low

High

Emergency

Printer

JnlPrint

\$HISELCT (R681 and later)

PV High Selective Cutout

Cutout

Allow

range = Cutout, Allow

default = Cutout

PVLOTP

PV Low Alarm Trip Point

|_.|

range = PVLLTP - PVHITP, & NaN

default = NaN

PVLOPR

PV Low Alarm Priority

NoAction

(R670 and later)

Journal

Low

High

Emergency

Printer

JnlPrint

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[illegible]

STRLEN	8	<i>range = Not applicable</i>	
Length of Displayed Strings	16	<i>default = 8</i>	
	32		
	64		
SEQSLTSZ		<i>range = >=0</i>	
Sequence Slot Size	_._._._	<i>default = 0</i>	
CNTLLOCK	Operator	Engineer	Lowest access lock that
Control Lock	Supervis	Program	permits certain stores

SEQPR	NoAction	Alarm is not reported to the system and not annunciated (R670 & later)
Sequence Alarm Priority	Journal	Alarm is historized, not reported to US, not annunciated
	Low	Alarm is historized, reported to US, and annunciated
	High	Alarm is historized, reported to US, and annunciated
	Emergency	Alarm is historized, reported to US, and annunciated Not
	Printer	historized, reported to the printer, not annunciated Alarm
	JnlPrint	historized, reported to the printer, not annunciated

APM Flag Point

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Unit:	Tagname:	Engineer:	Date:
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NAME Tag Name		<i>range</i> = Not applicable <i>default</i> = Blank
NODETYPE Node Type		An APM point is being configured If a PM, HPM, LM, or SM point is being configured, go to the appropriate forms for those devices.
PNTFORM Point Form	<i>Full</i> Componnt	Point is fully displayed and alarmed Pt. Is partially displayed but not alarmed

[illegible]

N/A if PNTFORM = Component

PRIMMOD Primary Module ID		<i>range</i> = Not Applicable <i>default</i> = Null
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APM Flag Point

Unit:	Tagname:	Engineer:	Date:
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Operating Configuration Display

STATETXT(1) State 1 Descriptor		<i>range</i> = Not applicable <i>default</i> = ON
STATETXT(0) State 0 Descriptor		<i>range</i> = Not applicable <i>default</i> = OFF

PV Current PV State	_._._._._._.__	<i>range</i> = Statetxt(0) – Statetxt(1) <i>default</i> = OFF
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BOXCLR (1) State 1 Box Color	RED CYAN	GREEN YELLOW	WHITE BLUE	BLACK MAGENTA
BOXCLR (0) State 0 Box Color	RED CYAN	GREEN YELLOW	WHITE BLUE	BLACK MAGENTA

Alarming Display – N/A if PNTFORM = Compnnt or SLOTNUM > 128

[illegible]

APM Timer Point

Unit:	Tagname:	Engineer:	Date:
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1 of 2 Timer Points - Point Assignment Display

NAME Tag Name		<i>range</i> = Not applicable <i>default</i> = Blank
NODETYP Node Type	APM	An APM point is being configured If a PM, HPM, LM, or SM point is being configured, go to the appropriate forms for those devices.
PNTFORM Point Form	Full	Point is fully displayed and alarmed
PTDESC Point Descriptor		<i>range</i> = Not applicable <i>default</i> = Blank
EUDESC Eng. Unit Descriptor		<i>range</i> = 1 to 8 characters <i>default</i> = Blank
KEYWORD Keyword Descriptor		<i>range</i> = Not applicable <i>default</i> = Blank
ASSOCDSP Associated Display		<i>range</i> = Not applicable <i>default</i> = Blank
\$CDETAIL (R620 and later) Point Custom Display		<i>range</i> = Not applicable <i>default</i> = Blank
UNIT Process Unit Point Is Assigned To	_ . _	<i>range</i> = Two Alphanumeric Characters <i>default</i> = Not applicable
NTWKNUM NIM's UCN That This Point Is In	_ . _	<i>range</i> = 1 to 20 <i>default</i> = Not applicable
NODENUM APMM's Address On The UCN	_ . _	<i>range</i> = 1 to 64 <i>default</i> = Not applicable
SLOTNUM Slot Number Within APMM	_ . _	<i>range</i> = 1-64 <i>default</i> = Not applicable

PRIMMOD	N/A if PNTFORM = Component.
Primary Module ID	range = Not Applicable default = Null

Operating Configuration Display

Operating Configuration Display		
TIMEBASE	SECONDS	<i>range = Seconds/Minutes</i>
Timer Base Period	MINUTES	<i>default = Seconds</i>
SP		<i>range = 0 - 32000</i>
Set Time	_.:_.:_.:_.	<i>default = 0</i>

PV				_ _ _ _ _ _ _	<i>range</i> = Not Applicable
Numeric PV Value					<i>default</i> = 0.0
PVFORMAT		D0	D1		<i>range</i> = D0 - D3
PV Decimal Format		D2	D3		<i>default</i> = D1

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USERID User ID Reservation	_____	<i>range</i> = Sixteen Character String <i>default</i> = Dashes
EXTDATA External Data Option	None IO_FL IO_NN IO_STR	No data Mapped from SI device Flags are mapped from SI device Numerics are mapped from SI device Strings are mapped from SI device

APM Array Point

Unit:	Tagname:	Engineer:	Date:
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Configuration Display – N/A if EXTDATA = None

IOPNUM SI IOP Number	_ _	<i>range = 1 to 40</i> <i>default = blank</i>
FTANUM SI FTA Number	1 2	<i>range = 1 or 2</i> <i>default = 1</i>
DEVADDR Serial Link Device Address	_ _ _ _	<i>range = 0 to 99999</i> <i>default = 0</i>
SCANPRI SI Data FTA Scan Priority	Low High	<i>range = Low or High</i> <i>default = Low</i>

FTA-Driver Program Auxiliary Data

AUXDATA1 Auxiliary Data 1	_ _ _ _
AUXDATA2 Auxiliary Data 2	_ _ _ _
AUXDATA3 Auxiliary Data 3	_ _ _ _
AUXDATA4 Auxiliary Data 4	_ _ _ _

range = Not applicable
default = NaN

FTA-Driver Program Allen-Bradley Data

AB_DATA1 A – B PLC type	_ _ _ _
AB_DATA2 A – B PLC file #	_ _ _ _
AB_DATA3 A – B Data type	_ _ _ _
AB_DATA3 A – B Scan Period	_ _ _ _

range = Not applicable
default = NaN

APM Array Point

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Unit:	Tagname:	Engineer:	Date:
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Configuration Display (Continued)

FLSTIX Flag Array Start Index	_ _ _ _ _ _ _	<i>range =</i> 0 to 99999 when EXTDATA = IO_FL 0 to 16384 all other EXTDATA options <i>default =</i> 0
NFLAG Number of Flags	_ _ _ _ _ _ _	<i>range =</i> 0 to 512 when EXTDATA = IO_FL 0 to 1023 all other EXTDATA options <i>default =</i> 0
NNSTIX Numeric Array Start Index	_ _ _ _ _ _ _	<i>range =</i> 0 to 99999 when EXTDATA = IO_NN 0 to 16384 all other EXTDATA options <i>default =</i> 0
NNUMERIC Number of Numerics	_ _ _ _	<i>range =</i> 0 to 64 dep. on device, if EXTDATA = IO_NN 0 to 240 all other EXTDATA options <i>default =</i> 0
STRSTIX String Array Start Index	_ _ _ _ _ _ _	<i>range =</i> 0 to 99999 when EXTDATA = IO_STR 0 to NSTRING for all other EXTDATA options (16384 max.) <i>default =</i> 0
STRLEN String Length in Characters	8 16 32 64	<i>range =</i> Not applicable <i>default =</i> 8
NSTRING Number of Strings	_ _ _ _	<i>range =</i> 0 to 240 when EXTDATA <> IO_STR 0 to 64/STRLEN when EXTDATA = IO_STR <i>default =</i> 0
TIMESTIX Time Array Start Index	_ _ _ _ _ _ _	<i>range =</i> 0 to 4096 <i>default =</i> 0
NTIME Number of Times	_ _ _ _	<i>range =</i> 0 to 240 <i>default =</i> 0
SPLOCK Data Array Setpoint Lock	Operator Supervis Engineer Program	<i>range =</i> Not applicable <i>default =</i> Operator

APM Array Point

Unit:	Tagname:	Engineer:	Date:
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Configuration Display (Continued)

Descriptors for Flag, Numeric String, and Time Arrays

FLDESC Flag Descriptor	<i>range = Sixty-four Character String</i> <i>default = Blank</i>
_____ _____	
NDESC Numeric Descriptor	<i>range = Sixty-four Character String</i> <i>default = Spaces</i>
_____ _____	
STRDESC String Descriptor	<i>range = Sixty-four Character String</i> <i>default = Spaces</i>
_____ _____	
TIMEDESC Time Descriptor	<i>range = Sixty-four Character String</i> <i>default = Spaces</i>
_____ _____	

APM Device Control Point

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Point Assignment Display

NAME Tag Name		<i>range = Not applicable</i> <i>default = Blank</i>
NODETYPE Node Type	APM HPM	An APM Point is Being Configured An HPM Point Is Being Configured
PNTFORM Point Form	FuLL Componnt	Point is fully displayed and alarmed Pt.is partially displayed but not alarmed

PTDESC Point Descriptor		<i>range = Not applicable</i> <i>default = Blank</i>
EUDESC Engineering Units Descriptor		<i>range = Not applicable</i> <i>default = Blank</i>
KEYWORD Keyword Descriptor		<i>range = Not applicable</i> <i>default = Blank</i>
ASSOCDSP (R510 and later) Associated Display		<i>range = Not applicable</i> <i>default = Blank</i>
\$CDETAIL (R620 and later) Point Custom Display		<i>range = Not applicable</i> <i>default = Blank</i>
UNIT Process Unit Point Is Assigned To		<i>range = Two Alphanumeric Characters</i> <i>default = Not applicable</i>
NTWKNUM NIM's UCN That This Point Is In		<i>range = 1 to 20</i> <i>default = Not applicable</i>
NODENUM APMM's Address On The UCN		<i>range = 1 to 64</i> <i>default = Not applicable</i>
SLOTNUM Slot Number		<i>range = 1-160</i> <i>default = Not applicable</i>

PRIMMOD Primary Module ID		N/A if PNTFORM = Componnt. <i>range = Not Applicable</i> <i>default = Null</i>
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USERID User ID Reservation		N/A if PNTFORM = Componnt. <i>range = Sixteen Character String</i> <i>default = dashes</i>
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NOSTATES Number of Digital States	2 3	2 digital switch states can be configured 3 digital switch states can be configured
NODINPTS # Of Digital Input Connections	0 1 2	0 digin connections can be configured 1 digin connections can be configured 2 digin connections can be configured
NODOPTS # Of Digital Output Connections	0 1 2 3	0 digout connections can be configured 1 digout connections can be configured 2 digout connections can be configured 3 digout connections can be configured

Unit:	Tagname:	Engineer:	Date:
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NOLINPTS Number of Logic Inputs	__._	<i>range</i> = 0 to 12 <i>default</i> = 0
LIABADOPT Logic Bad Input Handling Option	Off On Hold	<i>range</i> = "Off" state is substituted for bad input <i>default</i> = "On" state is substituted for bad input last good value is substituted for bad input

Refer to the following explanations when selecting the Primary Gate Algorithm :

- [illegible]

PIALGID (n)
Primary Input Gate # ____ Algorithm

NULL **INVERT** **IN_SET**

GT GE LT LE EQ NE GT2 GE2 LT2 LE2 EQ2 NE2

PINN (n)
Constant Numeric

PISRC (n)
Second Input State

Null
L1 L2 L3
L4 L5 L6
L7 L8 L9
L10 L11 L12

default = 0.0 range = <=> NaN

range = L1 through L12 correspond to the Logic Inputs LISRC (1) – (12)
default = Null

PIDEADBD (n)	1.	<i>range</i> = ≥ 0
Deadband		<i>default</i> = 0.0

APM Device Control Point

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Refer to the following definitions when selecting the Secondary Input Gate Algorithm and destination :

Algorithm		Destination	
DLY	- Delay	SI0, I0, I1, I2	- Interlocks
ONDLY	- On delay	P0, P1, P2	- Permissive Interlocks
OFFDLY	- Off delay	OPCMD	- Output Command
PULSE	- Pulse	SOCMD0, 1, 2	- Output Status Command 0, 1, 2
MAXPULSE	- Max pulse	SG1, SG2	- Secondary Gates 1, 2
MINPULSE	- Min pulse	PG1, 2, 3, 4	- Primary Gates 1 - 4

Input Connections Display (Continued)

SIALGID (n) Secondary Input											
Gate # _____	Algorithm =	NULL	DLY	ONDLY	OFFDLY	PULSE	MAXPULSE	MINPULSE			
SIDLYTIM (n)						range = 0 to 8000 seconds default = 0					
Delay or Pulse Width											
SIDSTN (n)		SI0	I0	I1	OPCMD	SOCMD0	SOCMD1	SOCMD2			
Destination		I2	P0	P1	P2	SG1	SG2	PG1	PG2	PG3	PG4

In_Set Algorithm Configuration Display

Applies if PIALGID(n)=In_Set

NNINSET (1)	_____	NNINSET (6)	_____
NNINSET (2)	_____	NNINSET (7)	_____
NNINSET (3)	_____	NNINSET (8)	_____
NNINSET (4)	_____	NNINSET (9)	_____
NNINSET (5)	_____	NNINSET (10)	_____
Numerics 1-10		range = 0 to 32, 767 default = 0	

APM Device Control Point

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Primary Gate Configuration Display

NOPGATE

Number of Primary Gates

range = 0 to 4
default = 0

Use a separate section of this form for each Primary Gate (n). Circle the Algorithm type (for example, AND, OR, etc.) and enter/select the parameters that apply for the algorithm. Up to four copies of this form are needed to configure all four possible Primary Gates and Destinations.

Refer to the following definitions when selecting the Primary Gate Algorithm and destination:

Algorithm

PAND - AND, pulsed output
POR - OR, pulsed output
PNAND - NAND, pulsed output
PNOR - NOR, pulsed output
PXOR - XOR, pulsed output

Destination

SI0, I0, I1, I2 - Interlocks
P0, P1, P2 - Permissive Interlocks
OPCMD - Output Command
SOCMD0,1,2 - Output Status Command 0, 1, 2
SG1, SG2 - Secondary Gates 1, 2

PGALGID (n)

Primary Gate # Algorithm

AND OR NAND NOR XOR

PAND POR PNAND PNOR PXOR

PGPLSWTH (n)

Primary Gate Pulse Width

range = 0 to 8000 seconds
default = 0

PGDSTN (n)

Primary Gate #n Destination

SI0	I0	I1	OPCMD	SOCMD0	SOCMD1	SOCMD2
I2		P0	P1	P2	SG1	SG2

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SGALGID (n)	Algorithm	AND	OR	NAND	NOR	XOR	PAND	POR	PNAND	PNOR	PXOR
Secondary Gate # __											
SGPLSWTH (n)											
Secondary Gate Pulse Width							<i>range = 0 to 8000 seconds</i> <i>default = 0</i>				
SGDSTN (n)		SI0	I0	I1		OPCMD	SOCMD0	SOCMD1	SOCMD2		
Secondary Gate #n Destination		I2		P0		P1 P2					

Unit:	Tagname:	Engineer:	Date:
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[illegible]

Numerics 1-8 are used as local variables. Entry is a real number. Default = NaN

NN1	_._._._._._.__	NN2	_._._._._._.__
NN3	_._._._._._.__	NN4	_._._._._._.__
NN5	_._._._._._.__	NN6	_._._._._._.__
NN7	_._._._._._.__	NN8	_._._._._._.__

Flags 7-12 are used as local variables

FL7	<i>Off</i>	On	FL8	<i>Off</i>	On
FL9	<i>Off</i>	On	FL10	<i>Off</i>	On
FL11	<i>Off</i>	On	FL12	<i>Off</i>	On

APM Device Control Point

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State Configuration Display

PVTXTOPT PV Text Option	☐ Off ☐ On	N/A if PNTFORM = Componnt. range = Not applicable default = "Off"
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MOVPTXT Moving PV Text		N/A if PNTFORM = Componnt. range = Eight character string default = "MOVING"
BADPVTXT Bad PV Text		range = Eight character string default = "BAD"

STATETXT (1) State 1-Description Text		range = Not applicable default = "On"		
STATETXT (0) State 0-Description Text		default = Not applicable default = "Off"		
STATETXT (2) State 2-Description Text		default = Not applicable default = "State2"		
BOXCLR (1) State 1 Box Color	Red Cyan	Green (Upper box default) Yellow	White Blue	Black Magenta
BOXCLR (0) State 0 Box Color	Red Cyan	Green Yellow (Lower box default)	White Blue	Black Magenta
BOXCLR (2) State 2 Box Color	Red (Upper box default) Cyan	Green Yellow	White Blue	Black Magenta

MOMSTATE Momentary Output States	None Mom_0 Mom_1 Mom_2 Mom_1_2	No momentary output states State 0 is valid if (NOSTATES = 2) State 1 is valid if (NOSTATES = 2 or 3) State 2 is valid if (NOSTATES = 3) State 1 or 2 is valid if (NOSTATES = 3)
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LOGICSRC Logic Source		N/A if PNTFORM = Componnt Logic Point driving the interlock to this Device Control Point
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APM Device Control Point

Unit:	Tagname:	Engineer:	Date:
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PV / Input Configuration Display

DISRC (Digital-Input-Connection Source)		
Input #	DISRC (point.parameter)	Destination
1	_____	D1
2	_____	D2

Only if NODINPTS = 1

D1_1 D1_One PV State	PVState0 PVState1	Use & display contents of STATETXT0 Use & display contents of STATETXT1
D2D1_00 D2_D1 Zero_Zero PV State D2 = Off, D1 = Off	PVState1 PVState0 PVState2 MovPV BadPV	Use & display contents of STATETXT1 Use & display contents of STATETXT0 Use & display contents of STATETXT2* Use & display contents of MOVPTXT Use & display contents of BADPTXT
D2D1_01 D2D1 Zero_One PV State D2 = Off, D1 = On	PVState1 PVState0 PVState2 MovPV BadPV	Use & display contents of STATETXT1 Use & display contents of STATETXT0 Use & display contents of STATETXT2* Use & display contents of MOVPTXT Use & display contents of BADPTXT
D2D1_10 D2D1 One_Zero PV State D2 = On, D1 = Off	PVState1 PVState0 PVState2 MovPV BadPV	Use & display contents of STATETXT1 Use & display contents of STATETXT0 Use & display contents of STATETXT2* Use & display contents of MOVPTXT Use & display contents of BADPTXT
D2D1_11 D2D1 One_One PV State D2 = On, D1 = On	PVState1 PVState0 PVState2 MovPV BadPV	Use & display contents of STATETXT1 Use & display contents of STATETXT0 Use & display contents of STATETXT2* Use & display contents of MOVPTXT Use & display contents of BADPTXT

*Only applies if NOSTATES = 3

N/A if NODINPTS = 0

PVSRCOPT PV Source Option	Only Auto	PV Source selection is not available and Field wiring or memory fetch supplies PV. Selection from PVSOURCE is available & Operator/CL-program may supply PV.
	All	
PVSOURCE PV Source	Auto Man Sub Track	Field wiring /memory -fetch supplies PV PV is supplied by Operator or program A value is substituted by a CL program PV tracks the output

APM Device Control Point

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Unit:	Tagname:	Engineer:	Date:
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Output Configuration Display – N/A if NODOPTS = 0

		<u>NOSTATES</u>	<u>NODOPTS</u>
ST1_OP1	Off	2 - 3	1 - 3
State 1 - Output 1	On		
ST1_OP2	Off	2 - 3	2 - 3
State 1 - Output 2	On		
ST1_OP3	Off	2 - 3	3
State 1 - Output 3	On		
ST0_OP1	Off	1 - 3	1 - 3
State 0 - Output 1	On		
ST0_OP2	Off	1 - 3	2 - 3
State 0 - Output 2	On		
ST0_OP3	Off	1 - 3	3
State 0 - Output 3	On		
ST2_OP1	Off	3	1 - 3
State 2 - Output 1	On		
ST2_OP2	Off	3	2 - 3
State 2 - Output 2	On		
ST2_OP3	Off	3	3
State 2 - Output 3	On		
DODSTN (Digital Output Connection Destination)			
<u>Output #</u>	<u>DODSTN (point.parameter)</u>		
1			
2			
3			

APM Device Control Point

Unit:	Tagname:	Engineer:	Date:
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Output Configuration Display (Continued)

PULSEWTH		range = 0 0 to 60.0
Pulse Width For DevCtl Point Outputs	_._._._	default = 1.0

Applies if NOSTATES = 3 and MOMSTATE = None

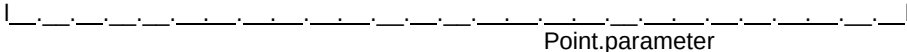
STCHGOPT State Change Option	None State 0		<i>range</i> = Not Applicable <i>default</i> = None
PAUSTIM State 0 Pause Time			<i>range</i> = 0 – 1000 seconds <i>default</i> = 0

N/A if PNTFORM = Componnt.

SEALOPT	None	<i>range</i> = Not Applicable
Seal-in Circuit Option	Sealin	<i>default</i> = None

LMREV Local Manual Reverse	Direct Reverse	<i>range</i> = Not Applicable <i>default</i> = Direct
LMSRC Local Manual Source		<i>range</i> = See Parameter Ref. Guide <i>default</i> = Null.null
Point.Parameter		

SV Configuration Display

SV Configuration Display		
SVSRC SV Input Connection		<i>range</i> = See Parameter Ref. Guide <i>default</i> = Null
SVDESC SV Descriptor		<i>range</i> = Eight Alphanumeric Characters <i>default</i> = Blank
SVEUDESC SV E.U. Descriptor		<i>range</i> = Eight Alphanumeric Characters <i>default</i> = Blank
SVEUHI SV EU Range High		<i>range</i> = >SVEULO <i>default</i> = NaN
SVEULO SV EU Range Low		<i>range</i> = <SVEUHI <i>default</i> = NaN
SVTV SV Target Value		<i>range</i> = SVEULO to SVEUHI <i>default</i> = NaN
\$REALARM (R660 and later) Re-arming Interval		<i>range</i> = 0 to 9999 minutes <i>default</i> = 0 minutes

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		<u>Maintenance Statistics</u>	
<u>Maximum Hours Allowed in Each State</u>			<u>Maximum Number of Transitions Allowed to State</u>
STATE 1 MAXTIM1H	_._._._._._.__	MAXTRAN1	_._._._._._.__
STATE 0 MAXTIM0H	_._._._._._.__	MAXTRAN0	_._._._._._.__
STATE 2 MAXTIM2H	_._._._._._.__	MAXTRAN2	_._._._._._.__
<i>range</i> = N/A <i>default</i> = 0		<i>range</i> = 0 (No Limit) <i>default</i> = 0.0	

APM Device Control Point

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Mode Configuration Display

N/A if NODOPTS = 0

NMODATTR Normal Mode Attribute	None Operator Program	Parameter NMODE sets "normal" mode Operator sets "normal" operating mode Program sets "normal" operating mode
MODEPERM Mode Permissive	Permit NotPerm	Operator can change this point's mode Optr. cannot change this point's mode
\$MODESEL (R630 and later) Mode Selection List	Man Prog	Man mode and Prog mode attribute are allowed for this data point.
OROPT Interlock Bypass Option	Off On	Optr. Cannot bypass this point's interlocks Operator can bypass this point's interlocks If BYPASS = ON
BYPASS Interlock Bypass State	Off On	

Alarming Display

N/A if PNTFORM = Componnt.

\$AUXUNIT Auxiliary Unit	_ _ _	<i>range</i> = Any configured LCN Unit ID <i>default</i> = NaN
PVNORMAL PV Value's Normal State Text	_ _ _ _ _ _ _ _ _	<i>range</i> = STATETXT(0) – STATETXT(2) <i>default</i> = Text equal to "None"
FBTIME Feedback Time	_ _ _ _	<i>range</i> = 0 to 1000 <i>default</i> = 0

N/A if FBTIME = 0

CMDFALTM Command Failure Timeout	_ _ _ _	<i>range</i> = 0 to 1000 seconds <i>default</i> = 0												
ALENBST Point Alarm Enable Status	Enable Disable Inhibit	<table> <tr> <td><u>Displayed</u></td><td><u>Logged</u></td><td><u>Reported to EIP</u></td></tr> <tr> <td>Yes</td><td>Yes</td><td>Yes</td></tr> <tr> <td>No</td><td>Yes</td><td>Yes</td></tr> <tr> <td>No</td><td>No</td><td>No</td></tr> </table>	<u>Displayed</u>	<u>Logged</u>	<u>Reported to EIP</u>	Yes	Yes	Yes	No	Yes	Yes	No	No	No
<u>Displayed</u>	<u>Logged</u>	<u>Reported to EIP</u>												
Yes	Yes	Yes												
No	Yes	Yes												
No	No	No												

N/A if PNTFORM = Componnt or if NODINPTS = 0

SIOALOPT Safety Override Alarm	None Auto_Rtn Cnfm_Rqd	No Override Alarming Return to normal when Override is Cleared Confirm to Clear after Interlock is Cleared SIOALPPR must not be set to NoAction or Journal.
OVRDALOP Override Alarm Option	None Auto_Rtn Cnfm_Rqd	No Override Alarming Return to normal when Override is Cleared Confirm to Clear after Interlock is Cleared OVRDALPR must not be set to NoAction or Journal.
BADPVPR Bad PV Alarm Priority	NoAction Journal Low High Emergency Printer JnlPrint (R670 and later)	

APM Device Control Point

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Alarming Display (Continued) N/A if FBTIME = 0.

CMDDISPR Command Disagree Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint
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N/A if SI0ALOPT = None

SI0ALPR Override SI0 Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint
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N/A if OVRDALOP = None.

OVRDALPR Override Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint
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EVTOPT Event Reporting Option	None Eip	No reporting Digital PV changes invoke Event- Initiated Processing
EIPPCODE Event Init Proc Destination	<i>range = Not applicable</i> <i>default = Null</i>	

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CTLALGID Control Algorithm Identifier	<i>Null</i> PID Pos Pr	No affect PID with Position Proportional
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APM PID with Position Proportional CTL Algorithm

Unit:	Tagname:	Engineer:	Date:
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PV Configuration Display

PVEUHI PV High Range In Engr Units	_ . _ . _ . _ .	range = PVEULO to PVEXEUHI default = 100.0	
PVEULO PV Low Range In Engr Units	_ . _ . _ . _ .	range = PVEXEULO to PVEUHI default = 0.0	
PVFORMAT PV Decimal Point Format	D0 D2 D1 D3	D0=9999 D2=99.99 D1=999.9 D3 = 9.999	

PV Configuration Display (Continued)

PVSRCOPT PV Source Option		PV Source selection is not available Selection from PVSOURCE is available
PVSOURCE PV Source		Field-wiring /memory -fetch supplies PV PV is supplied by Operator or program A value is substituted by a CL program

OVERAL Overview Display Value	_ . _ .	range = 0 to 100 default = 25
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Mode Configuration Display

BADCTLOP Bad PV Mode Shed Option	No_Shed Shed Hold Shed Low Shed High Shed Safe	range = See Parameter Reference Dictionary default = No_Shed
RCASOPT Remote Cascade Option	None Spc Rsp	Local cascade is the only valid cascade AM strategy stores to SP within SP limits. AM strategy stores to SP when this point is in Auto and is being initialized by its secondary. SP limits are applied.
SHEDTIME Remote Cascade Shed Time	_ . _ . _ . _ .	range = 0 to 1000 (seconds) default = 0 (mode shed disabled)
SHEDMODE (If Shedtime ≥ 1) Shed Mode	Man Auto Bcas	Supervisory point will shed to Manual, Automatic, or Backup Cascade mode.
NMODE RegCtl Point's Normal Mode	None Man Auto Cas Bcas	No configured "normal" operating mode Manual is the "normal" operating mode Automatic is "normal" operating mode Cascade is "normal" operating mode Backup Cascade is "normal" operating mode

N/A if NMODE =None

NMODATTR Normal Mode Attribute	None Operator Program	Parameter NMODE sets "normal" mode Operator sets "normal" operating mode Program sets "normal" operating mode
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Unit:	Tagname:	Engineer:	Date:
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MODEPERM Mode Permissive	<i>Permit</i> NotPerm	Operator can change this point's mode Optr. cannot change this point's mode
\$MODESEL (R630 and later) Mode Selection List	<i>Man</i> <i>Prog</i> <i>Auto</i> <i>Cas</i> <i>Bcas</i>	Man, Auto, Cas and Bcas mode, and Prog mode attribute are allowed for this data point.
EXTSWOPT External Mode Switching	<i>None</i> Ems	No external mode switching is allowed Ext. source can change point's mode

Setpoint Display

Setpoint Display		
SPHILM SP High Limit		<i>range</i> = SPLOLM – SPEUHI, and NaN <i>default</i> = NaN
SPLOLM SP Low Limit		<i>range</i> = SPEULO – SPHILM, and NaN <i>default</i> = NaN
\$SPTOL Set Point Tolerance		<i>range</i> = > = 0.0 <i>default</i> = 0.0
SP Set Point In Engineering Units Of The PV		<i>range</i> = SPLOLM to SPHILM <i>default</i> = 0.0
SPOPT Set Point Option Asp is invalid if PNTFORM=Componnt.	None Tv Asp	No specialized options are available Target value processing Advisory deviation alarming
RBOPT Ratio Bias Option	NoRatBi FixRatBi AutoRat AutoBi	No ratio/bias is used to calculate the SP Fixed ratio (R) and fixed bias (B) are used R is back calculated during initialization B is back calculated during initialization
RTHILM Ratio High Limit		<i>range</i> = RTLOLM – 100.0, & NaN <i>default</i> = 100.0
RTLOLM Ratio Low Limit		<i>range</i> = 0.0001 - RTHILM, & NaN <i>default</i> = 0.0001
BSHILM Bias High Limit		<i>range</i> = >=BSLOLM, & NaN <i>default</i> = 50.0
BSLOLM Bias Low Limit		<i>range</i> = <=BSHILM, & NaN <i>default</i> = -50.0
RATIO Ratio		<i>range</i> = RTLOLM to RTHILM <i>default</i> = 1.0
BIAS Bias Parameter		<i>range</i> = BSLOLM to BSHILM <i>default</i> = 0.0

APM PID with Position Proportional CTL Algorithm

Unit:	Tagname:	Engineer:	Date:
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Reg Control PidPosPr Algo Display

PIDFORM PID Controller Form	Interact Ideal	(Proportional + Integral) x (Derivative) (Proportional) +(Integral) + (Derivative)
CTLEQN (Pid) Control Equation Type	EqA EqB EqC EqD	P, I & D act on Error P & I act on Error, D acts on PV I acts on Error, P & D act on PV Integral-only control
PVTRACK PV Tracking Option	No Track Track	SP will never be set equal to PV Man mode or init. causes SP to track PV
CTLACTN Control Action	Reverse Direct	As PV increases, Output decreases As PV increases, Output increases

Reg Control Pid PosPr Algo Display (continued)

N/A if CTLEQN = EqD.

GAINOPT Gain (K) Option	Ext NonLin Gap Lin	K = KFF*KEXT (gain from external source) K = KFF*((NLFM)+(NLGAIN*[PVP-SPP]/[100]) K = KLIN*KGAP if PV>(SP-GAPLO) or if PV<(SP+GAPHI), else K = KLIN. K = the value entered by the user	
K Overall Gain In % Per %		range = 0.0 – 240.0 default = Not applicable	
KLIN Linear Gain Factor		range = 0.0 – 240.0 default = 1.0	
KGAP Gap Gain Factor		range = 0.0 to 1.0 default = 1.0	
GAPHI Gap High Limit		range = >=0.0 default = 0.0	
GAPLO Gap Low Limit		range = >=0.0 default = 0.0	
NLGAIN Non Linear Gain		range = 0.0 – 240.0 default = 0.0	
NLFM Non Linearity Form	0 1	K can go to zero K cannot go to < 1	
KEXT External Gain Modifier		range = 0.0 to 240.0 default = 1.0	

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T1			<i>range = 0.0 to 1440.0</i>
Integral Time In Minutes	_._._._._		<i>default = 0.0 (minutes)</i>
T2			<i>range = 0.0 to 1440.0</i>
Derivative Time In Minutes	_._._._._		<i>default = 0.0 (minutes)</i>
DEADBAND(PidPosProp)			<i>range = 0.0 to 100.0</i>
Dead Band In % Of Full Scale	_._._._._		<i>default = 5.0 %</i>
CYCLETIM			<i>range = 0.25 to 1000.0</i>
PidPosProp Output Cycle Time In Seconds	_._._._._		<i>default = 10.0 (seconds)</i>
K1			<i>range = 0.0 to 10.0</i>
Output Pulse Gain Value	_._._._._		<i>default = 1.0</i>

```
default = Null
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Unit:	Tagname:	Engineer:	Date:
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OPLOSRC		
OPLOFL Input Connection	(point.parameter)	<i>range</i> = See Parameter Ref.Guide <i>default</i> = Null
SAFOPCMD	<i>Idle</i>	<i>range</i> = Not Applicable
Safe OP Command	Raise Lower	<i>default</i> = Idle
LMSRC		
Local Manual Source	(point.parameter)	<i>range</i> = See Parameter Ref.Guide <i>default</i> = Null.null

Reg Control PidPosPr Algo Display (Continued)

NOCINPTS	_	<i>range = 1 to 2</i>
#Of Control Input Connections		<i>default = 1</i>
CISRC (Control-Input-Connection Source)		<u>Destination Para-</u>
<u>Input#</u>	<u>CISRC(n) (point.parameter)</u>	<u>-meter (CIDSTN)</u>
1		PVAUTO
2		SP

Alarming Display – N/A if PNTFORM = Component.

\$AUXUNIT Auxiliary Unit	_ _	<i>range</i> = Any configured LCN Unit ID <i>default</i> = NaN						
\$ALDLOPT (R620 and later) Alarm Delay Option	None OffDlyLo OffDlyHi OffDlyEm OnDlyLo OnDlyHi OnDlyEm	Off Delay Lo Off Delay Hi Off Delay Emergency On Delay Lo On Delay Hi On Delay Emergency						
\$DLBADAL (R681 and later) Delay for BADPV	None Offdelay	<i>range</i> = Offdelay, None <i>default</i> = None						
\$DLTIME (R620 and later) Delay Time	_ _ _	<i>range</i> = 0 to 9999 seconds <i>default</i> = 0 seconds						
\$REALARM (R660 and later) Re-alarms Interval	_ _ _	<i>range</i> = 0 to 9999 minutes <i>default</i> = 0 minutes						
\$BADCOPT BAD OP Conn Alarm Option	Off On	Bad Output Connection Alarm is not initially permitted Bad Output Connection Alarm is initially permitted						
\$BADOCPR BAD OP Conn Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint	

APM PID with Position Proportional CTL Algorithm

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Unit:	Tagname:	Engineer:	Date:
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Alarming Display N/A if PNTFORM = Compnnt (Continued)

	Half	One	Two	Three	Four	Five	EU	
PVALDB PV Alarm Deadband In %								
PVALDBEU PV Alarm Deadband in EU								<i>range</i> >= 0.0 <i>default</i> = 1.0
PVHITP PV High Alarm Trip Point								<i>range</i> = PVLOTP - PVHHTP, & NaN <i>default</i> = NaN
PVHIPR PV High Alarm Priority	NoAction (R670 and later)		Journal	Low	High	Emergency	Printer	JnlPrint
\$HISELCT (R681 and later) PV High Selective Cutout	Cutout Allow							<i>range</i> = Cutout, Allow <i>default</i> = Cutout
PVLOTP PV Low Alarm Trip Point								<i>range</i> = PVLLTP - PVHITP, & NaN <i>default</i> = NaN
PVLOPR PV Low Alarm Priority	NoAction (R670 and later)		Journal	Low	High	Emergency	Printer	JnlPrint
\$LOSELCT (R681 and later) PV Low Selective Cutout	Cutout Allow							<i>range</i> = Cutout, Allow <i>default</i> = Cutout
PVHHTP PV Hi Hi Alarm Trip Point								<i>range</i> = PVHITP to PVEUHI, & NaN <i>default</i> = NaN
PVHHPR PV High High Alarm Priority	NoAction (R670 and later)		Journal	Low	High	Emergency	Printer	JnlPrint
\$HHSELCT (R681 and later) PV High High Selective Cutout	Cutout Allow							<i>range</i> = Cutout, Allow <i>default</i> = Cutout
PVLLTP PV Low Low Alarm Trip Point								<i>range</i> = PVEULO – PVLOTP, & NaN <i>default</i> = NaN
PVSGCHTP PV Significant Change Alarm Trip Point								<i>range</i> = >=0.0, and NaN <i>default</i> = NaN
PVROCTP PV Pos Rate Of Change Trip Point								<i>range</i> = >=0.0, and NaN <i>default</i> = NaN
PVROCPPR PV Pos Rate of Chg Alarm Priority	NoAction (R670 and later)		Journal	Low	High	Emergency	Printer	JnlPrint
\$RPSELCT (R681 and later) PV Positive Rate of Change Selective Cutout	Cutout Allow							<i>range</i> = Cutout, Allow <i>default</i> = Cutout
PVROCNTP PV Neg Rate Of Change Trip Point								<i>range</i> = >=0.0, and NaN <i>default</i> = NaN

APM PID with Position Proportional CTL Algorithm

Unit:	Tagname:	Engineer:	Date:
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Alarming Display (Continued)

PVROCNP PV Neg Rate Of Chg Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint
\$RNSELECT (R681 and later) PV Negative Rate of Change Selective Cutout	Cutout Allow				<i>range = Cutout, Allow default = Cutout</i>		
PVLLPR PV Low Low Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint
\$LLSELECT (R681 and later) PV Low Low Selective Cutout	Cutout Allow				<i>range = Cutout, Allow default = Cutout</i>		
BADPVPR Bad-PV Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint
DEVHITP Deviation High Alarm Trip Point					<i>range = >=0.0, and NaN default = NaN</i>		
DEVHIPR Deviation High Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint
\$DHSELECT (R681 and later) Deviation High Selective Cutout	Cutout Allow				<i>range = Cutout, Allow default = Cutout</i>		
DEVLOTP Deviation Low Alarm Trip Point					<i>range = >=0.0, and NaN default = NaN</i>		
DEVLOPR Deviation Low Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint
\$DLSELECT (R681 and later) Deviation Low Selective Cutout	Cutout Allow				<i>range = Cutout, Allow default = Cutout</i>		
ADVDEVTP Advisory Deviation Alarm Trip Point (When SPOPT = ASP)					<i>range = >=0.0, and NaN default = NaN</i>		
ADVDEVPR Advisory Deviation Alarm Priority (When SPOPT = ASP)	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint
ALENBST Point Alarm Enable Status	Enable Disable Inhibit			<u>Displayed</u> Yes No No	<u>Logged</u> Yes Yes No	<u>Reported to EIP</u> Yes Yes No	

NIM Library Configuration

Unit:	Tagname:	Engineer:	Date:
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Network Number(NTWKNUM)

|_|_|

Library Number (LIBRYNUM)

|_|

range = 1 - 3

NIM Library Configuration Display

Index	Entry(LIBRYTXT)	Index	Entry(LIBRYTXT)	Index	Entry(LIBRYTXT)	Index	Entry(LIBRYTXT)
_01*	_ _ _ _ _ _ _	_02	_ _ _ _ _ _ _	_03	_ _ _ _ _ _ _	_04	_ _ _ _ _ _ _
_05	_ _ _ _ _ _ _	_06	_ _ _ _ _ _ _	_07	_ _ _ _ _ _ _	_08	_ _ _ _ _ _ _
_09	_ _ _ _ _ _ _	_10	_ _ _ _ _ _ _	_11	_ _ _ _ _ _ _	_12	_ _ _ _ _ _ _
_13	_ _ _ _ _ _ _	_14	_ _ _ _ _ _ _	_15	_ _ _ _ _ _ _	_16	_ _ _ _ _ _ _
_17	_ _ _ _ _ _ _	_18	_ _ _ _ _ _ _	_19	_ _ _ _ _ _ _	_20	_ _ _ _ _ _ _
_21	_ _ _ _ _ _ _	_22	_ _ _ _ _ _ _	_23	_ _ _ _ _ _ _	_24	_ _ _ _ _ _ _
_25	_ _ _ _ _ _ _	_26	_ _ _ _ _ _ _	_27	_ _ _ _ _ _ _	_28	_ _ _ _ _ _ _
_29	_ _ _ _ _ _ _	_30	_ _ _ _ _ _ _	_31	_ _ _ _ _ _ _	_32	_ _ _ _ _ _ _
_33	_ _ _ _ _ _ _	_34	_ _ _ _ _ _ _	_35	_ _ _ _ _ _ _	_36	_ _ _ _ _ _ _
_37	_ _ _ _ _ _ _	_38	_ _ _ _ _ _ _	_39	_ _ _ _ _ _ _	_40	_ _ _ _ _ _ _
_41	_ _ _ _ _ _ _	_42	_ _ _ _ _ _ _	_43	_ _ _ _ _ _ _	_44	_ _ _ _ _ _ _
_45	_ _ _ _ _ _ _	_46	_ _ _ _ _ _ _	_47	_ _ _ _ _ _ _	_48	_ _ _ _ _ _ _
_49	_ _ _ _ _ _ _	_50	_ _ _ _ _ _ _				

NIM Library Configuration Display

_51	_ _ _ _ _ _ _	_52	_ _ _ _ _ _ _	_53	_ _ _ _ _ _ _	_54	_ _ _ _ _ _ _
_55	_ _ _ _ _ _ _	_56	_ _ _ _ _ _ _	_57	_ _ _ _ _ _ _	_58	_ _ _ _ _ _ _
_59	_ _ _ _ _ _ _	_60	_ _ _ _ _ _ _	_61	_ _ _ _ _ _ _	_62	_ _ _ _ _ _ _
_63	_ _ _ _ _ _ _	_64	_ _ _ _ _ _ _	_65	_ _ _ _ _ _ _	_66	_ _ _ _ _ _ _
_67	_ _ _ _ _ _ _	_68	_ _ _ _ _ _ _	_69	_ _ _ _ _ _ _	_70	_ _ _ _ _ _ _
_71	_ _ _ _ _ _ _	_72	_ _ _ _ _ _ _	_73	_ _ _ _ _ _ _	_74	_ _ _ _ _ _ _
_75	_ _ _ _ _ _ _	_76	_ _ _ _ _ _ _	_77	_ _ _ _ _ _ _	_78	_ _ _ _ _ _ _
_79	_ _ _ _ _ _ _	_80	_ _ _ _ _ _ _	_81	_ _ _ _ _ _ _	_82	_ _ _ _ _ _ _
_83	_ _ _ _ _ _ _	_84	_ _ _ _ _ _ _	_85	_ _ _ _ _ _ _	_86	_ _ _ _ _ _ _
_87	_ _ _ _ _ _ _	_88	_ _ _ _ _ _ _	_89	_ _ _ _ _ _ _	_90	_ _ _ _ _ _ _
_91	_ _ _ _ _ _ _	_92	_ _ _ _ _ _ _	_93	_ _ _ _ _ _ _	_94	_ _ _ _ _ _ _
_95	_ _ _ _ _ _ _	_96	_ _ _ _ _ _ _	_97	_ _ _ _ _ _ _	_98	_ _ _ _ _ _ _
_99	_ _ _ _ _ _ _	_00	_ _ _ _ _ _ _				

Note that each library can contain upto 1000 entries; therefore, upto 5 copies of this form may be needed to configure each library.

*No entry allowed for Library 1, Index 001

NIM Library Configuration

Unit:	Tagname:	Engineer:	Date:
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Network Number(NTWKNUM)

|_|_|

Library Number (LIBRYNUM)

|_|

range = 1 - 3

NIM Library Configuration Display

Index	Entry(LIBRYTXT)	Index	Entry(LIBRYTXT)	Index	Entry(LIBRYTXT)	Index	Entry(LIBRYTXT)
_01*	_ _ _ _ _ _ _	_02	_ _ _ _ _ _ _	_03	_ _ _ _ _ _ _	_04	_ _ _ _ _ _ _
_05	_ _ _ _ _ _ _	_06	_ _ _ _ _ _ _	_07	_ _ _ _ _ _ _	_08	_ _ _ _ _ _ _
_09	_ _ _ _ _ _ _	_10	_ _ _ _ _ _ _	_11	_ _ _ _ _ _ _	_12	_ _ _ _ _ _ _
_13	_ _ _ _ _ _ _	_14	_ _ _ _ _ _ _	_15	_ _ _ _ _ _ _	_16	_ _ _ _ _ _ _
_17	_ _ _ _ _ _ _	_18	_ _ _ _ _ _ _	_19	_ _ _ _ _ _ _	_20	_ _ _ _ _ _ _
_21	_ _ _ _ _ _ _	_22	_ _ _ _ _ _ _	_23	_ _ _ _ _ _ _	_24	_ _ _ _ _ _ _
_25	_ _ _ _ _ _ _	_26	_ _ _ _ _ _ _	_27	_ _ _ _ _ _ _	_28	_ _ _ _ _ _ _
_29	_ _ _ _ _ _ _	_30	_ _ _ _ _ _ _	_31	_ _ _ _ _ _ _	_32	_ _ _ _ _ _ _
_33	_ _ _ _ _ _ _	_34	_ _ _ _ _ _ _	_35	_ _ _ _ _ _ _	_36	_ _ _ _ _ _ _
_37	_ _ _ _ _ _ _	_38	_ _ _ _ _ _ _	_39	_ _ _ _ _ _ _	_40	_ _ _ _ _ _ _
_41	_ _ _ _ _ _ _	_42	_ _ _ _ _ _ _	_43	_ _ _ _ _ _ _	_44	_ _ _ _ _ _ _
_45	_ _ _ _ _ _ _	_46	_ _ _ _ _ _ _	_47	_ _ _ _ _ _ _	_48	_ _ _ _ _ _ _
_49	_ _ _ _ _ _ _	_50	_ _ _ _ _ _ _				

NIM Library Configuration Display

_51	_ _ _ _ _ _ _	_52	_ _ _ _ _ _ _	_53	_ _ _ _ _ _ _	_54	_ _ _ _ _ _ _
_55	_ _ _ _ _ _ _	_56	_ _ _ _ _ _ _	_57	_ _ _ _ _ _ _	_58	_ _ _ _ _ _ _
_59	_ _ _ _ _ _ _	_60	_ _ _ _ _ _ _	_61	_ _ _ _ _ _ _	_62	_ _ _ _ _ _ _
_63	_ _ _ _ _ _ _	_64	_ _ _ _ _ _ _	_65	_ _ _ _ _ _ _	_66	_ _ _ _ _ _ _
_67	_ _ _ _ _ _ _	_68	_ _ _ _ _ _ _	_69	_ _ _ _ _ _ _	_70	_ _ _ _ _ _ _
_71	_ _ _ _ _ _ _	_72	_ _ _ _ _ _ _	_73	_ _ _ _ _ _ _	_74	_ _ _ _ _ _ _
_75	_ _ _ _ _ _ _	_76	_ _ _ _ _ _ _	_77	_ _ _ _ _ _ _	_78	_ _ _ _ _ _ _
_79	_ _ _ _ _ _ _	_80	_ _ _ _ _ _ _	_81	_ _ _ _ _ _ _	_82	_ _ _ _ _ _ _
_83	_ _ _ _ _ _ _	_84	_ _ _ _ _ _ _	_85	_ _ _ _ _ _ _	_86	_ _ _ _ _ _ _
_87	_ _ _ _ _ _ _	_88	_ _ _ _ _ _ _	_89	_ _ _ _ _ _ _	_90	_ _ _ _ _ _ _
_91	_ _ _ _ _ _ _	_92	_ _ _ _ _ _ _	_93	_ _ _ _ _ _ _	_94	_ _ _ _ _ _ _
_95	_ _ _ _ _ _ _	_96	_ _ _ _ _ _ _	_97	_ _ _ _ _ _ _	_98	_ _ _ _ _ _ _
_99	_ _ _ _ _ _ _	_00	_ _ _ _ _ _ _				

